



Dinutuximab beta

Mouse-human chimeric monoclonal anti-disialoganglioside (GD2) IgG1 antibody

Investigator's Brochure

CONFIDENTIAL

EUSA Pharma
Breakspear Way
Hemel Hempstead
HP2 4TZ
United Kingdom

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Signature

Date

(12 Feb 2021)

Darrel Cohen, MD, PhD
VP Head of Clinical Development

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Abbreviations

13-cis-RA	13-cis retinoic acid, Isotretinoin
ADCC	Antibody-dependent cellular cytotoxicity
ADA	Anti-drug antibodies
AE	Adverse event
ALT	Alanine transaminase
APN311	Former development code for dinutuximab beta
ASCT	Autologous stem cell transplantation
AST	Aspartate transaminase
AUC	Area under the time concentration curve
BuMel	Busulfan and melphalan
cCR	Continued complete response
CDC	Complement dependent cytotoxicity
ch14.18/CHO	Dinutuximab beta: a chimeric 14.18 anti-GD2 monoclonal antibody produced in Chinese hamster ovary cells
CHO	Chinese hamster ovary
C _{max}	Maximum observed serum concentration
COJEC	Cisplatin, vincristine, carboplatin, etoposide, and cyclophosphamide
CLS	Capillary leak syndrome
CR	Complete response
CT	Computed tomography
CTCAE	Common terminology criteria for adverse events
EC ₅₀	Half maximal effective concentration
ECG	Electrocardiogram
EEA	European Economic Area
EFS	Event-free survival
ELISA	Enzyme linked immunosorbent assay
FAS	Full analysis set
GD2	Disialoganglioside antigen GD2
GGT	Gamma glutamyl transferase
GvHD	Graft versus host disease
GM-CSF	Granulocyte macrophage colony-stimulating factor
HACA	Human anti-chimeric antibodies
Haplo-SCT	Haploidentical stem cell transplantation
HSCT	Hematopoietic stem cell transplant
IgG	Immunoglobulin G
IL-2 (6)	Interleukin 2 (6)
INSS	International Neuroblastoma Staging System
INN	International Nonproprietary Name
i.v.	Intravenous
IQR	Inter-quartile range
KUSS	The Childhood Discomfort and Pain Scale
LTI	Long-term infusion
mIBG	^{123/131} Iodine-meta-iodobenzylguanidine

MRI	Magnetic resonance imaging
MRD	Minimal residual disease
NK	Natural killer cells
OD	Optical density
ORR	Overall response rate
OS	Overall survival
PBSCR	Peripheral blood stem cell rescue
PD	Progressive disease
PFS	Progression-free survival
PK	Pharmacokinetics
PR	Partial response
PRES	Posterior reversible encephalopathy syndrome
PSUR	Periodic Safety Update Report
RECIST	Response evaluation criteria in solid tumors
SAE	Serious adverse event
s.c.	Subcutaneous
SD	Stable disease or Standard deviation
SmPC	Summary of Product Characteristics
SP2/0	Murine, non-secreting myeloma cells
STI	Short-term infusion
TVD	Topotecan, vincristine, doxorubicin
$t_{\max}$	Time to maximum serum concentration
VGPR	Very good partial response
vs	Versus
WHO	World Health Organization

1. Summary

Dinutuximab beta (ch14.18/CHO) is a mouse-human chimeric monoclonal immunoglobulin G1 (IgG1) antibody specific to the disialoganglioside GD2 antigen. The GD2 antigen is highly expressed by neuroblastoma cells and is a target for high-risk neuroblastoma treatment.

Dinutuximab beta is administered as either an 8-hour short-term intravenous (i.v.) infusion over 5 consecutive days, or as a long-term i.v. infusion over 10 consecutive days. The dose of dinutuximab beta depends on the body surface area or body weight of patients and is equal to a total dose of 100 mg/m²/course.

1.1 Non-clinical studies

Dinutuximab beta specifically binds to GD2 *in vitro*. It mediates both complement dependent and antibody-dependent cell-mediated cytotoxicity (CDC and ADCC) against GD2 positive neuroectodermal tumor cell lines. In a neuroblastoma mouse model, dinutuximab beta demonstrated dose-dependent and antigen specific suppression of metastases. Dinutuximab beta is metabolized as all large proteins via unspecific proteolysis to small peptides and amino acids.

Administration of dinutuximab beta to guinea pigs resulted in drug-related toxicities comprised of clinical signs (slightly reduced motility), a decrease in body weight and food consumption, changes in hematology (decreases in white blood cells, reticulocytes and platelets), changes in blood chemistry (increases in activated partial thromboplastin time, plasma cholesterol, urea, alpha-amylase and glutamate dehydrogenase activity, decreases in plasma creatinine and glucose), increased urine volume, reduction in absolute and relative liver organ weight and bone marrow findings (suppression). These observations are well in accordance with the expected toxicities of dinutuximab beta, i.e. induction of pain by binding to peripheral nerve tissue structures (resulting in the observed, dose-dependent clinical signs, food and body weight decreases) and interference with the hematopoietic system/bone marrow. Repeated administration of dinutuximab beta resulted also in the induction of anti-drug antibodies, potentially interfering with the pharmacokinetics (PK) of dinutuximab beta as well as possibly compensating dinutuximab beta-induced clinical signs and hematological changes.

Similar toxicities were observed in cynomolgus monkeys after repeated administration of dinutuximab beta including reduced overall activity, body weight and food consumption decrease, bone marrow findings and changes in thymus size and weight which are all in line with the expected toxicities of dinutuximab beta. The findings partially or fully recovered within about 3 weeks. Two animals died. The cause of the 2 deaths is unknown but may have been dinutuximab beta-related due to a worsened general condition (pain caused wasting syndrome). No observed adverse effect level could be determined, and the maximum tolerated dose was 2.5 mg/kg/day (30 mg/m²/day).

1.2 Effects in humans

Pharmacology, safety and therapeutic efficacy of dinutuximab beta is available from 6 studies performed in patients with relapsed/refractory neuroblastoma or minimal residual disease (MRD). Dinutuximab beta was administered in combination with subcutaneous (s.c.) interleukin (IL-2) or as a single therapy, and either as a 10-day continuous long-term infusion

(LTI) or as a short-term infusion (STI) regimen (8-hour infusion over 5 days). Overall more than 1200 patients have received dinutuximab beta.

1.2.1 Pharmacology

Cytotoxic effector functions in patient sera treated with dinutuximab beta correlated with dinutuximab beta infusion. ADCC and CDC increased strongly from start till end of the ch14.18/CHO infusion. Dinutuximab beta combined with IL-2 treatment appeared to increase the number of activated natural killer (NK) cells and the number of cytotoxic T-cells. In addition, dinutuximab beta appeared to induce monocytes and all types of granulocytes. Human anti-chimeric antibodies (HACA) were observed in about 47% of patients (65 patients out of 137 tested positive) (see Section 5.2.2).

Concerning PK data, maximum dinutuximab beta concentrations were achieved at the end of the infusion, i.e. after about 220 hours under a 10-day 24-hour infusion and at about 80 hours after an 8-hour interval infusion. The mean half-life of dinutuximab beta was longer with a 10-day 24-hour infusion than with an 8-hour interval infusion (8 days vs 4 days). Area under the time concentration curve (AUC) values were higher with the 10-day 24-hour infusion than with an 8-hour interval infusion, while maximum concentrations were higher with the 8-hour interval infusion regimen (see Section 5.2.1).

1.2.2 Efficacy

Dinutuximab beta either given as STI (5 × 8 hours) or LTI (10 days, 24 hours) demonstrated significant clinical benefits to patients with high-risk neuroblastoma in terms of overall response rates (ORR), event-free survival (EFS), and overall survival (OS). ORR, EFS and OS across studies are presented in Table 1. In historical control groups, OS at 2 years was between 20% and 30%.

Table 1: Efficacy of dinutuximab beta in clinical studies in European patients

Study APN311-							
	303 ^a	202 ^a	201 ^b	101 ^{bc}	304 ^{ad}	302 ^b	
	(N=54)	(N=124)	(N=68)	(N=15)	(N=38)	w/o IL-2 (N=466)	with IL-2 (N=378)
Evaluable (N) for response	37	77	20	12	38	188	204
Response (CR, PR, VGPR)	41%	57.1%	50%	33%	36.8%	44%	51%
Evaluable (N) for EFS+OS	53	118	68	-	-	466	378
EFS rates at 3 years	31%	51.7%	47%	ND	-	55%	61%
EFS rates at 5 years	-	48.2%	-	-	-	42%	57%
OS rates at 3 years	66%	67.0%	57.9%	ND	-	64%	69%
OS rates at 5 years	-	57.8%	-	-	-	50%	64%

^a Continuous 24-hour infusion.

^b 8-hour infusion. The exposure data for the 302 study was based on the final approved clinical study report for the original study, but additional exposure has been disclosed by the study Sponsor to EUSA Pharma, and additional patients have been included in the publication by Ladenstein et al. [1]

^c No IL-2 co-treatment

^d Of the 38 patients, 1 patient died 36 days after the start of treatment. For the remaining 37 patients, the observations were censored. Because of the large number of censored observations, no reliable mean OS could be determined.

CR = complete response, EFS = event-free survival, IL-2 = interleukin 2, N = number of patients, ND = not determined, OS = overall survival, PR = partial response, VGPR = very good partial response, w/o = without.

The efficacy was independent of disease state at baseline. However, compared to multimodality treatment for relapsed/refractory neuroblastoma without immunotherapy, treatment including dinutuximab beta showed significant advantages. Also, the effects were independent of IL-2 co-administration. Monotherapy with dinutuximab beta showed similar survival outcomes as combination treatment with IL-2 when used as part of the maintenance phase of first-line therapy for high-risk neuroblastoma patients. SIOPEX recommend the use of dinutuximab beta without cytokine (s.c. IL-2) according to the LTI schedule. [11] The results are also comparable to therapy with ch14.18/SP2/0 (ch14.18 monoclonal antibody produced in murine SP2/0 cells) plus i.v. IL-2 and granulocyte macrophage colony-stimulating factor (GM-CSF) although this asset was assessed in a population of complete responders only. [2]

1.2.3 Safety

The most prominent adverse reaction with dinutuximab beta is severe neuropathic pain, especially during the first courses, due to the binding of dinutuximab beta to GD2 on nerve cells. Standard pain management including opioids, gabapentin, and nonsteroidal anti-inflammatory drugs should thus be established by experienced physicians when

dinutuximab beta is administered. Hypersensitivity reactions are commonly observed with monoclonal antibody treatment and were also frequently observed with dinutuximab beta treatment including pyrexia, pruritus, tachycardia, and hypertension. Capillary leak syndrome is a common adverse reaction of IL-2 and has also occurred frequently with dinutuximab beta. Other frequently reported adverse events (AEs) varied across studies but commonly included constipation, nausea, vomiting, cough, increased liver enzymes, anemia, decreased neutrophils, and decreased platelets. Most AEs were Grade 1 (mild) to Grade 3 (severe) in severity; Grade 4 events occurred with incidences between 6% and 45% across studies. A continuous treatment regimen led to a clearly improved pain-toxicity profile compared with the 8-hour infusion regimen. The omission of s.c. IL-2 as co-treatment with the 8-hour infusion regimen greatly reduced side effects. To ensure patient's safety, administration of dinutuximab beta must be performed by a healthcare professional well experienced in managing severe allergic reactions including anaphylaxis. Infusion of dinutuximab beta must be initiated in an environment where full resuscitation services are immediately available.

2. Introduction

2.1 Description

Dinutuximab beta is a mouse-human chimeric monoclonal IgG1 antibody specific to the disialoganglioside GD2 antigen. The GD2 antigen is highly expressed by neuroblastoma cells.

The structure of the human/mouse chimeric anti-ganglioside GD2 antibody dinutuximab beta corresponds to a chimeric monoclonal antibody with approximately 30% mouse and 70% human sequences. Constant regions are derived from human IgG1. The 30% mouse sequences were used from murine IgG2a monoclonal antibody 14.G2a.

2.2 Disease background

Neuroblastoma commonly refers to a spectrum of neuroblastic tumors (including neuroblastoma, ganglioneuroblastoma, and ganglioneuroma). Neuroblastomas have a rather heterogeneous character and may vary in terms of location, histopathologic appearance, and biologic characteristics as well as in their clinical behavior, which can cover a broad spectrum from spontaneous regression, to maturation to a benign ganglioneuroma, or aggressive disease with metastatic dissemination leading to death.

Neuroblastoma can be classified based on the degree of neuroblastic differentiation (undifferentiated, poorly differentiated, and differentiating) and the mitosis-karyorrhexis index (low, intermediate, or high). Histologically, neuroblastoma is characterized by limited Schwannian cell production, stroma-poor cells, and abundant neuroblasts. [3]

High-risk neuroblastoma, a mainly pediatric disease, remains one of the major challenges in pediatric oncology and is characterized at diagnosis by metastatic disease beyond 18 months of age and unfavorable biological features like e.g. v-myc myelocytomatosis viral related oncogene (MYC-N) amplification as the most prominent one. [4] Despite the introduction of novel treatment strategies, including high dose chemotherapy followed by autologous bone marrow or stem cell transplantation, the outcome of these high-risk patients remains poor. Development of effective adjuvant therapeutic strategies appears to be the only realistic way to further improve outcome in this disease. In this regard, there has been growing interest in major areas of adjuvant therapy of malignant diseases including tumor-specific immunotherapy. The outcome of patients with refractory/relapsing high-risk neuroblastoma is particularly poor and equally requires new innovative therapies. [5]

2.3 Rationale for clinical investigation of dinutuximab beta

In the development of novel immunotherapies for high-risk cancers, one goal is to find tumor targets that are not widely shared by normal cells. One such target is the carbohydrate disialoganglioside antigen GD2. Several high-risk tumors frequently express GD2, making it an attractive target for relatively tumor-specific therapies such as antibody therapy. Expression of GD2 in normal tissues is mostly restricted to neurons that are protected from the effects of i.v. monoclonal antibodies by the blood-brain barrier. However, GD2 is also expressed to a limited extent on peripheral nerve fibers. In a National Cancer Institute Pilot Project for the Acceleration of Translational Research, ganglioside GD2 was prioritized at position 12 of 76 ranked cancer antigens. [6]

Anti-GD2 antibodies target disialoganglioside antigen GD2 which is expressed on virtually all neuroblastoma cells. “First generation” antibodies were entirely mouse products causing a quick immunological response with the production of human anti-mouse antibodies which neutralized the action of the murine antibody. Dinutuximab beta belongs to the “second generation” anti-GD2 antibodies, using the murine variable genes of 14.18 and the human constant IgG1 and κ genes, known to effectively mediate ADCC and to maintain CDC.

The first chimeric anti-GD2 antibody was produced in murine, non-secreting myeloma cells (ch14.18/SP2/0, dinutuximab) [7] and was used in several academic driven clinical trials either as monotherapy [8-13] or in combination with cytokines such as IL-2 and GM-CSF. [2, 14-17] Later, the dinutuximab beta (ch14.18) construct was recloned from a murine cell line into Chinese hamster ovary (CHO) cells leading to dinutuximab beta (ch14.18/CHO). Preclinical testing revealed no relevant difference between ch14.18/SP2/0 (dinutuximab; Unituxin) and dinutuximab beta concerning binding to disialoganglioside GD2. However, the glycosylation pattern varies between the different production cell lines, i.e. dinutuximab beta shows considerably lower glycosylation (α -gal epitopes) compared to dinutuximab. The absence of α -gal epitopes is a guarantee to avoid possible α -gal epitope associated anaphylactic reactions, which may have a fatal outcome for patients. It can also help to reduce the severity and frequency of treatment-associated allergic reactions.

Dinutuximab beta was approved by the EMA on 8 May 2017 for the treatment of high-risk neuroblastoma in patients aged 12 months and above, who have previously received induction chemotherapy and achieved at least a partial response (PR), followed by myeloablative therapy and stem cell transplantation, as well as patients with history of relapsed or refractory neuroblastoma, with or without residual disease.

Clinical studies to particularly evaluate the use of dinutuximab beta as single agent therapy as part of the first-line therapy (maintenance phase) as well as in patients with relapsed or refractory neuroblastoma are currently ongoing.

3. Physical, chemical, and pharmaceutical properties and formulation

International Nonproprietary Name (INN): Dinutuximab beta

Company code: APN311

Other compound code: Ch14.18/CHO

Chemical name: Mouse/human chimeric anti-GD2 monoclonal antibody

CAS number: 1613303-02-8

Molecular formula: C₆₄₂₂H₉₉₇₈N₁₇₂₂O₂₀₀₆S₄₈ (without glycosylation)
C₆₅₄₆H₁₀₁₈₂N₁₇₃₀O₂₀₉₄S₄₈ (with glycosylation: evaluated on the basis of the most abundant glycoside pattern)

CAS name: Immunoglobulin G1, anti- (ganglioside GD2) (human-Mus musculus monoclonal dinutuximab beta heavy chain), disulfide with human-Mus musculus monoclonal dinutuximab beta light chain, dimer

Molecular weight: Approximately 150 kDa

Immunoglobulin subclass: IgG1
produced in CHO cells

Structure: Heterotetramer consisting of 2 heavy chains (each 443 amino acids) and 2 light chains (each 220 amino acids).
30% mouse sequences and 70% human sequences. Constant regions are derived from human IgG1. The 30% mouse sequences are the ones from the parent antibody, the monoclonal murine GD2 specific IgG2a antibody 14.G2a.

Trade name: QARZIBA[®]

Formulation and excipients: Concentrate for solution for infusion. The concentrate is a colorless to slightly yellow clear solution provided in 6 mL sterile glass vials with rubber stoppers and caps containing 4.5 mL.

Concentration: 4.5 mg/mL

Excipients: 20 mM histidine, 5% sucrose, 0.01% Polysorbate 20

The drug product has to be diluted before administration.

Storage conditions:

Refrigerated, 2°C to 8°C, protected from light

Shelf-life:

Concentrate: 36 months

Solution: Dinutuximab beta solution for infusion is for immediate use only. Post-reconstitution/dilution of the product, the solution for infusion must not be stored for more than 4 hours at 2°C to 8°C before initiation of infusion. From a physico-chemical point of view, in-use stability has been demonstrated for up to 48 hours at 25°C (50 mL syringe) and for up to 7 days at 37°C (250 mL infusion bag).

Posology and method of administration:

Dinutuximab beta is pre-diluted aseptically to the patient-specific concentration/dose with 0.9% sodium chloride solution, containing 1% human albumin. After dilution, the dinutuximab beta-containing solution is administered via i.v. infusion, using a commercially available mini-pump.

Treatment with dinutuximab beta consists of 5 consecutive cycles, each cycle comprising 35 days. Dinutuximab beta may be administered continuously over the first 10 days of a cycle at the daily dose of 10 mg/m² and can also be administered as 5 daily discontinuous infusions of 20 mg/m², each over 8 hours, on the first 5 days of each course.

Infusion times and rate may be extended or lowered, respectively, to increase tolerability.

The dose of dinutuximab beta is calculated individually, depending on the body surface area/body weight of patients. The recommended total dose of dinutuximab beta per cycle is 100 mg/m² body surface area.

4. Non-clinical studies

4.1 Non-clinical pharmacology

4.1.1 Primary pharmacology

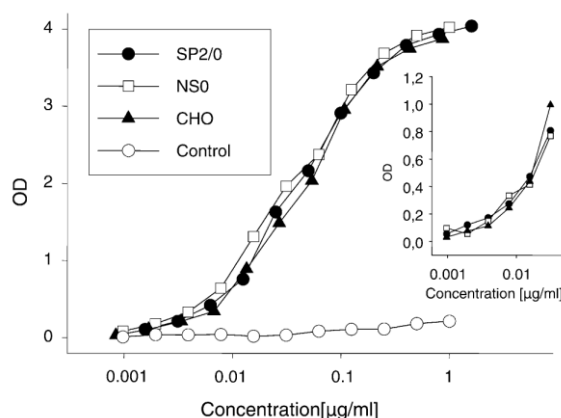
In vitro studies

Several investigations were conducted to elucidate dinutuximab beta's mechanism of action and pharmacological activities. Some experiments have been conducted with similar antibodies, i.e. ch14.18/SP2/0 and ch14.18/NS0, comprising the same amino acid sequence as dinutuximab beta, but produced in a different expression system, i.e. murine SP2/0 cells and murine NS0 cells, respectively. [18-20] In addition, 14G.2a, a fully murine antibody, was used for comparison.

Receptor binding

Dinutuximab beta specifically binds to the disialoganglioside antigen GD2 (dissociation constant = 22 nM) as evidenced by binding assays using bovine GD2 coated enzyme linked immunosorbent assay (ELISA) plates, surface plasmon resonance technology, as well as by a competition ELISA analysis (half maximal effective concentration [EC₅₀] = 25 ng/mL). The binding affinity was similar to that of 14G.2a, ch14.18/SP2/0, and ch14.18/NS0 (Figure 1) (Study 01-2014 and [20]).

Figure 1: Binding of dinutuximab beta, ch14.18/SP2/0, and ch14.18/NS0 to ganglioside GD2 in solid phase ELISA



Rituximab was used as a negative control. Error bars are covered by the symbols. The insert represents a second experiment, covering the low concentration range.

CHO = Chinese hamster ovary, ELISA = enzyme linked immunosorbent assay, OD = optical density, SP2/0 = Murine, non-secreting myeloma cells.

Source: [20]

The binding was GD2 specific, because precursor- and metabolite-gangliosides of GD2 were not recognized by dinutuximab beta. [20]

In addition, binding of the Fc part of dinutuximab beta to CD16 (FcγRIII) was also demonstrated via surface plasmon resonance technology.

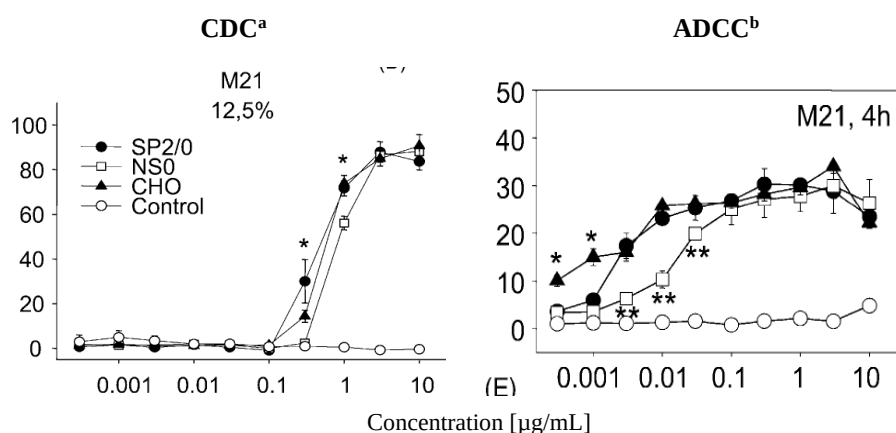
Specific binding of dinutuximab beta on GD2-positive tumor cell lines, M21 (melanoma) and LAN-1 (neuroblastoma), was demonstrated using flow cytometry analysis.

Effector function: CDC, ADCC, and whole blood cytolytic activity

Dinutuximab beta showed strong cytolytic activities on the GD2-positive cell lines M21 and LAN-1 in a CDC assay (EC_{50} values 138 and 78 ng/mL), in an ADCC assay (EC_{50} value 4.4 ng/mL), and in a whole blood test (EC_{50} values 140 and 15 ng/mL).

CDC-mediated specific tumor cell lysis of dinutuximab beta was similar to that with ch14.18 produced in other cell lines (Figure 2). With respect to ADCC, dinutuximab beta triggered a significantly stronger ADCC activity than other dinutuximab beta (ch14.18) products at low concentrations (0.0005 and 0.001 μ g/mL), most probably due to the different glycosylation pattern (Figure 2).

Figure 2: CDC and ADCC of dinutuximab beta, ch14.18/SP2/0, and ch14.18/NS0 against M21 cells



Rituximab was used as a negative control.

Different antibody dilutions in healthy blood serum (CDC) or healthy human peripheral blood mononuclear cells (ADCC) were incubated with ^{51}Cr -labeled target cells (LAN-1). ^{51}Cr (radioactivity) release was measured, which is proportional to cell lysis.

^a CDC measured in 12.5% serum at concentrations between 0 - 10 μ g/mL. * = $p < 0.01$.

^b Using 4 hours of incubation, at concentrations between 0 - 10 μ g/mL.

* = $p < 0.01$ (between dinutuximab beta and ch14.18/SP2/0 or ch14.18/NS0).

** = $p < 0.01$ (between ch14.18/NS0 and ch14.18/SP2/0 or ch14.18/CHO).

ADCC = antibody-dependent cytotoxicity, CDC = complement dependent cytotoxicity, Cr = chromium.

Source: [20]

In vivo studies

In a neuroblastoma mouse model, dinutuximab beta demonstrated dose-dependent and antigen specific suppression of metastases. *In vivo* depletion of natural killer (NK) cells completely abrogated this effect, suggesting that the mechanism involved in the dinutuximab beta-induced anti-neuroblastoma effect in this model is mediated by NK-dependent ADCC. [20]

4.1.2 Secondary pharmacology

No secondary pharmacology studies were performed.

4.1.3 Safety pharmacology

No standalone safety pharmacology studies have been performed. However, a screening of neurological and respiratory functions included into repeat-dose toxicity studies in guinea pigs

and cynomolgus monkeys (see Section 4.3.2 for more details) showed no treatment-related effects.

4.2 Pharmacokinetics and product metabolism in animals

No classical absorption, distribution, metabolism, excretion studies were performed for dinutuximab beta. The PK profile of dinutuximab beta was investigated in guinea pigs after single i.v. 24-hour and a 10-day continuous infusion of dinutuximab beta and, comparable to ch14.18/SP2/0, in mice after a single i.v. injection. Dinutuximab beta serum concentrations were measured using ELISA methods.

4.2.1 Pharmacokinetics

Single-injection pharmacokinetics of ch14.18/SP2/0

The elimination half-life of 1 mg/kg (= 3 mg/m²) ch14.18/SP2/0 after single i.v. injection was ~60 hours in athymic mice and 118 hours in BALB/c mice. The AUC for ch14.18/SP2/0 was 500 µg/mL/h after a single i.v. injection (1 mg/kg) in BALB/c mice. [18,21]

Pharmacokinetics of dinutuximab beta after 24-hours and 10-day i.v. administration in guinea pigs

Serum PK characteristics of dinutuximab beta after a continuous 24-hour i.v. infusion in guinea pigs at 1.25, 6.25, and 12.25 mg/kg body weight (N = 9 per group) are shown in Table 2. Dinutuximab beta maximum observed serum concentration (C_{max}) and exposure exhibited dose-dependent, linear PK over the administered dose range. The terminal half-lives (t_{1/2}) were similar between doses (~40 hours) with a slight trend towards increasing values with increasing dose.

Table 2: Pharmacokinetic characteristics of dinutuximab beta after a 24-hour continuous i.v. infusion in serum of male guinea pigs

Dose (mg/kg BW [d])	C _{max} ^a (µg/mL)	t _{max} ^a (h)	t _{1/2} (h)	AUC ₀₋₁₄₄ (µg·d/mL)	AUC _{0-∞} (µg·d/mL)	CL (mL/d/kg)	V _z (mL/kg)
1.25	18.1	24.0	36.7	926.8	1001.7	1.25	66.0
6.25	82.4	24.0	38.4	4529.8	4933.0	1.27	70.2
12.25	173.4	24.0	43.4	10035.7	11142.2	1.12	70.2

^a Values obtained from analytical results, all other values were calculated by pharmacokinetic analysis.

AUC = area under the time concentration curve, BW = body weight, CL = serum clearance, C_{max} = maximum observed serum concentration, t_{max} = time to maximum observed serum concentration, t_{1/2} = half-life, V_z = volume of distribution.

Serum PK characteristics of dinutuximab beta after a 10-day continuous i.v. infusion in guinea pigs at 1.25, 6.25, and 12.25 mg/kg body weight are shown in Table 3 and also indicated as dose-dependent, linear PK over the administered dose range. No clear sex differences in exposure levels were observed. Compared with a 24-hour infusion (Table 2), C_{max} levels were similar while exposure (AUC) levels were 3- to 4 fold lower after the 10-day infusion regimen. The lower AUC values may indicate an early anti-dinutuximab beta antibody response (see Section 4.1.3), which compensates for dinutuximab beta accumulation resulting in an increased clearance or which may interfere with the ELISA method.

Table 3: Pharmacokinetic characteristics of dinutuximab beta after a 10-day continuous i.v. infusion in serum of male and female guinea pigs

Sex	Dose (mg/kg BW [d])	C _{max} ^a (µg/mL)	t _{max} ^a (h)	AUC _{0-∞} (µg·d/mL)	t _{1/2α} (h) ^c	t _{1/2β} (h) ^d
Male	1.25	9	8	291	11.5	31.9
	6.25	117	8	2200	5.9	95.9
	12.25	213	8	4045	6.4	61.8
Female	1.25	4	8	82 ^b	3.8	n.c.
	6.25	104	8	2027	4.5	54.8
	12.25	184	8	3382	6.1	38.0

Data were obtained in a repeat-dose toxicity study described in detail in Section 4.3.2.

^a Measured on Day 10. Values obtained from analytical results, all other values were calculated by PK analysis.

^b AUC₀₋₃₆ value.

^c t_{1/2α} elimination phase values were calculated based on time points ranging from 24 to 48 hours after end of infusion.

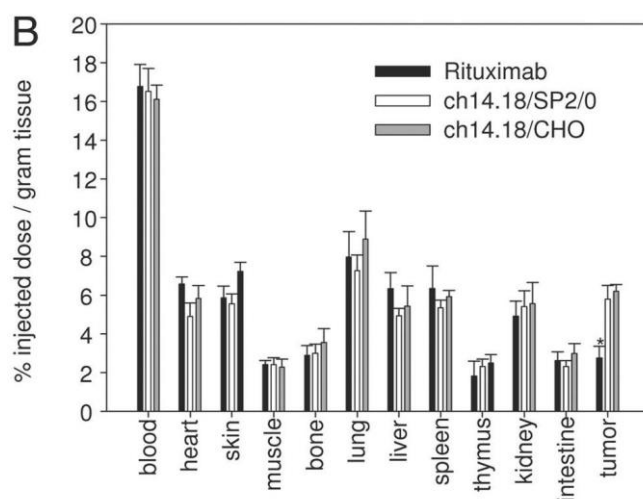
^d t_{1/2β} elimination phase values were calculated based on time points ranging from 48 to t_{last} hours after end of infusion.

AUC = area under the time concentration curve, BW = body weight, CL = serum clearance, C_{max} = maximum observed serum concentration, d = day, h = hour, n.c. = value could not be calculated, t_{max} = time to maximum observed serum concentration (the 8-hour time point on Day 10 was the last sampling time point before the infusion was ended), t_{1/2} = half-life.

Pharmacokinetics and biodistribution of dinutuximab beta and ch14.18/SP2/0 in mice after single i.v. injection

Dinutuximab beta and ch14.18/SP2/0 showed comparable PK profiles, with half-lives of the alpha and beta phases of 23 hours and 136 hours (dinutuximab beta) vs 25 hours and 135 hours (ch14.18/SP2/0), respectively. Overall exposure (AUC) of BALB/c mice to dinutuximab beta was slightly lower than to ch14.18/SP2/0 at equivalent doses, possibly due to the different glycosylation pattern of the proteins produced in CHO or SP2/0 cells. Biodistribution of ¹²⁵I-labeled dinutuximab beta and ch14.18/SP2/0 was comparable, indicating similar GD2 targeting activity of the 2 products *in vivo* (Figure 3).

Figure 3: Biodistribution of ^{125}I -labeled dinutuximab beta, ch14.18/SP2/0, and rituximab (control) after single i.v. injection into female neuroblastoma bearing A/J mice



The figure shows tissue radioactivity (in % of injected dose per gram tissue, +standard deviation) of ^{125}I -labeled dinutuximab beta, ch14.18/SP2/0, and rituximab (control) after single i.v. injection into female neuroblastoma bearing A/J mice 8 hours after injection.

Source: [22].

Pharmacokinetics of dinutuximab beta after repeated-dose administration over 10 days in cynomolgus monkeys

Serum levels and AUC values after repeated administration of dinutuximab beta for 10 days (4-hour infusion/day) at doses of 30 and 100 mg/m²/day revealed a dose-related, linear exposure with no sex differences in cynomolgus monkeys. Elimination half-lives were similar between doses, ranging from 18 to 33 hours. Pharmacokinetic data are summarized in Table 4.

Table 4: Pharmacokinetic characteristics of dinutuximab beta after repeated-dose administration (4-hour infusion/day over 10 days) in cynomolgus monkeys (Day 1-15)

Sex	Dose (mg/m ² [d])	C _{max} (µg/mL)	t _{max} (h)	AUC _{0-tlast} (ng·h/mL)	AUC _{0-∞} (ng·h/mL)	t _{1/2} (h)
Male	30	149	220	20235	20473	22.4
	100	375	180	55938	60115	32.8
Female	30	103	179	14522	14733	17.8
	100	339	118	49247	50311	23.4

Data were obtained in a repeat-dose toxicity study described in detail in Section 4.3.2.

Means are shown.

AUC_{0-tlast(∞)} = area under the time concentration curve from time 0 to the last quantifiable plasma concentration (extrapolated to infinity), C_{max} = maximum observed serum concentration, d = day, h = hour, t_{max} = time to maximum observed serum concentration, t_{1/2} = half-life.

Source: LPT Report 31743

4.2.2 Metabolism

No studies on metabolism of dinutuximab beta have been performed since the metabolic pathway of antibodies is well understood.

Unbound immunoglobulins are metabolized as all large proteins via unspecific proteolysis in lysosomes, proteasomes and by liver enzymes to small peptides and amino acids, which are

recycled. Another mechanism, which leads to degradation occurs via phagocytosis of respective immune complexes which also leads to the degradation of the antibody into small peptides and amino acids that are used for the *de-novo* protein synthesis.

4.3 Toxicology

4.3.1 Single-dose studies

No single-dose toxicity studies have been performed.

4.3.2 Repeated-dose studies

2-Cycle, 10-day continuous i.v. infusion toxicity study with dinutuximab beta in guinea pig

Methods

Juvenile guinea pigs received 1.25, 6.25, and 12.5 mg/kg/day (10, 50 and 100 mg/m²/day) dinutuximab beta as a 10-day continuous i.v. infusion in repeat 35-day cycles. A total of 5 cycles were planned. The main study group included 10 animals/sex/group. The vehicle control group and high dose group comprised in addition a “recovery” group of 5 animals/sex (animals scheduled to be followed up for 4 weeks after the end of the last cycle). Satellite groups of 3 animals per sex were used for toxicokinetic analysis (described in Section 4.2.1). Animals of the main group were sacrificed on Day 69/71 (end of Cycle 2), approximately 25 days after the last dinutuximab beta administration. Animals of the recovery groups were sacrificed on Day 99 (ca. 4 weeks after end of Cycle 2).

Animals were evaluated for clinical signs, body weight, food consumption, neurological readout, hematology, blood chemistry, urinalysis, ophthalmoscopy, organ weight, myeloid/erythroid ratio, gross pathology, and histopathology.

Results

Mortality

The study was prematurely terminated at the end of Cycle 2 because 50 to 75% of the animals died or had to be prematurely terminated across all treatment groups. All deaths were judged to be not test item-related and no influence of dinutuximab beta administration on mean survival time was noted compared with the vehicle control group. The causes of death or preliminary termination were valvular/thrombotic endocarditis (as a result of the LTI induced injection site changes and/or post-surgical trauma/infection) as well as rupture of the infusion catheter.

Clinical signs

Clinical observations included slightly reduced motility, partly with short-lasting lateral position, starting about 2 minutes after infusion start on Day 1 lasting for approximately 8 hours for all animals treated at 6.25 or 12.5 mg/kg/day. Slightly (6.25 mg/kg/day) or slightly to moderately (12.5 mg/kg/day) reduced motility was also noted during the off-dose period of Cycle 1. All surviving male animals were affected on Days 7 to 13 or 4 to 15 for the intermediate and high dose, respectively, and 4 of 10 female animals each of the intermediate

and high dose on up to 7 or 12 days starting on Day 6 or 5, respectively. No test item-related changes of behavior or external appearance were noted during the course of Cycle 2.

Local tolerance

Microscopy revealed abscess, fibrosis, granulation tissue, hemorrhage, granulomatous inflammation, or chronic active inflammation at the infusion site without a clear dose-relationship. Significant changes were observed at the infusion site in males dosed with 6.25 mg/kg dinutuximab beta (slight fibrosis for 3/3 and granulomatous inflammation 2/3) and in females dosed with 1.25 mg/kg dinutuximab beta (marked chronic/active inflammation for 4/7 animals). Due to difference in number of animals observed and the lack of a dose-relationship, these changes were not considered to be test item-related.

Body weight and food consumption

Body weights and food consumption were statistically significantly reduced (up to -30% and 47%, respectively) in a dose-dependent manner as compared with controls during Cycle 1. During the off-dose phase of each cycle higher gains were apparent but the overall body weight changes for all treated groups during each cycle were either less than or similar to the control. Body weights at necropsy were reduced accordingly. This reduction in food consumption was not observed during Cycle 2.

Neurological screening

Parameters evaluated for neurological screening (salivation, respiration, breathing, convulsion, piloerection, diarrhea, lacrimation, stereotypy, tremors, startle response, auditory function, toe pinch, urination, limb rotation, pupil size/response) were not affected by dinutuximab beta administration on Days 4 and 39.

Hematology and blood chemistry

Statistically significant decreases in the numbers of white blood cells, reticulocytes, and platelets compared with control animals were observed on Day 12 of Cycle 1. Females seemed to be stronger affected than males. These findings are well in line with respective toxicities observed during clinical trials and might be a result of dinutuximab beta also targeting the cells of the bone marrow. Additional observations included: increases in activated partial thromboplastin time, plasma cholesterol, urea, alpha-amylase, and glutamate dehydrogenase activity at all doses; decreases in plasma creatinine and glucose; and a dose-dependent increase in urine volume in male animals.

Ophthalmological and auditory examinations

No dinutuximab beta-related influence.

Macroscopic post-mortem findings

All macroscopic post-mortem examinations revealed changes that were either correlated with post-surgical trauma/infection and/or were considered either coincidental or to lie within the normal background alterations seen also in untreated guinea pigs of this age and strain. No relation to the test item could be observed. These observations included axilla grazes abrasions, enlarged or reduced hearts; endocarditis; enlarged livers, pericardium, or spleens, and dark-red discolored lungs.

Organ weight

In dinutuximab beta-treated male animals decreases in relative (up to 13%) and absolute (up to 22%) liver organ weights were observed without a clear dose-response. After the recovery period, the liver weights of male animals previously treated with 12.5 mg/kg dinutuximab beta were still decreased.

Histopathology (including bone marrow, myeloid:erythroid ratio)

Relevant and dinutuximab beta-related changes were observed for bone marrow which showed myeloid hyperplasia with significant decreases in the higher dose groups in female animals. The myeloid hyperplasia consisted of an increased cellularity of the myeloid compartment, most often due to increased tissue demand or utilization of leukocytes related to inflammation, e.g. (valvular/thrombotic) endocarditis or tissue injury. In addition, there was an increased myeloid/erythroid ratio in animals with this change. As GD2 is known to be expressed on mesenteric stem cells of the bone marrow [23], these findings might be a result of decreased cellularity/bone marrow suppression caused by dinutuximab beta treatment.

All other histopathology findings were not considered dinutuximab beta-related since these occurred also in the control group without any dose-relationship. Such findings included chronic active (valvular/thrombotic) endocarditis often in combination with chronic fibrous pericarditis; abscess, fibrosis, granulation tissue, hemorrhage, granulomatous inflammation or chronic active inflammation at the infusion site; liver Oil Red O staining; and significantly decreased mammary acinar development. However, spontaneous background changes following infection site findings, in combination with high morbidity and mortality with associated autolysis observed in a number of organs could have masked a possible dinutuximab beta-related effect.

Conclusions

Administration of 1.25, 6.25, or 12.5 mg/kg dinutuximab beta was safe. Most changes observed were either correlated with post-surgical trauma/infection and/or were considered either coincidental or to lie within the normal background alterations seen in untreated guinea pigs of this age and strain. Drug-related toxicities comprised clinical signs (slightly reduced motility), a decrease in body weight and food consumption, changes in hematology (decreases in white blood cells, reticulocytes, and platelets), changes in blood chemistry (increases in activated partial thromboplastin time, plasma cholesterol, urea, alpha-amylase and glutamate dehydrogenase activity, decreases in plasma creatinine and glucose), increased urine volume, reduction in absolute and relative liver organ weight and bone marrow findings (suppression). These observations are well in accordance with the expected toxicities of dinutuximab beta, i.e. induction of pain by binding to peripheral nerve tissue structures (resulting in the observed, dose-dependent clinical signs, food and body weight decreases) and interference with the hematopoietic system/bone marrow. Repeated administration of dinutuximab beta resulted also in the induction of anti-drug antibodies which may interfere with the PK of dinutuximab beta and may compensate dinutuximab beta-induced clinical signs and hematological changes. At the end of the recovery period all changes noted during the treatment period had subsided with

the exception of reduced liver organ weights and an increased myeloid:erythroid ratio in male animals dosed with 12.5 mg/kg.

10-day toxicity study with dinutuximab beta, administered as 4-hour i.v. infusion, in cynomolgus monkeys

Methods

Cynomolgus monkeys received 0, 2.5, or 8.33 mg/kg/day (0, 30 and 100 mg/m²/day) dinutuximab beta as a 4-hour infusion for 10 days. The main study group included 3 animals/sex/group. The vehicle control group and high dose group comprised in addition a “recovery” group of 2 animals/sex (animals followed up for about 3 weeks after the last dinutuximab beta administration). Animals of the main group were sacrificed on Day 14, 4 days after the last dinutuximab beta administration. Animals of the recovery groups were sacrificed on Day 36. All animals were used for toxicokinetic analysis (described in Section 4.2.1).

Animals were evaluated for clinical signs, body weight, food consumption, electrocardiogram (ECG), blood pressure, hematology, blood chemistry, urinalysis, ophthalmoscopy, histopathology, organ weight, macroscopic post-mortem examinations, and bone marrow evaluations.

Results

Mortality

Two unscheduled deaths occurred in the high dose group: one animal (male) showed reduced overall activity and died on Day 14, another animal (female) also showed signs of toxicity including reduced overall activity for at least 4 test days and was prematurely sacrificed on Day 14. Both animals showed a reduced thymus size. The male animal had enlarged thyroid, while the female animal showed reduced size of ovaries and thyroid. Multiple reddenings were found in the caecum.

Clinical signs

While no signs of behavioral changes were observed in the low dose group, pronounced changes were noted in the high dose group (i.e. slightly to severely reduced overall activity). All clinical signs subsided in the recovery period.

Local tolerance

No dinutuximab beta-related signs of local intolerance reactions were noted.

Body weight

Compared with the control group, dinutuximab beta-related differences in body weight were observed for male and female animals and in both dose groups. Differences were, however, not statistically significant, but sustained during the recovery period.

Food consumption

Food consumption was significantly decreased in all animals treated with dinutuximab beta. The reduction was dose-dependent in both sexes. Changes in food consumption were reversible in the recovery period.

ECG and blood pressure

No dinutuximab beta-related changes in ECGs or blood pressure were observed.

Hematology (including coagulation), blood chemistry, urinalysis

No dinutuximab beta-related changes in hematology, blood chemistry, or urinalysis were observed.

Ophthalmological and auditory examinations

No dinutuximab beta-related changes were observed.

Macroscopic post-mortem findings

Dinutuximab beta-related reductions in organ size of the thymus were observed in all female animals of the low dose group, and in all male and female animals in the high dose group.

Organ weight

A dinutuximab beta-related reduced thymus weight was observed in both dose groups and in both sexes. Reduced thymus weights were still observed during the recovery period.

Histopathology (including bone marrow)

Histopathological examination revealed dinutuximab beta-related changes in the bone marrow (atrophy) in all animals of the low and high dose group. Changes consisted of hypoplasia of bone marrow affecting both the myeloid and the erythroid precursor lines with loss and/or reduction in cellularity of granulocytic and erythrocytic precursors. This was sometimes accompanied by dilation of marrow sinusoids and extravasation of red blood cells into the marrow space as part of the atrophy.

Recovery period

No dinutuximab beta-related delayed effects were noted for any parameter evaluated within the recovery period for any animal.

Conclusions

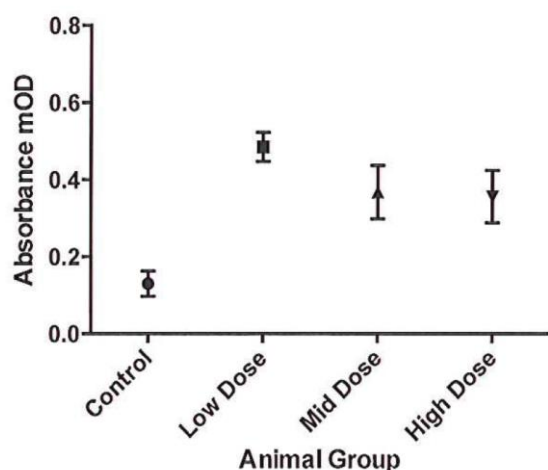
The observed, drug-related toxicities comprising reduced overall activity, body weight and food consumption decrease, bone marrow findings, changes in thymus size and weight are in line with the expected toxicities of dinutuximab beta targeting GD2, i.e. induction of pain by binding to peripheral nerve tissue structures (resulting in the observed, dose-dependent clinical signs, food and body weight decreases) and effects on bone marrow (via cytokine release), as also observed in clinical trials. Partial to full recovery of the observed findings were documented during or at the end of the recovery period. Two animals of the 100 mg/m²/day group died prematurely 4 days after the end of dosing. Even though the exact cause of death could not be determined, the deaths were judged to be test item-related and may be related to worsening of overall condition (reduced body weight gain/food consumption; wasting syndrome). No observed adverse effect level could be determined and the maximum tolerated dose was judged to be 2.5 mg/kg/day (30 mg/m²/day).

4.3.3 Immunotoxicity

The development of antibodies capable of binding to dinutuximab beta was evaluated in the above described, repeated-dose toxicity study in guinea pigs. Antibodies were detected using an ELISA assay.

The development of anti-drug antibodies was observed in most animals. Anti-drug antibodies (Figure 4) were observed at end of treatment Cycle 1 (Day 34, earlier time points were not assessed for antibodies) as well as at the end of treatment Cycle 2 (Day 59 or 61).

Figure 4: Analysis of anti-drug antibodies in guinea pigs (ELISA assay)



Antibody analyses were done in serum samples taken from control group, low dose (1.25 mg/kg/day), mid dose (6.25 mg/kg/day) and high dose (12.5 mg/kg/day) animals from the repeated-dose toxicity study (Section 4.3.2) taken on Days 59 or 61. The used ELISA method was not validated.

ELISA = enzyme linked immunosorbent assay.

4.4 Genotoxicity and carcinogenicity

Antibodies cannot interact directly with deoxyribonucleic acid or other chromosomal material. Additionally, there is no evidence that monoclonal antibodies have the potential to support or induce proliferation of transformed cells. Thus, no genotoxicity or mutagenicity studies have been performed. The dinutuximab beta formulation contains compendial excipients only.

4.5 Reproductive toxicity

No studies on fertility, early embryonic development, or pre- and postnatal toxicity have been conducted. No effects of dinutuximab beta on reproductive organs were observed in the repeat-dose toxicity study in guinea pigs (Section 4.3.2). No embryofetal toxicology studies have been conducted.

Based on the mode of action of dinutuximab beta (induction of CDC and ADCC after binding), the expression of GD2 on neuronal tissues especially during embryofetal development, and the potential of placental transfer of antibodies, dinutuximab beta has the potential to cause fetal harm when administered to pregnant women.

4.6 Local tolerance

No standalone local tolerance studies have been performed. Local tolerance testing was performed as part of the repeated-dose toxicity study in guinea pigs (see Section 4.3.2).

4.7 Cross-reactivity studies

Cross-reactivity of dinutuximab beta or 14G.2a (murine analogue of dinutuximab beta) with guinea pig or human disialoganglioside GD2, respectively, was tested by immunohistochemistry using frozen tissue sections, biotinylated secondary antibodies, and staining with a complex of avidin-horseradish peroxidase and diaminobenzidine.

When human tissues were used, 14G.2a was used as a surrogate for the detection of potential tissue cross-reactivity of dinutuximab beta. The use of dinutuximab beta would have required a human IgG specific detection antibody, which would have unspecifically cross-reacted with human tissue.

Cross-reactivity of dinutuximab beta with selected guinea pig tissues

Peripheral nerves (sciatic nerve) of the guinea pig showed specific tissue structure-related positive cross-reactivity with 14G.2a. Furthermore, GD2-positive cross-reactivity was found for the nervous structures of most of the organs and tissues examined: adrenal gland, eye, heart, jejunum, kidney, lung, lymph node, mammary gland, esophagus, pancreas, prostate, salivary gland, stomach, thymus, ureter, urinary bladder, and uterus (cervix). This positive nervous tissue staining was related to external and internal innervation structures. The cerebrum exhibited GD2-positive periventricular tissue and some neurons reacted positive in the cerebellum and spinal cord.

The cross-reactivity pattern showed the expected expression pattern of GD2 on central and peripheral nerves and organ innervations structures. The observed cross-reactivity pattern was similar to the one observed for human tissues (see below). Using 14G.2a as detection antibody, very similar expression patterns of GD2 were reported for mouse, rat, rabbit, dog, and human tissues. [24]

Cross-reactivity of 14G.2a (murine analogue of dinutuximab beta) with selected human tissues

In human tissues, unequivocal positive cross-reactivity was virtually limited to neuronal tissues, including cerebrum (neuropil), cerebellum (Purkinje cells), and peripheral nerves of most of the organs and tissues examined, including striated muscle, thymus, ureter, uterus, parotid gland, esophagus, stomach, breast, colon, eye (retina, ciliary body), fallopian tube, nervus opticus, lung, bronchus, ovary, pancreas, parathyroid gland, prostate, skin, spinal cord, spleen, thyroid gland. This positive nervous tissue staining was related to external and internal innervation structures. Unequivocal positive staining was also found in few non-neuronal tissues, including eccrine glands of the skin and Hassall's corpuscles in the thymus. In addition, low cross-reactivity was observed in connective tissues and smooth muscle cells of most tissue types. Since this faint staining was not found with the isotype control antibody, or with the negative control lacking any primary antibody, it must be considered specific for 14G.2a. Strong cross-reactivity was also observed for bone marrow. However, as a similar pattern was observed also in the respective controls, no clear judgment could be drawn.

5. Effects in humans

5.1 Overview on clinical studies with dinutuximab beta

Currently, information on pharmacology, safety, and clinical efficacy of dinutuximab beta is available from 6 completed or ongoing studies performed in Europe, Israel, Hong Kong, and Australia. These studies evaluated dinutuximab beta in the first-line treatment setting of high-risk neuroblastoma as well as relapsed/refractory neuroblastoma patients. Patients received dinutuximab beta treatment in combination with IL-2 or as a monotherapy. Three of these studies investigated a continuous LTI regimen (over 10 days to reduce side effects) while 3 used a STI regimen (8-hour infusion over 5 consecutive days). During clinical development, the safety profile of dinutuximab beta was characterized using doses ranging from 10 to 30 mg/m²/day in up to 9 treatment cycles of 4 or 5 weeks in length. All studies included mainly pediatric and a few adolescent and adult patients. As most studies were uncontrolled, control data were retrieved from retrospective data collection from comparable neuroblastoma patients who did not receive immunotherapy. All studies were investigator-initiated trials.

Study APN311-304 is still ongoing with availability of an interim study report (last patient visit included in analysis (cut-off date: 01 Apr 2019)). The study was being conducted in patients with primary relapsed/refractory neuroblastoma utilizing the LTI treatment regimen without IL-2 co-medication. Further, studies APN311-201 (V1.2), APN311-202 (V1 and V2), and APN311-302 (V1.5) were completed.

An overview of the completed and ongoing studies is provided in [Table 5](#).

Table 5: Overview of studies with dinutuximab beta

Study code Phase (Status) EudraCT no	NB Setting	Design	APN Scheme (Dose(s) [mg/m²/cycle])	Co- treatment	Cycles	Patients Treated/ planned Age
APN311-303 (completed) (Comp. Use)	R/R ^a	OL, uncontrolled, single-center	24h/10d (100)	IL-2, RA	Up to 6 (each 35d)	54/54 >1 y to ≤45 y
APN311-101 Phase I (completed) 2005-001267-63	R/R	OL, uncontrolled, multicenter, dose- escalation	8h/5d (50, 100, 150)	none	1-3 (each 28d)	15/16 >1 y to ≤21 y
APN311-201 (Haplo) Phase II (Ongoing. Protocol 1.2 study part completed ^b) 2009-015936-14	Relapsed	OL, uncontrolled, multicenter	8h/5d (100)	none (Cycl 1- 3), IL-2 (Cycl 4-9)	Up to 9 (each 28d)	68/68 ^b ≤21 y
APN311-202 Phase I/II (Ongoing. Protocol 1 [V1] and 2 [V2] study parts completed ^c) 2009-018077-31.	R/R ^a	OL, uncontrolled, multicenter, dose- escalation, dose schedule finding	24h/10d (100, 150, 210)	IL-2, 13- cis, RA	5 (each 35d)	122/124 ^c >1 y to ≤21 y
APN311-302 Phase III (Ongoing. Protocol 1.5 part completed ^d) 2006-001489-17, Patients are in long-term follow-up.	First-line therapy	OL, randomized, controlled, multicenter	8h/5d (100)	13-cis RA with or w/o IL-2	5 (each 35d)	378/466 ^d <21 y
APN311-304 Phase II (ongoing: interim safety report available with last patient visit included in analysis (cut-off date): 1 Apr 2019) ^e 2014-000588-42	R/R	OL, uncontrolled, multicenter	24h/10d (100)	none	5 (each 35d)	40 (planned) ≥1 y to ≤21 y
Historical control analyses						
Prof. Lang (University of Tübingen)	R/R Haplo	Retrospective	NA	NA	NA	17 historical control patients/NA
Prof. Garaventa (Giannina Gaslini Children's Hospital)	R/R	Retrospective	NA	NA	NA	29 historical control patients /NA
Neuroblastoma patients from study HR-NBL- 1/SIOPEL R1 randomization (patients without DB treatment)	High-risk	Retrospective	NA	NA	NA	450 historic al control patients /NA

Study code Phase (Status) <i>EudraCT no</i>	NB Setting	Design	APN Scheme (Dose(s)) [mg/m ² /cycle]	Co- treatment	Cycles	Patients Treated/ planned Age
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^aAlso patients with adequate response to first-line therapy only (in addition to R/R patients) have been accrued to a certain extent.

^bData cut-off: 28 Jul 2020. A total of 35 patients were planned according to protocol version 1.2. Twenty-five additional patients were added in an amendment to protocol version 1.3.

^cCSR APN311-202 [V1]and [V2] dated 21 Dec 2020 with Last Patient Visit Included in Analysis (cut-off date): 11 Oct 2018. The study consisted of a dose schedule finding phase (completed, N = 24), followed by a confirmatory phase (N = 100). Data from the 24 patients of the dose finding phase as well as from the 100 patients in the confirmatory cohort were analyzed and reported (based on protocol version 2). Amendment 2 to protocol 2 [V3] added an additional 160 patients and included a second treatment arm of dinutuximab beta without concomitant IL-2.

^dData cut-off: 5 Sep 2016. A total of 406 patients were randomized based on the protocol including Amendment 5 (HR-NBL-1.5), but only 370 patients were treated and included in the analysis of the clinical study report (overall data from 385 patients were included in the clinical study report but 19 patients did not receive dinutuximab beta treatment). From Aug 2013 to Sep 2017, 3 additional amendments were implemented in the ongoing study. In Amendment 6 the DMC recommended to only assign patients to the dinutuximab beta single agent arm (excluding IL-2). In Amendment 7, the treatment regimen was switched to long-term infusion (LTI). Amendment 8 excluded treatment with dinutuximab beta as study question and recommended it as standard treatment. The exposure data for the 302 study is based on the final approved clinical study report for the original study, but additional exposure has been disclosed by the study Sponsor to EUSA Pharma, and additional patients have been included in the publication by Ladenstein et al [11].

^eThe total number of patients screened in accordance with study protocol V6.0 consisted of 46 patients. 6 patients were excluded due to screening failures. A total of 40 patients are analyzed in this interim CSR for safety and a total of 38 patients for efficacy.

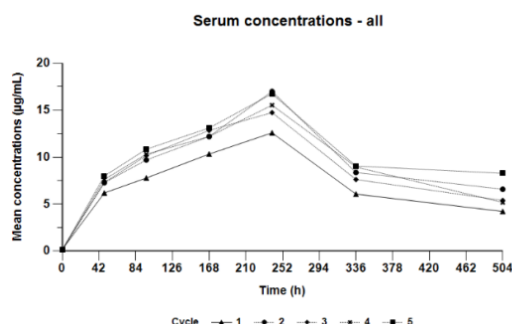
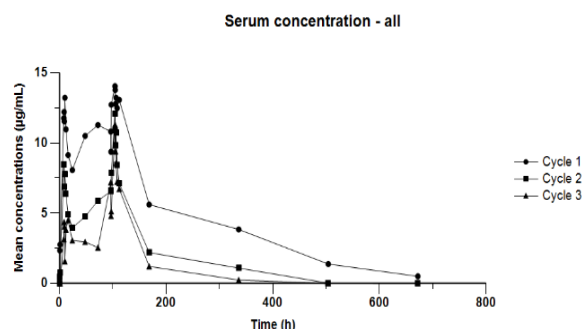
Comp. = compassionate, CSR = clinical study report, cycl = cycle, d = day, Haplo = haploidentical stem cell transplantation, IL-2 = interleukin 2, NA = not applicable, NB = neuroblastoma, OL = open-label, R/R = relapsed/refractory, w/o = without, y = year.

5.2 Clinical pharmacology

5.2.1 Pharmacokinetics

Dinutuximab beta serum concentrations were measured using ELISA methods and were assessed under LTI and STI infusion regimens.

With LTI mean dinutuximab beta serum concentrations increased until 240 hours after treatment start, in line with the 10-day duration of the continuous infusion ([Figure 5a](#)). Under STI (5×8-hour infusion, 5 consecutive days) serum antibody concentrations fluctuated in line with the daily 8-hour infusion scheme, which was applied for 5 consecutive days ([Figure 5b](#)). Only a slight accumulation in maximum serum concentrations was seen from cycle to cycle.

Figure 5: Mean serum concentration-time profiles of dinutuximab beta**a) Continuous infusion (APN311-202)****b) 8-hours infusion (APN311-101)**

Source: Report 140669001 (Prolytic), Report 150669004 (Prolytic), Report 140669002 (Prolytic).

Pharmacokinetic data are summarized in [Table 6](#). As expected, maximum dinutuximab beta concentrations were achieved at the end of the infusion, i.e. after about 220 hours under a 10-day 24-hour infusion and at about 80 hours after an 8-hour interval infusion. The mean half-life of dinutuximab beta was longer with a 10-day 24-hour infusion than with an 8-hour interval infusion (8 days vs 4 days). AUC values were higher with the 10-day 24-hour infusion than with an 8-hour interval infusion, while maximum concentrations were higher with the 8-hour interval infusion regimen.

Excluding patients with HACA (see Section [5.2.2](#)), slightly increased maximum concentration, AUC, and half-life values were observed.

Table 6: Pharmacokinetic parameters of Cycle 1

	C_{max} (µg/mL)	t_{max} (h)	AUC_{0-last} (h·ug/mL)	AUC_{0-xxx} (h·ug/mL)	t_{1/2} (h)
Long-term infusion					
Study APN311-303 (cycle time 0-504 h)					(time 0-768 h)
N	46	46	46	30	15
Mean	12.86	216.00	3066.94	3758.72	180.90
SD	4.62	87.34	1555.90	1422.50	65.35
Study APN311-202 V1 (cycle time 0-840 h)					(time 0-840 h)
N	39	39	39	39	19
Mean	11.91	236.92	3602.65	3053.83	197.71
SD	4.25	104.06	1436.63	1155.52	79.03
Short-term infusion					
Study APN311-101 (cycle time 0-672 h)					(time 0-672 h)
N	7	7	7	6	6
Mean	16.47	77.93	2349.84	2596.91	96.56
SD	6.48	47.15	1035.90	1014.52	63.71

AUC_{0-last} = area under the time concentration curve from administration till the last sampling time point, AUC_{0-xxx} = area under the time concentration curve from administration till end of Cycle 1, C_{max} = maximum observed serum concentration, h = hour, N = number of patients, SD = standard deviation, t_{max} = time to maximum observed serum concentration, t_{1/2} = half-life.

Source: Report 140669001 (Prolytic), Report 150669004 (Prolytic), Report 140669002 (Prolytic).

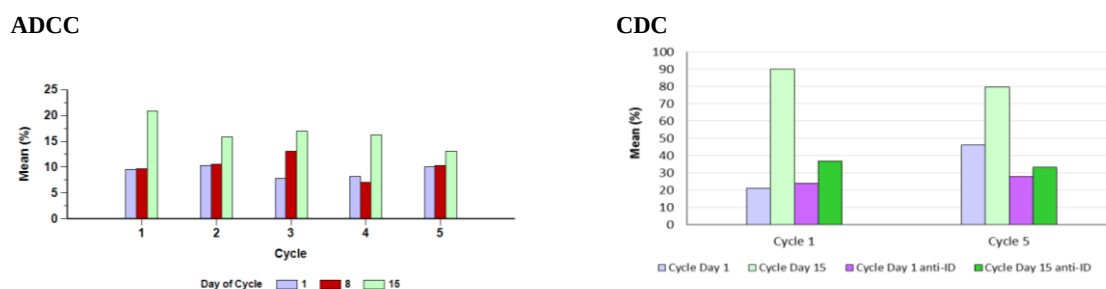
5.2.2 Pharmacodynamics

Cytotoxic effector functions

Ganglioside GD2 is a glycolipid antigen devoid of an intracellular signal transduction domain and the mechanisms of action of anti-GD2 monoclonal antibodies thus mostly rely on immune effector functions like CDC and ADCC. CDC is induced through binding of a serine protease complex C1 to the Fc domains of 2 or more monoclonal antibodies which are attached to antigens expressed on tumor cells. This classical complement pathway results in an activation cascade resulting in a membrane attack complex disrupting the target cell. ADCC is a result of Fc-gamma receptor mediated interaction with effector immune cells such as NK cells, macrophages, and granulocytes. The binding of the FcγR to the Fc domain induces both release of granzymes and perforin from effector cells leading to a target cell lysis and Fc-dependent tumor cell phagocytosis.

Patient-specific CDC and ADCC were assessed with cytotoxicity assays based on the release of an acetomethoxy derivate of calcein (calcein-AM), which is a membrane-permeable live-cell labeling dye. For ADCC, target cells (LAN-1) labeled with the membrane-permeable fluorescent dye calcein-AM were incubated with effector cells and serum of treated patients. For CDC, patient serum was incubated with calcein-AM-labeled target cells (LAN-1) without effector cells.

Cytotoxic effector functions correlated well with dinutuximab beta infusion, with ADCC and CDC increasing strongly from start till end of the dinutuximab beta infusion of each cycle (Day 8 to Day 15 in [Figure 6](#)). With ganglidiomab (anti-idiotypic antibody which neutralizes the binding of dinutuximab beta to GD2), CDC activity at Day 15 was strongly reduced, confirming the role of dinutuximab beta activity in triggering CDC.

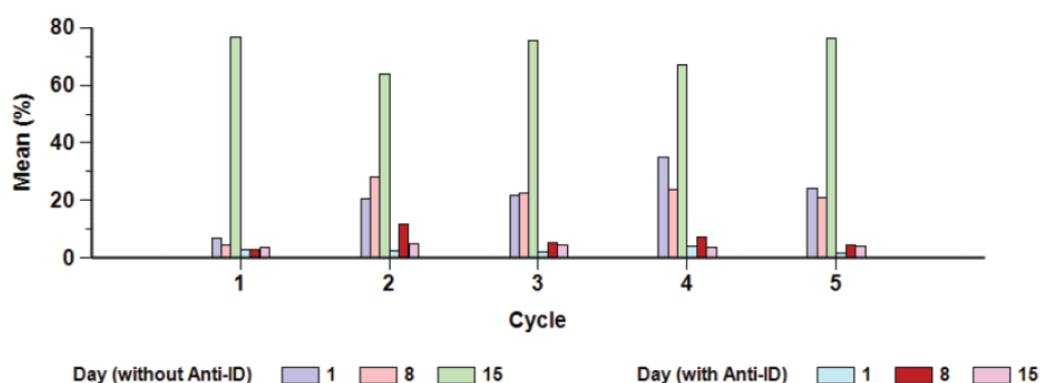
Figure 6: Mean ADCC and CDC activation (Study APN311-303)

Interleukin-2 was administered from Days 1 to 5 and 8 to 12 of each cycle, dinutuximab beta continuously from Days 8 to 18. Patient serum was incubated with calcein-AM-labeled target cells (LAN-1) with effector cells for ADCC, and without effector cells for CDC. Calcein-AM release was fluorometrically measured, which is proportional to cell lysis.

ADCC = antibody-dependent cellular cytotoxicity, calcein-AM = acetomethoxy derivate of calcein, CDC = complement dependent cytotoxicity.

Report 140669001 (Prolytic), Figure 12 and Table 21

A whole blood lysis test, in which whole blood from dinutuximab beta-treated patients was incubated with the GD2-positive and radiolabeled LAN-1 human neuroblastoma cell line, confirmed the strong lysis capacity of dinutuximab beta. Following dinutuximab beta infusion (starting Day 8 in Figure 7), the lysis capacity of patient's blood increased from about 5% at Days 1 and 8 to about 75% at Day 15. The lysis capacity did not return to pre-treatment baseline levels until the start of the next treatment cycle but showed a remaining lysis capacity of about 20% to 30%. These results indicate that following the initial antibody treatment, a constant lytic activity was maintained over the entire course of the 5- or 6-cycle treatment period. An anti-idiotypic antibody which neutralizes the binding of dinutuximab beta to GD2 strongly reduced the dinutuximab beta-induced lysis (Figure 7).

Figure 7: Whole blood lysis capacity of dinutuximab beta (Study APN311-303)

Whole blood from dinutuximab beta-treated patients was incubated with the GD2-positive and ⁵¹chromium labeled LAN-1 human neuroblastoma cell line. Percentage of lysed cells was determined by released radioactivity.

Interleukin-2 was administered from Days 1 to 5 and 8 to 12 of each cycle, dinutuximab beta continuously from Days 8 to 18.

Report 140669001 (Prolytic), Figure 32

Concomitant IL-2 treatment had no relevant effect on the cell lysis capacity.

Hematology parameters

To assess the effect of dinutuximab beta treatment on hematological parameters, NK cells, T-cells, granulocytes (including eosinophils, neutrophils, and basophils), leukocytes, lymphocytes, monocytes, and soluble IL-2 receptor were assessed in whole blood of dinutuximab beta-treated patients.

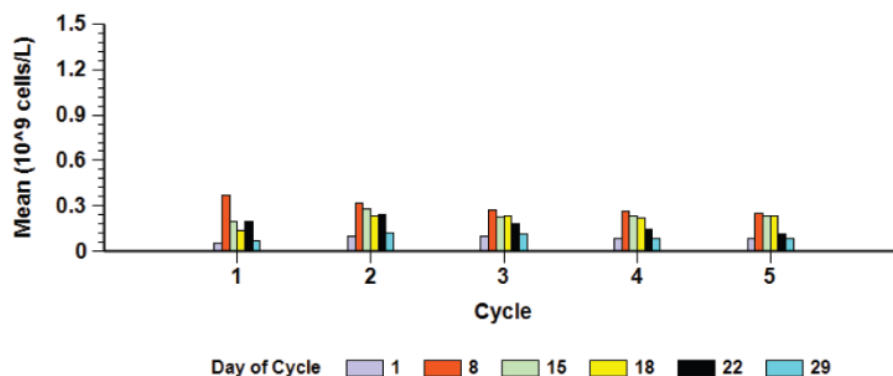
Natural killer cells

Natural killer cells are involved in direct ADCC and help to build up memory NK cells (as part of the adaptive immune response for future tumor attacks).

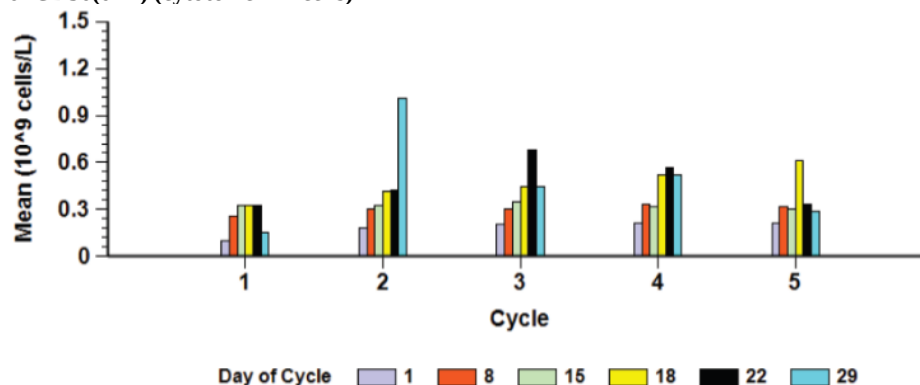
In the dinutuximab beta studies, IL-2 given prior to dinutuximab beta infusion (Days 1 to 8, [Figure 8](#)) increased the number of activated NK cells overall as well as the number of cytotoxic or cytokine producing NK cells. During the subsequent dinutuximab beta treatment until Day 18 (concomitant with IL-2 until Day 15), the numbers of cytotoxic NK cells tended to increase, while the numbers of overall activated NK cells and cytokine producing NK cells remained stable or decreased slightly.

Figure 8: Changes in mean NK cell numbers with dinutuximab beta treatment (Study APN311-303)

CD16+CD56(bright) (cytokine producing NK cells)



CD16+CD56(dim) (cytotoxic NK cells)



Interleukin-2 was administered from Days 1 to 5 and 8 to 12 of each cycle, dinutuximab beta continuously from Days 8 to 18.

NK = natural killer cells. Report 140669001 (Prolytic), Figures 26, 27 and 28

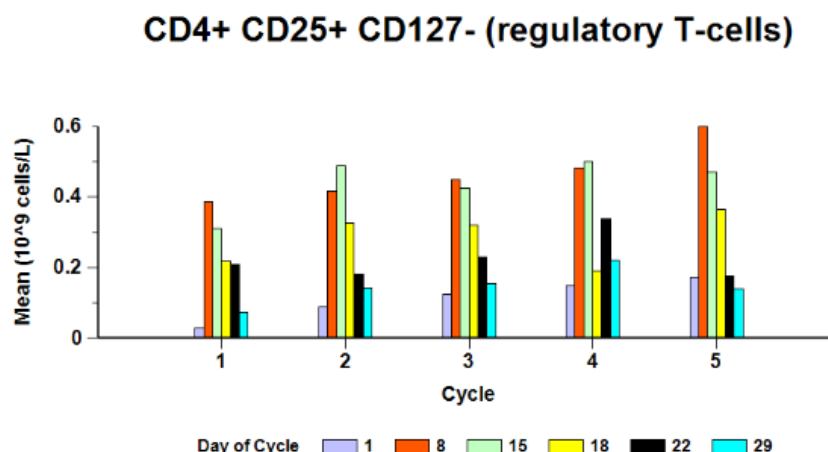
The percentage of activated CD69+ NK cells increased with dinutuximab beta treatment independent of IL-2 administration.

Confidential

Regulatory T-cells

Regulatory T-cells (CD4+/CD25+/CD27- cells) are found in tumor tissue and suppress effector T-cell activation. The intent during immunotherapy is to reduce regulatory T-cells and enhance tumor-specific effector T-cells. IL-2 is known to increase regulatory T-cell function [25], and such an increase was also observed in dinutuximab beta studies (Figure 9). During dinutuximab beta treatment regulatory T-cells remained stable or decreased until end of dinutuximab beta infusion.

Figure 9: Changes in regulatory T-cells with dinutuximab beta treatment (Study APN311-202)



Interleukin-2 was administered from Days 1 to 5 and 8 to 12 of each cycle, dinutuximab beta continuously from Days 8 to 18.

Absolute T-cell, cytotoxic T-cell, and T-helper cell numbers also increased from Day 1 to Day 8 and tended to decrease afterwards.

Granulocytes and soluble IL-2 receptor

Dinutuximab beta appears to induce all types of granulocytes (absolute, neutrophil, eosinophil, basophil), with maximum cell counts usually seen at end of infusion. IL-2 may also induce activated granulocytes and monocytes as observed in one study. Since granulocytes may increase the immune response by directing the migration of effector cells to tumor cells [26], a general increase during antibody treatment is beneficial.

Dinutuximab beta showed only little if any effect on the number of activated, CD64 expressing monocytes, which were predominately increased during IL-2 treatment days.

Soluble IL-2 receptors were mainly upregulated with IL-2 treatment but upregulation may be enhanced with dinutuximab beta treatment.

Human anti-drug antibodies

The development of anti-drug antibodies (ADA) is a class effect of monoclonal chimeric antibodies. Antibody generation may interfere with the function of the antibody and limit dinutuximab beta efficacy. ADA titers were assessed in 4 studies: APN311-101, APN311-201, APN311-202, and APN311-303. Overall, measurable ADA titers were noted in 65 of the patients (62%) examined (N = 105). Eleven out of 15 (73.3%) patients with data available were ADA positive in APN311-101, 27 out of 42 (64.3%) patients in APN311-202 (V1), and 27 out of 48 (56.3%) patients in APN311-303. No patient (out of the 32 analyzed) was ADA positive

in study APN311-201. Given the limitation of the bioanalytical methods, data are currently insufficient to properly evaluate the impact of the formation of ADA on PK and pharmacodynamic parameters, as well as on the efficacy and safety of dinutuximab beta.

5.3 Clinical efficacy and safety

5.3.1 Efficacy and safety with other anti-GD2 antibodies

The monoclonal antibody dinutuximab beta (ch14.18) has been used in the management of neuroblastoma for more than 20 years. Accordingly, several studies have been initiated and were completed or are ongoing.

Initial studies were performed with the murine parental antibody 14.G2a, indicating beneficial effects in patients with neuroblastoma as well as melanoma and osteosarcoma. The most prominent toxicities observed with the antibody included pain and fever. The studies showed an expected immunogenicity of that murine antibody at high frequency which triggered the development of a human-mouse chimeric variant, i.e. dinutuximab beta (ch14.18) produced in SP2/0 cells.

Several clinical trials have been performed with that chimeric ch14.18/SP2/0 antibody either administered as monotherapy [8-13] or in combination with cytokines such as IL-2 and GM-CSF. [2,14-17] In Phase I trials, the maximum tolerated dose for dinutuximab beta (ch14.18) therapy in combination with GM-CSF was established at 40 mg/m²/day (4 days). In combination with IL-2 and GM-CSF, a safe dose of 25 mg/m²/day (4 days) of dinutuximab beta (ch14.18) was defined and applied. This therapy demonstrated a significant increase in EFS and OS rates compared to standard therapy (13-cis-RA only) in a multicenter, open-label, randomized Phase III trial including high-risk neuroblastoma patients with a complete response (CR) after induction therapy and myeloablative consolidation with stem cell rescue. In this trial, patients received either 6 cycles of 13-cis-RA (standard therapy) or 6 cycles of 13-cis-RA with 5 concomitant cycles of ch14.18/SP2/0 (25 mg/m²/day, 10 to 20-hour i.v. infusion on 4 consecutive days) combined with GM-CSF and i.v. IL-2 in alternating cycles (immunotherapy). During Cycles 1, 3, and 5, GM-CSF was given daily at a dose of 250 µg/m²/day for 14 days, starting 3 days before ch14.18/SP2/0 was started. During Cycles 2 and 4, IL-2 was given as continuous infusion for 4 days during Week 1 at a dose of 3.0×10^6 IU/m²/day, as well as for 4 days during Week 2 at a dose of 4.5×10^6 IU/m²/day, concurrent with ch14.18/SP2/0. At a median follow-up of 2.1 years after randomization, EFS was significantly higher for patients randomized to immunotherapy ($p = 0.01$), with 2-year estimates of $66\% \pm 5\%$ vs $46\% \pm 5\%$. Furthermore, OS was also significantly higher for immunotherapy ($p = 0.02$; $86\% \pm 4\%$ vs $75\% \pm 5\%$ at 2 years without adjustment for interim analysis). An analysis of OS conducted 3 years later also documented an improvement in OS in the immunotherapy arm compared to the 13-cis-RA arm (Hazard ratio 0.58 [95% confidence interval {CI}: 0.37, 0.91]). [2,27] Since the results of this study have become available, adjuvant dinutuximab beta (ch14.18) antibody-based immunotherapy is considered to be a “standard of care” for patients suffering from high-risk neuroblastoma. The combination therapy (including Unituxin[®] from United Therapeutics, IL-2 and GM-CSF) has been approved by the FDA in March 2015 for the treatment of high-risk neuroblastoma in pediatric patients with at least a PR to prior first-line multiagent, multimodality therapy.

The most common adverse drug reaction with immunotherapy was pain followed by pyrexia, thrombocytopenia, lymphopenia, hypotension, infusion reactions, hyponatremia, increased alanine aminotransferase (ALT), and anemia (all reported by more than 50% of patients). The most common serious adverse reactions (greater than or equal to 5%) were infections, infusion reactions, hypokalemia, hypotension, pain, fever, and capillary leak syndrome (CLS). [27]

5.3.2 Efficacy and safety with dinutuximab beta

5.3.2.1. Study APN311-303

(Source: CSR APN311-303 dated 17 Apr 2015, CSR Amendment APN311-303 with cut-off 18 Dec 2015; APN311-303 Historical control report dated 23 Apr 2015)

APN311-303 was an open-label, uncontrolled study based on compassionate use treatment of patients with high-risk, relapsed or refractory neuroblastoma. For analysis, data were collected retrospectively. The primary objective of the study was to evaluate the safety and pain-toxicity profile of prolonged continuous infusion of dinutuximab beta in combination with s.c. IL-2 in patients with high-risk, relapsed or refractory neuroblastoma.

Design

Patients: patients with high-risk, relapsed or refractory neuroblastoma.

Treatment: dinutuximab beta was administered as continuous infusion for 10 days, in combination with fixed doses of s.c. IL-2 (6×10^6 IU/m²/day over 5 consecutive days 1 week prior to dinutuximab beta start, as well as concurrently with dinutuximab beta for the first 5 dinutuximab beta infusion days) for up to 6 cycles (cycle length 28 to 35 days). Target daily dose was 10 mg/m², corresponding to a total dose of 100 mg/m²/cycle. Patients ≤ 12 kg received dinutuximab beta at 3.3 mg/kg per 10 days.

Premedication: intravenous morphine; bolus injection followed by continuous infusion, to be reduced on a daily basis as possible.

Safety and efficacy endpoints: primary endpoint was safety and tolerability evaluated by pain intensity and morphine use, AEs, vital signs, and changes in clinical laboratory assessments. Further endpoints: response rate, durability of response, OS, EFS.

Results - patients

Fifty-four patients were enrolled; 61% of the patients were male, 39% female (APN311-303 CSR, Table 11-3). 83.4% of the patients presented with relapsed or refractory neuroblastoma and had received additional, second-line treatment; 11.1% of patients had received first-line therapy only (APN311-303 CSR, Table 11-5). The majority of patients (72.2%) completed 5 or 6 cycles (APN311-303 CSR, Table 11-11). 40 patients (74.1%) completed the study as planned; 14 patients (25.9%) did not complete dinutuximab beta therapy. Study discontinuation was due to progressive disease (PD) in 92.9% (13 patients) and 7.1% for other reasons (APN311-303 CSR, Table 10-3).

Results - efficacy**Response**

In patients with evidence of disease at baseline, 40.5% of patients responded to treatment (CR and PR) as judged by the investigator using bone marrow examinations, computed tomography (CT), magnetic resonance imaging (MRI), $^{123/131}$ iodine-meta-iodobenzylguanidine (mIBG) scan, and/or response evaluation criteria in solid tumors (RECIST) ([Table 7](#)).

Table 7: APN311-303 - Best response in patients with evidence of disease at baseline

Category	Number (%) of patients	
	Best Response ^a (N = 37)	
Evaluable	37	(100.0%)
CR	5	(13.5%)
PR	10	(27.0%)
SD/no response	12	(32.4%)
PD	10	(27.0%)

^a Assessed by the investigator using bone marrow examinations, mIBG, and/or RECIST.

^b CR = complete response, mIBG = 123/131iodine-meta-iodobenzylguanidine, MR = minimal response, N = number of patients, PD = progressive disease, PR = partial response, RECIST = response evaluation criteria in solid tumors
SD = stable disease.

Source data: CSR APN311-303, Table 11-13

In the full analysis set (FAS) population, which included all 54 patients, best response was CR in 5 patients (11.9%) and PR in 10 patients (23.8%), resulting in an ORR of 35.7% (CSR APN311-303, Table 11-14).

When patients with evidence of disease at baseline were stratified according to relapsed or refractory disease status at treatment start, a best response of CR or PR was determined in 38% (8/21) patients with relapsed baseline disease, and in 100.0% (13/13) of patients with refractory baseline disease. Three had received first-line therapy only, i.e. did not present with relapsed or refractory baseline disease. Out of these, 2 (66.7%) patients achieved CR or PR (CSR APN311-303, Table 11-16).

In 14 of the 15 patients with response, the median duration of response¹ was 313 days (range: 71 to 847 days) for the patient population with evidence of disease at baseline (CSR Amendment APN311-303, Table 4-1; cut-off date 18 Dec 2015).

Event-free survival

EFS, i.e. the number of days from starting study treatment until relapse or disease progression as detected by MRI/CT, mIBG scan, or bone marrow examination was analyzed for 53 patients; at the cut-off date of 18 Dec 2015, EFS was 52.4% at 1 year, 34.9% at 2 years, and 30.9% at 3 years (CSR Amendment APN311-303, Table 4-2). Median time to progression was 589 days (CSR Amendment APN311-303, Table 14.2.3.1).

Event-free survival was 44.8%, 31.0%, and 24.1% at 1, 2, and 3 years, respectively in patients with relapsed neuroblastoma (n = 29), and 58.2% at 1 year and 29.1% at 2 and 3 years in patients with refractory neuroblastoma (n = 15) (CSR Amendment APN311-303, Table 4-2).

Overall survival

OS in all patients, i.e. regardless of disease status at baseline, was 92.3% at 1 year after immunotherapy start, 74.7% at 2 years, and 66.1% at 3 years (CSR Amendment APN311-303,

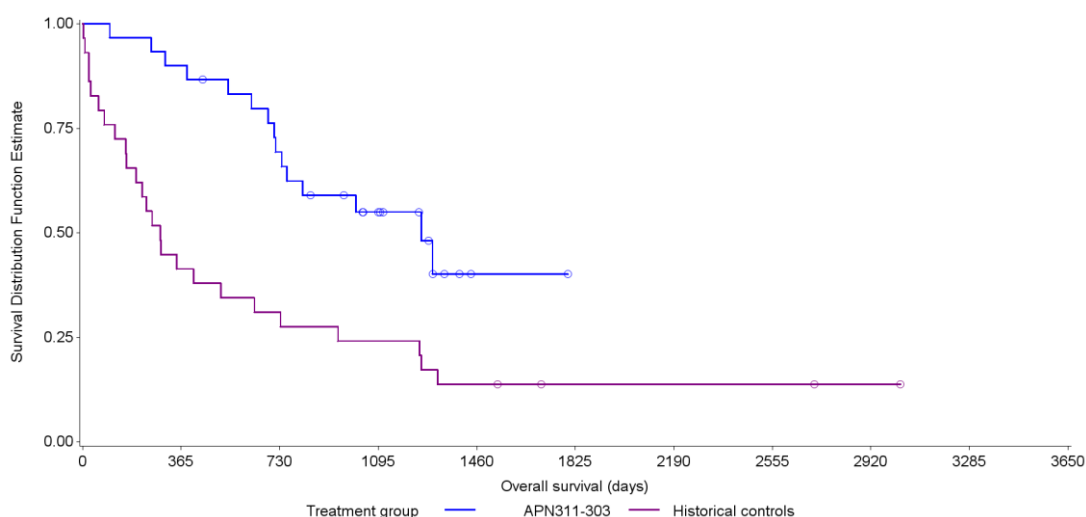
¹ Duration of response was defined as time from the first PR or CR reported after start of immunotherapy until progression of disease in patients who progressed and until the latest date from any examination (MRI/CT, mIBG or bone marrow) contributing to the latest assessment of overall response for patients without progression.

Table 4-4). The median survival time in the overall population was 1291 days, or about 42 months (CSR Amendment APN311-303, Table 14.2.5.1).

In patients with relapsed neuroblastoma at baseline (n = 29), OS was 89.7% at 1 year, 69.0% at 2 years, and 54.7% at 3 years. In patients with refractory neuroblastoma at baseline (n = 15), OS was 92.9% at 1 year, 69.8% at 2 years, and 69.8% at 3 years (CSR Amendment APN311-303, Table 4-4).

In a historical control analysis, OS data of patients with relapsed neuroblastoma from study APN311-303 (n = 30) were compared to a similar patient population from the Italian Neuroblastoma Registry (n = 29); (APN311-303 Historical Control Report, dated 18 Feb 2016). These patients experienced relapse of neuroblastoma after first-line therapy, but did not receive dinutuximab beta treatment. In the historical control group, OS was 41% at 1 year, 31% at 2 years, and 24% at 3 years. Overall survival in patients from the APN311-303 study was 90% at 1 year, 69% at 2 years, and 55% at 3 years. The difference in OS between APN311-303 and historical controls was statistically significant (p = 0.0009) (APN311-303 Historical Control Report, Table 5-2). Kaplan-Meier curves are provided in [Figure 10](#).

Figure 10: APN311-303 versus historical control - overall survival



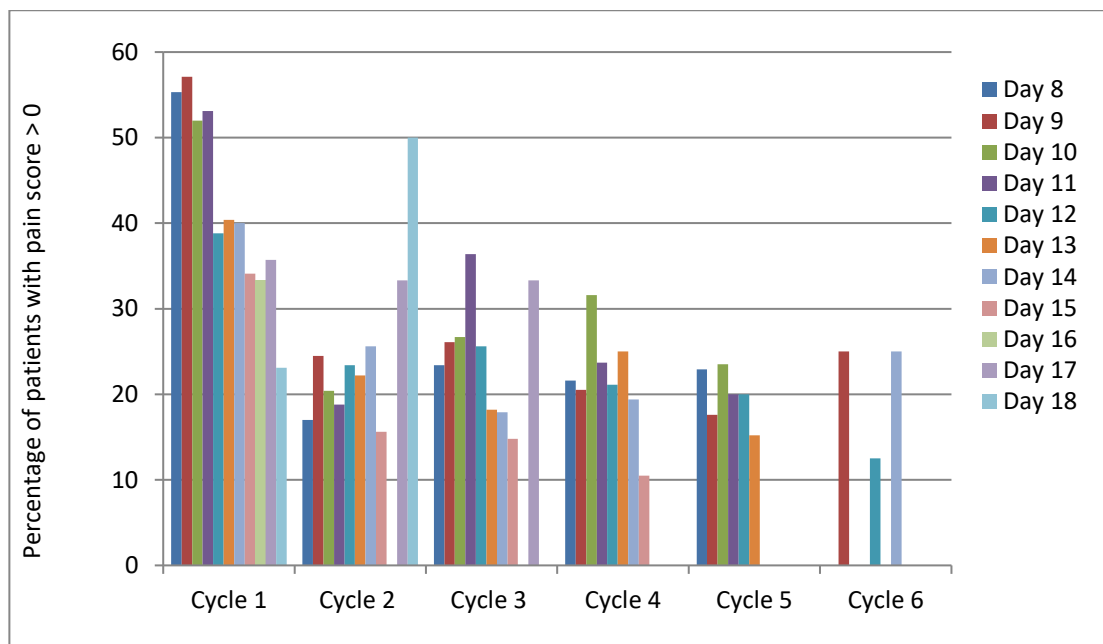
APN311-303 Historical Control Report, Figure 5-1

Results - safety

Pain

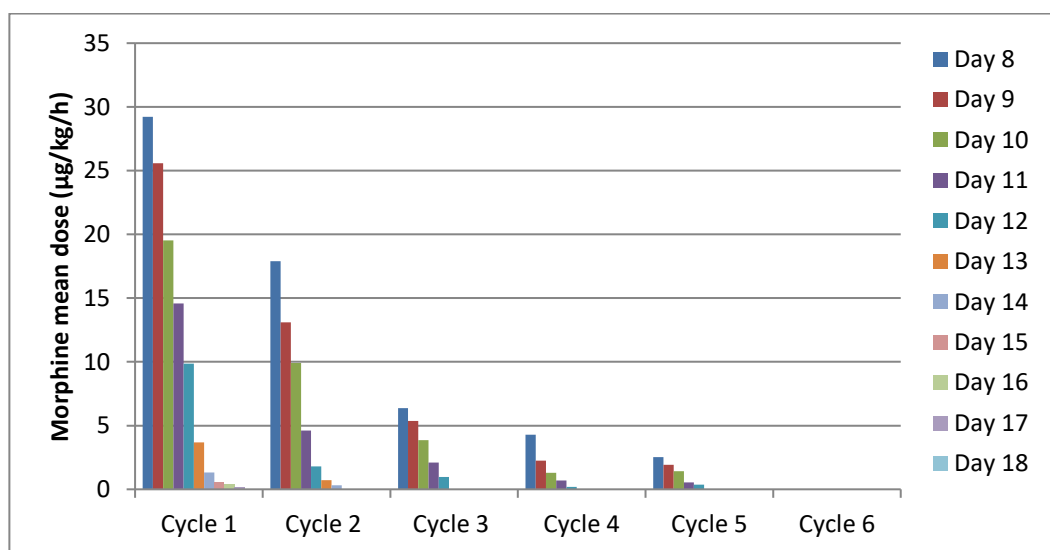
The numbers of patients with moderate to severe pain (pain score ≥ 7 on a scale of 0 to 10) decreased within each treatment cycle and also from cycle to cycle. Overall, the percentage of patients with pain decreased markedly from about 50% of patients during the first treatment days in Cycle 1 to about 20% in Cycle 5 ([Figure 11](#)). Similarly, the pain intensity in patients with pain decreased. At the same time, the number of patients in need of i.v. morphine infusion declined from Cycle 1 to Cycle 6, with no patient receiving i.v. morphine in Cycle 6 ([Figure 12](#)). Within a given cycle, the required i.v. morphine doses were highest on treatment Day 1, with a substantial decrease during the following 3 to 5 days. From treatment Day 6 of each cycle onwards, no i.v. morphine was required for almost all patients.

Figure 11: APN311-303 - Percentage of patients with pain (based on patients with pain evaluation)



Source data: CSR APN311-303; Table 14.3.14.1.1

Figure 12: APN311-303 - IV Morphine individual dose changes by cycle (all patients)



Source data: CSR APN311-303; Table 14.1.12.4.1

Adverse events

All patients reported at least one AE and all patients experienced at least one event judged by the investigators to be related to dinutuximab beta ([Table 8](#)). The incidence of serious adverse events (SAEs) was low. No patient died during the active study period.

Table 8: APN311-303 - Overview of adverse events

Patients with		N (%) patients (N=54)
Any AE	54	(100.0)
Any AE possibly related to dinutuximab beta	54	(100.0)
Any serious AE	12	(22.2)
Any serious AE possibly related to dinutuximab beta	6	(11.1)
Any AE leading to discontinuation of study drugs	5	(9.3)
Higher CTCAE Grade		
Grade 3	32	(59.3)
Grade 4	19	(35.2)
Any AE leading to death	-	
Deaths *	22	(40.7)

* All documented deaths, including deaths during follow-up period

Possibly related adverse events (AEs): AEs with relationship coded as 'Possible', 'Probable', 'Definite' or with missing relationship.

N = number of patients, CTCAE = Common Terminology Criteria for Adverse Events.

Source data: CSR APN311-303, Table 14.3.1.1; CSR Amendment APN311-303, Table 14.3.1.1

The most frequent AE by preferred term was pyrexia (98.1%), followed by pruritus (92.6%), pain in extremity (85.2%), constipation (83.3%), and CLS (83.3%) ([Table 9](#)).

Most AEs assessed as related to dinutuximab beta treatment were also assessed as related to IL-2 treatment, and included most frequently pyrexia, pruritus, and CLS. AEs considered more frequently related to dinutuximab beta treatment than to IL-2 treatment mainly concerned pain symptoms (pain, pain in extremity, back pain) and, to a lesser degree, also gastrointestinal symptoms (abdominal pain [upper], vomiting, nausea) (CSR Amendment APN311-303, Table 5-3).

A total of 45 patients (83.3%) experienced CLS. Since the event most frequently occurred during treatment with both dinutuximab beta and IL-2, the investigator assigned an at least possible relationship to dinutuximab beta as well as to IL-2 treatment for the majority of cases. Capillary leak syndrome tended to resolve shortly after cessation of IL-2 treatment, despite ongoing dinutuximab beta treatment.

Table 9: APN311-303 - Frequent adverse events (reported in more than 50% of patients)

System organ class Preferred term	N (%) patients (N=54)	
Overall	54	(100.0)
Gastrointestinal disorders	54	(100.0)
Constipation	45	(83.3)
Vomiting	40	(74.1)
Abdominal pain upper	33	(61.1)
Diarrhea	32	(59.3)
General disorders and administration site conditions	54	(100.0)
Pyrexia	53	(98.1)
Pain	40	(74.1)
Fatigue	28	(51.9)
Skin and subcutaneous tissue disorders	54	(100.0)
Pruritus	50	(92.6)
Dry skin	41	(75.9)
Vascular disorders	51	(94.4)
Capillary leak syndrome	45	(83.3)
Hypotension	34	(63.0)
Musculoskeletal and connective tissue disorders	49	(90.7)
Pain in extremity	46	(85.2)
Respiratory, thoracic and mediastinal disorders	49	(90.7)
Cough	43	(79.6)
Blood and lymphatic system disorders	45	(83.3)
Cardiac disorders	40	(74.1)
Tachycardia	40	(74.1)
Infections and infestations	39	(72.2)
Investigations	39	(72.2)

Source data: CSR Amendment APN311-303, Table 14.3.2.1

The highest Common Terminology Criteria for Adverse Events (CTCAE) grade was Grade 3 in 32 patients (59.3%) and Grade 4 in 19 patients (35.2%) (CSR APN311-303, Table 14.3.1.1). Most frequent (>15%) AEs of Grade 3 were neutropenia (37.0%), anemia (33.3%), pain (25.9%), pain in extremity (20.4%), and cough and thrombocytopenia (each: 16.7%). CTCAE Grade 4 events reported in more than 2 patients were: thrombocytopenia (7 patients, 13.0%), neutropenia (6 patients, 11.1%), pain in extremity (4 patients, 7.4%), and abdominal pain upper (3 patients, 5.6%) (CSR APN311-303, Table 14.3.2.2).

Twenty-two deaths occurred, mostly due to progression of disease (CSR Amendment APN311-303, Table 14.3.1.1); none was judged by the investigator to be treatment-related.

Twelve patients (22%) experienced SAEs. The only preferred term reported in more than 1 patient was vomiting, reported in 2 patients (CSR APN311-303, Table 14.3.7.1). SAEs possibly related to dinutuximab beta were reported in 6 patients (11.1%) and comprised enteritis, vomiting, pain, pyrexia, convulsion, neuropathy peripheral, peripheral motor neuropathy, and respiratory distress in 1 patient each (CSR APN311-303, Table 14.3.10.1).

Laboratory, vital signs, and other safety assessments

Out-of-range chemistry and hematology values occurred in the majority of patients, which is not unusual for the severely affected and heavily pre-treated patient population of study APN311-303 and the known effect of the study treatment on hematology and chemistry

parameters. Clinically relevant changes included mainly decreases in neutrophil, platelet, red, and white blood cell counts as well as increases in C-reactive protein and liver transaminases and low serum calcium levels.

No unexpected clinically significant changes were observed in physical examination, vital signs, or performance status.

Conclusion

Long-term infusion of dinutuximab beta (24 hours for 10 consecutive days) in combination with IL-2 was tolerable with an improved pain-toxicity profile compared with STI (8 hours for 5 consecutive days), enabling an outpatient setting. Best overall response in patients with evidence of disease at baseline was 40.5%, indicating anti-tumor activity. 1- and 2-year OS rates after LTI of dinutuximab beta were higher than those previously reported for these patient populations. Overall, the results indicate a clinically meaningful therapeutic effect for continuous infusion treatment of dinutuximab beta, both based on response rates and OS results.

5.3.2.2. Study APN311-101

(Source: APN311-101 CSR, dated 17 Apr 2015)

APN311-101 was an open-label, uncontrolled, dose-escalation, bridging study to evaluate the safety, clinical toxicity and in vivo immunological effects of dinutuximab beta in patients with neuroblastoma. The primary objective of the study was to reassess the toxicity profile of one treatment cycle with dinutuximab beta (ch14.18), recloned in CHO cells (dinutuximab beta), when administered as daily 8-hour infusions and accompanied by supportive care measures in particular to prevent pain, fever, and allergic reactions according to previously established standards.

Design

Patients: patients with relapsed or recurrent neuroblastoma.

Treatment: dinutuximab beta at doses of 10, 20, or 30 mg/m²/day was administered as 8-hour i.v. infusions, daily during the first 5 days of a 28-day cycle.

Premedication: intravenous morphine; bolus injection followed by continuous infusion.

Safety and efficacy endpoints: no specific endpoints defined; primary endpoint: safety; further evaluations: anti-tumor responses.

Results - patients

15 patients received treatment: 7 male (46.7%) and 8 female (53.3%) patients (CSR APN311-101 Table 14.1.5). Median age at baseline was 7 years (range: 3 to 17 years) (CSR APN311-101 Section 11.2.1).

International Neuroblastoma Staging System (INSS) stage was 4 in 14 patients (93.3%) and 2B in 1 patient (6.7%) (CSR APN311-101 Table 14.1.6). At study entry, 6 patients presented with refractory disease and 9 with relapsed disease. All patients had received initial intensive multimodality treatment (CSR APN311-101 Table 14.1.6). At the start of dinutuximab beta

treatment, 12 of 15 patients (80%) had evidence of disease and 3 patients (20.0%) had no evidence of disease (CSR APN311-101 Table 14.1.9).

Three patients received 10 mg/m², 8 received 20 mg/m², and 4 received 30 mg/m² (CSR APN311-101 Table 14.1.12.2). One patient completed only 1 cycle, 6 patients completed 2 cycles, and 8 patients completed 3 cycles (CSR APN311-101 Table 14.1.12.1). Two patients discontinued the study prematurely; one following the completion of 2 cycles due to PD and the other patient rejected further treatment and assessments following completion of one cycle (CSR APN311-101 Section 10.1).

Results - efficacy

Response

For the majority of patients, only 1 post-baseline assessment was available during the active study period. Therefore, assessment of response was based on the tumor status at the off-study evaluation visit. The response rate (CR+PR) in evaluable patients (n = 12) was 33.3% as judged by the investigator using bone marrow examinations, mIBG, and MRI/CT (Table 10).

Table 10: APN311-101 - Response at off-study evaluation visit

Category	Number (%) of patients	
	Overall response ^a (N=15)	
Evaluable	12	(100.0)
CR	3	(25.0)
PR	1	(8.3)
SD/no response	3	(25.0)
MR	-	
PD	5	(41.7)
Not evaluable ^b	3	

^a Response at off-study evaluation (i.e. at or up to 27 months after end of treatment), assessed by the investigator using bone marrow examinations, mIBG, and MRI/CT.

^b No evidence of disease at baseline.

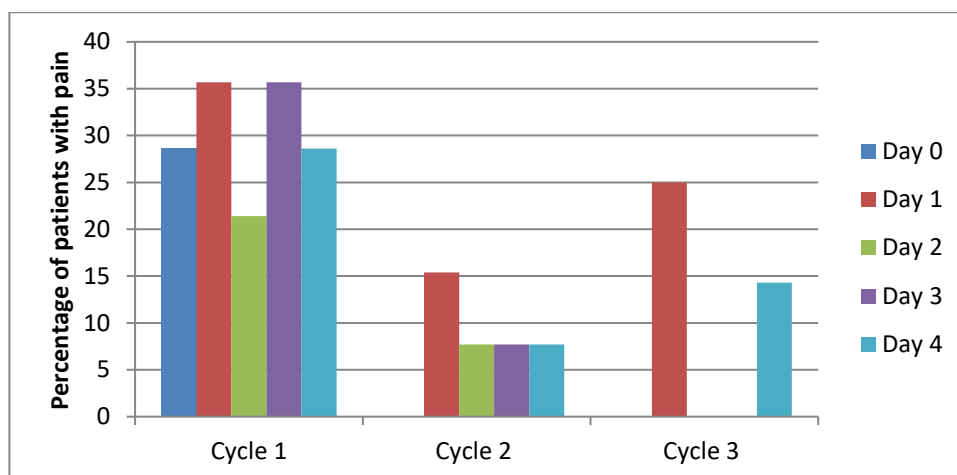
CR = complete response, CT = computed tomography, mIBG = ¹²³I/131I-iodine-meta-iodobenzylguanidine, MRI = magnetic resonance imaging, MR = minimal response, N = number of patients, PD = progressive disease, PR = partial response, SD = stable disease.

Source data: APN311-101, Table 14.2.1

Results - safety

Pain

About 30% of patients experienced pain during the first treatment cycle, which declined to about 8% during the second cycle from Day 2 onwards. No additional decrease in the percentage of patients with pain was seen during Cycle 3 (Figure 13).

Figure 13: APN311-101 - Percentage of patients with pain (based on patients with pain evaluation)

Source data: APN311-101, Table 14.3.8

All patients received i.v. morphine during each treatment cycle and during each day of dinutuximab beta treatment (APN311-101, Table 14.1.12.5.1).

Adverse events

All patients reported at least one AE and all patients experienced at least one event judged to be related to dinutuximab beta (Table 11). The incidence of SAEs was low. No patient died due to an AE.

Table 11: APN311-101 - Overview of adverse events

Patients with	N (%) patients (N=15)	
Any AE	15	(100.0)
Any AE possibly related to dinutuximab beta	15	(100.0)
Any SAE	2	(13.3)
Any SAE possibly related to dinutuximab beta	1	(6.7)
Any AE leading to discontinuation of study drugs	-	
Higher CTCAE Grade		
Grade 3	13	(86.7)
Grade 4	2	(13.3)
Any AE leading to death	-	
Deaths *	10	(66.7)

* All documented deaths, including deaths during follow-up period.

Possibly related AEs: AEs with relationship coded as 'Possible', 'Probable', 'Definite' or with missing relationship.

AE = adverse event, N = number of patients, CTCAE = Common Terminology Criteria for Adverse Events, SAE = serious adverse event.

Source data: APN311-101 Table 14.3.1.1

The most frequently reported AE was pyrexia, followed by constipation, pruritus, abdominal pain, and weight increased (Table 12).

Table 12: APN311-101 - frequent adverse events (reported in more than 25% of patients)

System organ class Preferred term	N (%) patients (N=15)	
Overall	15	(100.0)
Gastrointestinal disorders	15	(100.0)
Constipation	11	(73.3)
Abdominal pain	7	(46.7)
Nausea	6	(40.0)
Vomiting	4	(26.7)
Diarrhea	4	(26.7)
General disorders and administration site conditions	14	(93.3)
Pyrexia	13	(86.7)
Pain	6	(40.0)
Skin and subcutaneous tissue disorders	10	(66.7)
Pruritus	9	(60.0)
Rash	5	(33.3)
Vascular disorders	5	(33.3)
Hypotension	4	(26.7)
Musculoskeletal and connective tissue disorders	10	(66.7)
Pain in extremity	4	(26.7)
Investigations	11	(73.3)
Weight decreased	7	(46.7)

N = Number of patients.

Source data: APN311-101 Table 14.3.2.1

For 13 patients (86.7%) the most severe grade was CTCAE Grade 3 and for 2 patients (13.3%) it was Grade 4. The most frequent CTCAE grades were Grade 1 or 2. AEs of Grade 3 reported in more than 1 patient were pyrexia, AST increased (in 2 patients), transaminases increased, hypoxia, and abdominal pain (each in 3 patients). AEs of Grade 4 were leukocytopenia, metastases to central nervous system, pancreatitis, and paralysis (each: 1 patient) (APN311-101 Table 14.3.2.2).

Ten deaths occurred up until the cut-off date. All were linked to progression of the pre-existing cancer. None of the deaths occurred during the treatment period or was related to an AE reported during the treatment period. No death was judged to be treatment-related (APN311-101 Section 12.3.1.1).

In total, there were 4 SAEs in 2 patients i.e. Grade 4 pancreatitis, Grade 3 gallbladder obstruction, Grade 1 proctalgia, and Grade 1 pruritus. The SAE of gallbladder obstruction was judged to be dinutuximab beta-related (APN311-101 Tables 14.3.4.1 and 14.3.5.2).

Laboratory, vital signs, and other safety assessments

Out-of-range clinical laboratory values occurred in all patients, with no trends regarding a dose-relationship. Three patients experienced clinically relevant out-of-range hematology values during a treatment cycle, namely eosinophilia, thrombocytopenia, and lymphopenia. Five patients experienced clinically relevant out-of-range laboratory values for chemistry parameters, most frequently increased hepatic transaminases (CSR APN311-101, Section 12.4.2.1.1).

No significant changes were reported in chest X-ray, vital signs, or physical examinations (CSR APN311-101, Section 12.5).

Conclusion

Dinutuximab beta applied as STI (8 hours for 5 consecutive days per cycle) is an active immunotherapeutic agent with a manageable safety profile.

5.3.2.3. Study APN311-201

(Source: APN311-201 CSR dated 28 Jul 2020, APN311-201 Historical control report dated 16 Feb 2016)

APN311-201 was an investigator-initiated, multicenter, multinational, open-label Phase II study. The primary objective of the study was to determine the safety and feasibility of dinutuximab beta in combination with s.c. IL-2 in patients who had received allogeneic haploidentical stem cell transplantation (haplo-SCT) for neuroblastoma relapse. A total of 68 patients were planned for this report and 68 patients were analyzed according to the protocol version 1.2.

Design

Patients: patients who had received allogeneic haplo-SCT for neuroblastoma relapse.

Treatment: dinutuximab beta (20 mg/m²/day) was administered as an 8-hour i.v. infusion for 5 consecutive days (Days 1 to 5), for six 4-week cycles in combination with s.c. IL-2 (1x10⁶ IU/m²/day), administered in Cycles 4 to 6 only (at Days 6, 8 and 10). Patients could receive another 3 courses of dinutuximab beta, if there was evidence of SD or response after 6 cycles.

Premedication:

Patients were premedicated with an i.v. antihistamine and ranitidine within approximately 30 minutes prior and during the infusion of the study drug. Pain as an anticipated side effect of dinutuximab beta was managed by a standard pain prophylaxis with morphine hydrochloride (CSR dated 28 Jul 2020).

Safety and efficacy endpoints: Primary endpoint: "success of treatment", defined as a patient receiving the full protocol treatment and still alive 180 days after treatment without: unacceptable toxicities; acute graft vs host disease (GvHD) ≥Grade 3 or extensive chronic GvHD. Further endpoints: disease status, toxicities, AEs, rate and type of dose limiting toxicity, incidence of GvHD, progression during treatment, EFS, and OS.

Results - patients

Overall, 68 patients were evaluated. Median age at study entry was 6.5 years (range 3 to 20 years). Forty-six patients were male (67.6%) and 22 patients were female (32.4%). All patients suffered from a Stage IV relapse of a histologically confirmed neuroblastoma; most patients also had metastatic disease at initial diagnosis (94.1%) (APN311-201, Table 10-1).

Thirty-nine patients (57.4%) received full protocol treatment, i.e. at least 510 mg/m² dinutuximab beta in Cycles 1 to 6, and 21 patients (30.9%) completed 6 treatment cycles. Eighteen patients (26.5%) completed 9 treatment cycles (CSR dated 28 Jul 2020).

Results - efficacy

Treatment success

The primary efficacy endpoint was "success of treatment", defined as a patient receiving the full protocol treatment, alive 180 days after treatment without progression and without unacceptable toxicity and acute GvHD $\geq$ Grade 3 or extensive chronic GvHD. Forty patients (58.8%) were considered to have reached the primary efficacy endpoint and achieved "success of treatment" (CSR dated 28 Jul-2020) (APN311-201, Table 10-2).

Response

Three-year OS rate for all patients (n = 68) was 57.9% (95% CI 49.6% – 66.2%). For patients with CR prior haploidentical stem cell transplantation, 3-year OS rate was 62.8% (95% CI 48.3% - 77.3%), for patients with PR, OS was 69.2% (95% CI 69.6%-43.6%), (p = 0.58), patients with either PD, non-response or mixed response to relapse therapy had a 3-year OS rate of 20% (95% CI 47.4% - 3.1%), median survival was 1.35 years (p = 0.03 [CR/PD], p = 0.0002 [PR/PD]) (CSR dated 28 Jul 2020).

Three-year OS rate for MYC-N amplified (n = 19) was 66.8% (95% CI 84.5% – 52.0%) with a median OS of 5.5 years, for patients with a MYC-N non-amplified tumor (n = 46), 3-year OS was 53.7% (95% CI 84.6% – 37.9%) (p = 0.73) (unknown n = 3).

For patients with CR prior haploidentical stem cell transplantation, 3-year EFS rate was 61.4% (95% CI 80.5% – 33.3%), for patients with PR 53.2% (95% CI 68.1%-33.5%), median EFS rate 3.3 years (p = 0.78), patients with PD, non-response or mixed response had a 3-year EFS rate of 20% (95% CI 47.5% –3.1%), and median EFS rate was 1.05 years (p = 0.02 [CR/PD], p = 0.006 PR/PD).

There was no significant difference in EFS or OS between CR before or after hematopoietic stem cell transplant (HSCT), 3-year EFS for patients with CR before HSCT was 62.8% and 60% for patients with CR after HSCT (p = 0.9), for patients with PR before HSCT 3-year EFS was 69.2%, for patients with PR after HSCT 3-year EFS rate was 61.8% (p = 0.5) (Table 13) (CSR dated 28 Jul 2020).

Table 13: APN311-201 - Overall response at end of treatment in patients with evidence of disease at baseline

Category	Number (%) of patients	
Disease status at end of treatment	N = 20	
Evaluable	20	(100.0)
CR	10	(50.0)
PR	3	(15.0)
MR	-	-
NR	1	(5.0)
PD	6	(30.0)

CR = complete response, MR = mixed response, N = number of patients, NR = no response, PD = progressive disease, PR = partial response.

Source data: APN311-201 Table 14.2.2.3

Event-free survival

Three-year EFS rate was 47% (95% CI 59.1% – 33.7%), with a median EFS of 2.3 years. The EFS rate was 72.6% at 1 year, and 51.8% and 47.5% at 2 and 3 years, respectively (APN311-201 Figure 10-2) (CSR dated 28 Jul 2020).

Overall survival

During the study, 25 patients died. The median survival time was 5.5 years, which exceeded historical median control OS of 1 year. This study data compared with literature results suggest an advantage in OS. The OS rate was 86.3% at 1 year, 68.6% at 2 years, and 57.9% at 3 years (APN311-201, Figure 10-1). This represents an improvement over recently published data. London et al [5] (ASCO 2014) reported 1-year and 4-year OS rates of 56% and 14% respectively in patients with relapsed and refractory disease treated without antibody therapy during 2002 to 2014, Illhardt et al published OS of 61.5% at 1 year and a 3-year OS rate of 26.9% with haploidentical transplantation alone (n = 26) [30] (CSR dated 28 Jul 2020).

Historical control analysis

In a historical control analysis, EFS and OS for relapsed or refractory neuroblastoma patients in study APN311-201 was compared to a similar patient population (N = 17) that received haploidentical stem cell transplantation as relapse therapy only, but no immunotherapy. The historical control patients were derived from a Phase I/II study conducted at the University Hospital Tübingen, Germany.

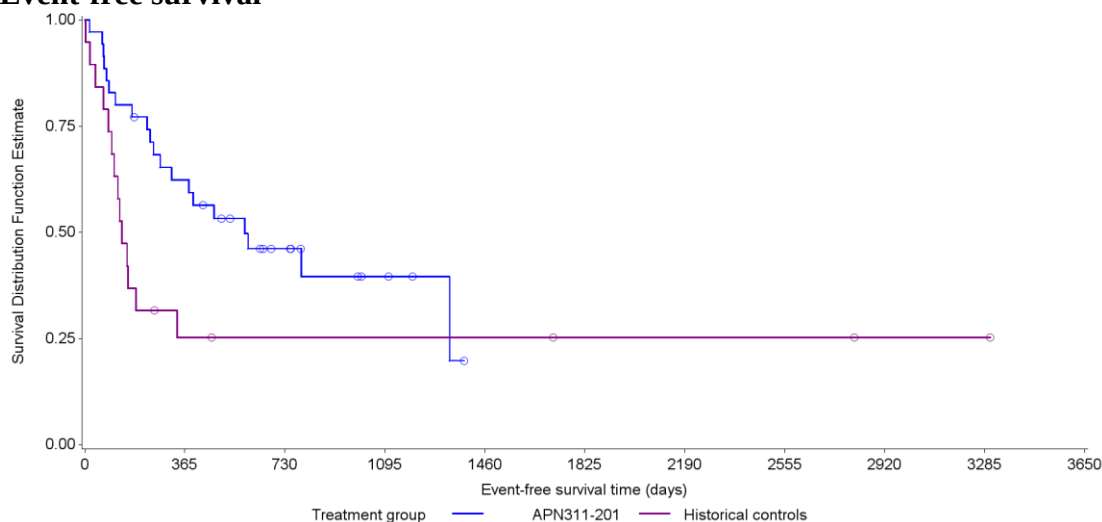
EFS was numerically higher in the APN311-201 group than in the historical control group (Figure 14) at all time points analyzed, with an EFS rate of 62%, 46% and 40% at 1, 2, and 3 years, respectively, (CSR dated 28 Jul 2020) vs 24% at 1 year, 2 years, and 3 years, respectively, in the historical control group. The p-value for this comparison was 0.0570 by log-rank test (Historical Control Report Table 3.1.1).

The mean OS time in APN311-201 was about twice as long as the OS time in the historical control group (25 months vs 13 months) (median OS in the APN311-201 study was not estimable; it was 368 days in the control group). OS in relapsed APN311-201 patients was 80%

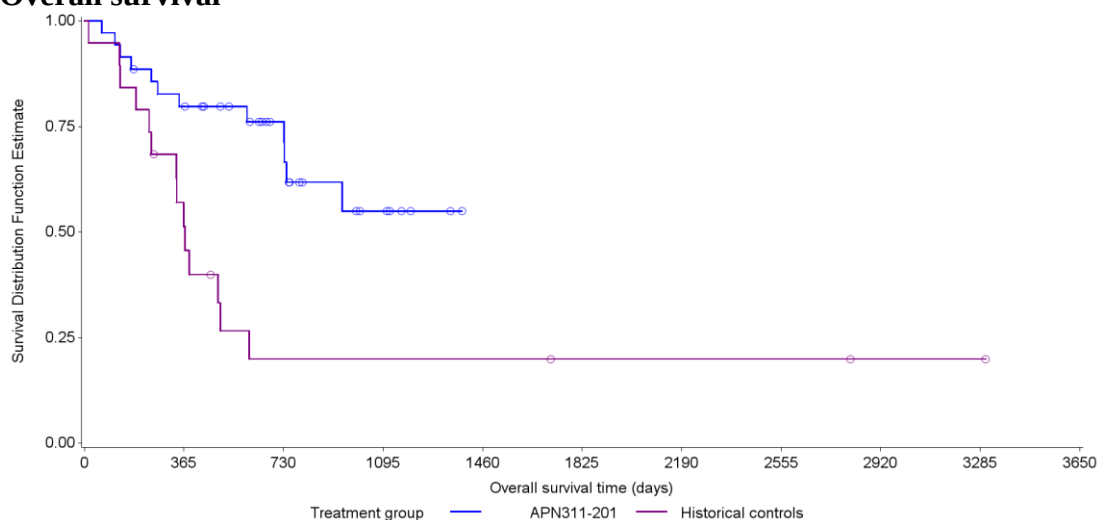
at 1 year, 76% at 2 years, and 55% at 3 years, compared with 53%, 21%, and 21% in relapsed historical control patients. At each time point analyzed, OS was thus at least 27% higher in the APN311-201 group than in the historical control group, reaching statistical significance for the between-treatment comparison ($p = 0.0026$) (Historical Control Report Table 3.2.1).

Figure 14: APN311-201 versus historical control - event-free survival and overall survival

Event-free survival



Overall survival



Source data: Historical Control Report Figures 3.1.1 and 3.2.1

Results - safety

Pain

Pain was observed in all patients and was most frequent during Cycle 1, occurring in 96% ($n = 65$) of patients, and decreasing to 68.4% ($n = 46$) during Cycle 6. The most common sites of pain were the abdomen and the legs.

As pain is a known and expected side effect with administration of ch14.18/CHO mAb, concomitant analgesic treatment was given throughout the study as needed. More than 90% of patients needed opioid analgesics; the median dose was slightly higher during the first 3 cycles

(0.02 to 0.03 mg/kg/h) however, it was constant throughout all treatment cycles (approximately 0.02 mg/kg/h) (CSR dated 28 Jul 2020).

Toxicities

Study APN311-201 documented AEs mainly as National Cancer Institute Common Toxicity Criteria for Adverse Events (NCI CTCAE). The APN311-201 protocol specified to only report AEs if not already documented on toxicity forms.

All patients experienced pain, fever, and general condition toxicities. Other frequent toxicities were hematologic, gut toxicities (nausea/vomiting, constipation), skin toxicity, and liver toxicity ([Table 14](#)).

Table 14: APN311-201 - Frequent toxicities

	Grade 1/2	Grade 3/4
Courses (n)		
Hematological toxicity		
Hemoglobin	306 (83.0%)	53 (14.3%)
White blood cell	217 (59.0%)	112 (30.4%)
Granulocytes (ANC)	182 (49.3%)	94 (25.5%)
Platelets	56 (15.2%)	35 (9.4%)
Cardinal toxicities		
General condition	296 (80.4%)	26 (7.1%)
Skin toxicity	148 (40.1%)	4 (1.1%)
Allergy	114 (29.4%)	52 (13.5%)
Pulmonary toxicity	59 (15.6%)	3 (0.8%)
Gastrointestinal		
Nausea/vomiting	142 (38.7%)	3 (0.8%)
Diarrhea	94 (25.5%)	12 (3.3%)
Constipation	163 (44.5%)	0 (0%)
Stomatitis	32 (8.8%)	0 (0%)
Cardiac		
Cardiac function	0 (0%)	0 (0%)
QTc prolongation	2 (1.5%)	1 (0.8%)
ECHO: LV/SF	2 (2.0%)	0 (0%)
Hypotension	54 (14.7%)	6 (1.7%)
Hypertension	7 (1.9%)	4 (1.1%)
Renal		
Creatinine	48 (13.2%)	1 (0.3%)
Proteinuria	29 (9.1%)	0 (0%)
Hematuria	5 (1.6%)	1 (0.3%)
Glomerular filtration rate	8 (7.6%)	0 (0 %)
Tubular phosphate reabsorption	0 (0 %)	1 (25.0%)
Hemorrhagic Cystitis	0 (0%)	0 (0%)
Neurotoxicity		
Central*	20 (5.5%)	10 (2.7%)
Peripheral neurotoxicity	9 (2.5%)	1 (0.3%)
Liver		
Bilirubin	11 (3.1%)	8 (2.2%)
SGOT/SGPT	195 (53.5%)	43 (11.8%)
(veno-occlusive disease) VOD ¹	0 (0%)	0 (0%)
Pulmonary Toxicity¹	59 (15.6%)	3 (0.8%)
GvHD²		
Skin	3 (0.8%)	0 (0%)
Gastrointestinal	1 (0.3%)	1 (0.3%)
Liver / CNS	0 (0%)	0 (0%)
Adverse Event - no grading		
	Courses n (%)	
Water retention³	236 (60.8%)	
Accommodation disturbances (Pupillotonia)	42 (10.8%)	
Fever during AB-cycle: received antibiotics	31 (8.0%)	

Grade 1/2	Grade 3/4
Courses (n)	

Abbreviations: ANC = absolute neutrophil count, ECHO = echocardiogram, LV/SF = left ventricular shortening fraction, SGOT = serum glutamic-oxaloacetic transaminase, SGPT = Serum glutamic pyruvic transaminase, GvHD = graft versus host disease, CNS = central nervous system, NIH = National institute of health, PRES = Posterior reversible encephalopathy syndrome.

¹ Bearman Toxicity (only Grades 1 to 3)

² NIH Grading

³ Patient received diuretics - no capillary leak

*One patient developed grade 5 central neurotoxicity and died with signs of brain atrophy/PRES

Source data: APN311-201, Section 11.1.

Adverse events

Severe neuropathic pain is an expected adverse drug reaction with dinutuximab beta mAb short-term administration that occurred in all patients and remedial therapy with opioids was needed for all of them. Skin reaction and allergy (including those of severe intensity) occurred frequently with ch14.18/CHO mAb which is expected for chimeric antibody treatment. The type and frequency of undesirable effects of ch14.18/CHO mAb treatment is consistent with previous studies ([Error! Reference source not found. Table 15](#)) (CSR dated 28 Jul 2020).

Table 15 APN311-201 - Overall summary of treatment-emergent adverse events

Severe Adverse Events	Total n (%) number of patients	Life-threatening /disabling/ fatal n (%)
Anaphylactic reaction	3 (4.4%)	1 (1.5%)
Capillary leak Syndrome/SIRS	7 (10.3%)	5 (7.4%)
Hemolytic anemia	7 (10.3%)	3 (4.4%)
Thrombosis	3 (4.4%)	1 (1.5%)
Eye	2 (2.9%)	
Anaphylactic reaction	3 (4.4%)	1 (1.5%)
Infections associated with pathogens		
Fever of unknown origin	22 (32.4%)	2 (2.9%)
Bacterial Sepsis: E. coli	3 (4.4%)	2 (2.9%)†
Salmonella infection	1 (1.5%)	-
Viral Infections		
Adenovirus associated enteritis	7 (10.3%)	-
BK virus infection	1 (1.5%)	-
Cytomegalovirus infection	5 (7.4%)	-
Human herpes virus	2 (2.9%)	1 (1.5%)††
Influenza	2 (2.9%)	-
Rota associated enteritis	1 (1.5%)	1 (1.5%)
Respiratory syncytial virus	2 (2.9%)	-
Varicella Zoster virus infection	1 (1.5%)	-
Pneumocystis jiroveci pneumonia	3 (4.4%)	1 (1.5%)

† Two patients died of E. coli blood stream infection

†† One patient died of HHV6-associated encephalitis, and pneumonitis

Source data: APN311-201 Table 11-1

The effects of most interest related to the immunotherapy were fever, pain, tachycardia, hypotension, hypersensitivity reactions, and CLS/fluid retention. Fever (>38°C) was experienced by nearly all patients, mostly Grade 2 or 3 in severity, and was frequent during all

cycles. Three patients with high grade fever had generalized seizures (CSR dated 28 Jul 2020). Overall, combination immunotherapy after haploidentical SCT was manageable in most patients. Sixteen out of 68 patients (23.5%) experienced life-threatening, disabling, or fatal adverse events (APN311-201 Table 11-1). The most frequent undesirable effects of dinutuximab beta treatment combined with aldesleukin were pain, CLS, hypotension, and hypersensitivity reactions. Undesirable effects on red and white blood cells as well as changes in transaminases were common laboratory adverse events.

Of the 68 patients, 5 patients died, resulting in a cumulative incidence of non-relapse mortality of 7.4%. Three patients died while receiving immunotherapy: Two (3%) patients because of infections, 1 (1.5%) patient with HHV-6 infection complicated by encephalitis and pneumonitis 134 days after HSCT, and another (1.5%) patient with fatal bacterial infection (*E. coli*) 463 days after HSCT. One patient (1.5%) died 3 weeks after the second antibody cycle (189 days after HSCT) with signs of encephalitis and/or posterior reversible encephalopathy syndrome. One patient (1.5%) died of an acute lymphoblastic leukemia 648 days after HSCT/230 days after the end of the antibody treatment and another patient (1.5%) died 656 days after HSCT because of an *E. coli* sepsis.

The fatal outcome of this treatment-emergent adverse event (TEAE) was most likely a consequence of immune suppression after myeloablative chemotherapy. The reported spectrum of fatal SAEs in this study is expected in patients with advanced cancer and following allogeneic SCT (CSR dated 28 Jul 2020).

Acute graft vs host disease (GvHD) was closely monitored in this study as both antibody and IL-2 therapy might alter the susceptibility for this complication after receiving a stem cell transplant. Acute GvHD was experienced by 22.4% (n = 15) of the patients, mostly Grade 1/2 acute GvHD of the skin, 2 patients developed Grade 2/3 acute GvHD of the gut. Induction of late-onset acute GvHD (during antibody treatment) was seen in 5 patients (7.4%), 1 patient (1.5%) developed a steroid sensitive Grade 3 GvHD of the GI-tract, 1 patient (1.5%) developed Grade 2 GvHD of the GI-tract, both resolved quickly. Three patients (4.4%) developed Grade 1/2 GvHD of the skin, without the need of systemic (steroid) therapy. No Grade 4 GvHD or liver GvHD occurred. (APN311-201 Section 11.1, CSR dated 28 Jul 2020).

Other safety assessments

No effect on QTc was observed in the ECG. Neither was there any appreciable effect on cardiac output in the echocardiography.

Conclusion

Dinutuximab beta showed an overall favorable human toxicity profile at the selected dose-level in pediatric patients with relapsed neuroblastoma after haploidentical SCT and a >50% response rate in this patient population. In addition, immunotherapy did not seem to enhance the risk for GvHD after haploidentical SCT over a treatment period of 9 cycles of dinutuximab beta and 6 cycles of combination therapy with IL-2.

5.3.2.4. Study APN311-202

(Source: APN311-202 [V1] and [V2- draft version] dated 21 Dec 2020 with Last Patient Visit Included in Analysis (Cut-Off Date): 11 Oct 2018)

APN311-202 was an investigator-initiated, multicenter, multinational, open-label, dose-escalation study. The primary objective was to find a tolerable treatment schedule that reduces the pain-toxicity profile of dinutuximab beta whilst maintaining immunomodulatory efficacy in patients with primary refractory or relapsed neuroblastoma. The study consisted of a dose schedule finding phase (completed, N = 24) followed by a confirmatory phase. Data from the 24 patients of the dose finding phase as well as from the 100 patients in the confirmatory cohort are analyzed and reported in the APN311-202 V1 CSR (based on protocol version 2).

Design

Patients: patients with primary refractory or relapsed neuroblastoma who had received at least one previous high dose treatment followed by stem cell rescue after conventional therapy.

Treatment: dinutuximab beta was administered as continuous infusion for 10 days (Days 8-18), in combination with fixed doses of s.c. IL-2 (6×10^6 IU/m²/day (Days 1-5 and 8-12) for 5 cycles (cycle length 28 to 35 days). Several dose regimens were scheduled, however, the starting level of 10 mg/m²/day dinutuximab beta over 10 days (total dose of 100 mg/m²) was considered the suitable dose based on the pain-toxicity profile and sufficient immunomodulatory efficacy (activated NK cells, dinutuximab beta level). This was the first and only dose used, i.e. all patients in the 'dose schedule' phase of the study received this dose. In the confirmatory phase, patients were treated with 10 mg/m²/day dinutuximab beta over 10 days and concomitant IL-2, as described above, for 5 cycles (each 35 days).

Premedication: intravenous morphine; bolus injection followed by continuous infusion to be reduced on a daily basis as possible. Prior to each cycle of dinutuximab beta the patient was to be dosed with oral gabapentin.

Safety and efficacy endpoints of the dose schedule phase: primary endpoints: a) Pain-toxicity endpoint: i.v. morphine-free dinutuximab beta infusion schedule after the first 5 days during the first cycle in $\geq 80\%$ of patients. b) Efficacy endpoint: On Day 15 of the first cycle in $\geq 80\%$ of patients: an increase of 500% and/or an absolute minimum increase to ≥ 100 cells/ μ L of the CD16/CD56 positive activated NK cells, AND a measurable dinutuximab beta level of at least 1 μ g/mL. Further endpoints: activated NK cell concentrations above baseline levels in $\geq 80\%$ of patients, appearance of soluble IL-2 receptor, increase of absolute lymphocyte counts by 50% over baseline, increase of absolute NK cell numbers > 1000 cells/ μ L in $\geq 80\%$ of patients, anti-tumor response in patients with measurable disease (bone marrow, skeletal lesions, soft tissue lesions, lymph nodes and/or primary tumor site) as measured by immunocytology, mIBG imaging, CT and/or MRI. The objective of the confirmation phase was to confirm the primary and secondary endpoints in the expansion cohort.

Results – 202 V1 and V2 patients

Twenty-four patients were included in the dose schedule cohort and 100 patients were included in the confirmatory cohort (based on protocol version 2). All 122 patients were evaluated together. Of these, 85 patients (69.7%) completed the study treatment as planned; 20 patients (16.4%) terminated dinutuximab beta therapy due to PD, 9 patients (7.4%) discontinued prematurely due to toxicity. In addition, 3 patients (2.5%) discontinued treatment based on 'physician/parent/patient decision' and 4 patients (3.3%) discontinued due to 'other' reasons. For 1 patient, the reason for study termination was missing; the patient received 8 instead of

5 cycles of treatment and had no reason for treatment termination documented (APN311-202 V1 and V2, Table 10-1).

Almost all female patients (94.2%) and majority of the male patients (78.6%) were children between 2 to 11 years of age. The mean age of the study population at baseline was 6.0 years, with a minimum of 1 year and a maximum of 27 years. The majority of patients had received rapid COJEC (cisplatin, vincristine, carboplatin, etoposide, and cyclophosphamide) induction chemotherapy followed by high-dose busulfan and melphalan (BuMel) + autologous stem cell transplantation (ASCT), therapy as first-line therapy, often in combination with radiotherapy, surgery, and 13-cis-RA maintenance therapy (APN311-202 V1 and V2 Table 11-3). However, 54 patients (44.3%) had relapsed and 67 patients (54.9%) had refractory disease at baseline. Forty-four patients (36.1%) had no evidence of disease at baseline while disease was evaluable at baseline for 78 patients; for 43 patients (35.3%) disease was evaluable only by mIBG and/or BM histology and for 35 patients (28.7%) disease was measurable by MRI and/or CT (APN311-202 V1 and V2, Table 11-2).

Overall, 87 of the 118 patients (73.7%) completed all 5 treatment cycles (APN311-202 V1 and V2, Table 11-5).

Results - efficacy – 202 V1 and V2 patients

Pain-toxicity endpoint

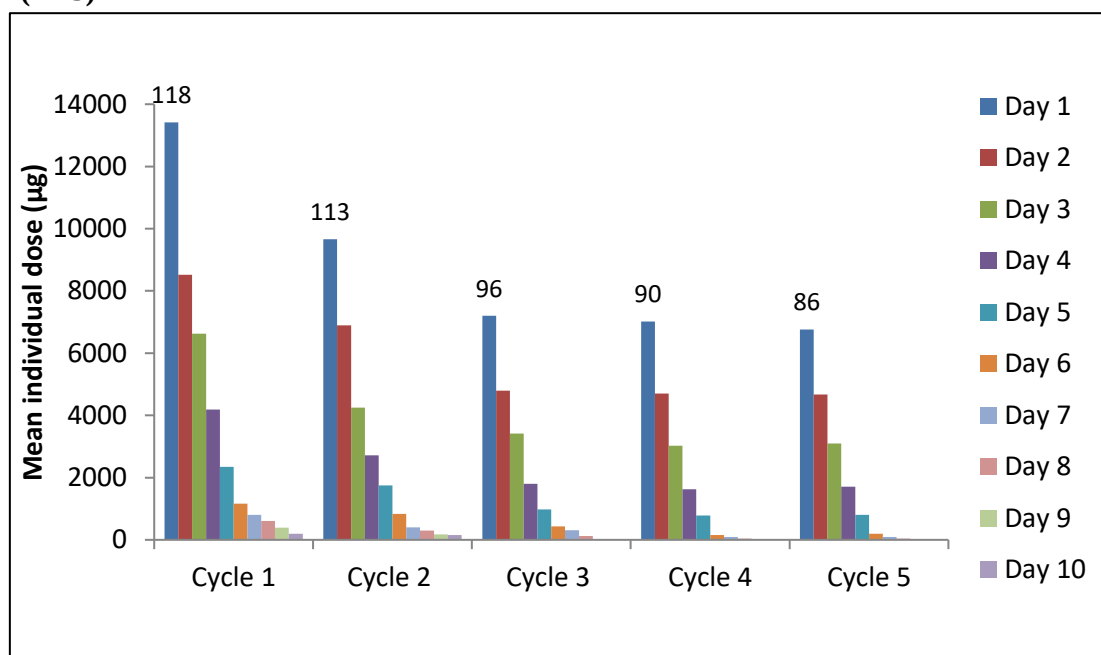
In dose finding Cohort 1 (first 10 patients), 40% of the patients still received i.v. morphine after the first 5 days for organizational reasons. Centers were going through a learning curve with administering dinutuximab beta as continuous infusion. After additional training the percentage of patients receiving i.v. morphine after Day 5 decreased substantially in the second dose schedule finding cohort. Hence, more than 90% of the patients were i.v. morphine-free after 5 days in Cycle 1, hence meeting the pain-toxicity criterion.

In accordance with this result, the acceptable dose schedule established during the dose schedule finding stage of the study (10 mg/m² over 10 days) was used for the first confirmatory phase cohort. In this confirmatory cohort, 83% of the patients were i.v. morphine-free after the first 5 days, hence confirming the primary pain-toxicity endpoint. In the second confirmatory cohort, added with protocol amendment 1, the percentage of morphine-free patients decreased to 67%. Across all patients included in the study, the percentage of patients with an i.v. morphine-free dinutuximab beta infusion schedule after the first 5 days during the first cycle was 72.0% (APN311-202 V1 and V2, Table 14.1.3-10).

The percentage of patients with an i.v. morphine-free dinutuximab beta infusion schedule after the first 5 days during the first cycle was 79.17% in the dose finding cohort (1+2) and 70.2% in the confirmatory cohort (1+2) (APN311-202 V1 and V2, Table 11-7).

An overview of treatment with i.v. morphine (mean treatment dose) over treatment days and cycles is provided in [Figure 15](#) for all patients and in [Figure 16](#) only for patients receiving i.v. morphine.

Figure 15: APN311-202 V1 and V2- i.v. morphine individual dose by cycle – all patients (FAS)

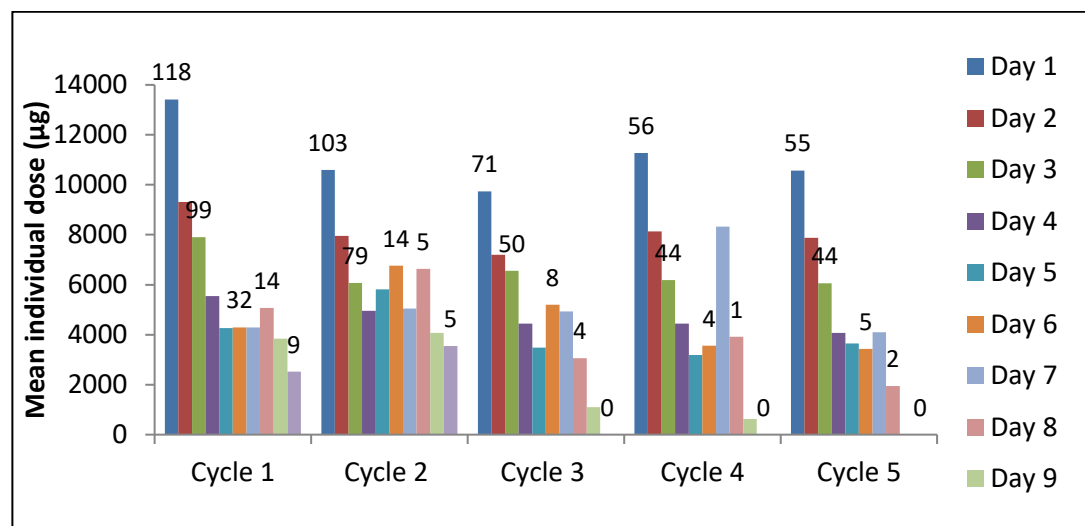


Abbreviation: FAS = full analysis set; i.v. = intravenous.

Column numbers = number of patients with data available.

Source data: APN311-202 V1 and V2, Table 14.3.3-8

Figure 16: APN311-202 V1 and V2 - i.v. Morphine individual dose by cycle – only patients receiving i.v. morphine (FAS)



Abbreviation: FAS = full analysis set; i.v. = intravenous.

Column numbers = number of patients with data available.

Source data: APN311-202 V1 and V2, Table 14.3.3-9

Proposed Primary efficacy endpoint

In the dose schedule cohort, 87.5% of 24 patients with data available fulfilled the response criteria for the primary efficacy endpoint i.e. had an increase of 500% and/or an absolute minimum increase to ≥ 100 cells/ μ L (0.1×10^9 cells/ μ L) of the CD16/CD56 positive activated

NK cells, and a measurable dinutuximab beta level of at least 1 µg/mL on Day 15 of the first treatment cycle (APN311-202 V1 Section 11.4.2.2).

Secondary efficacy endpoint

Anti-tumor response

Best overall response was evaluated based on immunocytology, MRI/CT, mIBG imaging, and/or BM in all patients (n = 77). 45 out of 118 evaluable patients (38.1%) in the FAS had no evidence of disease and 36 out of 118 patients (30.5%) had PD (APN311-202 V1 and V2 Table 14.2.1-1).

The overall treatment response is provided for all patients with evidence of disease at baseline in **Error! Reference source not found.**Table 16.

For 34.8% of the patients (41 out of 118 patients) no evidence of disease had been observed at baseline (APN311-202 V1 and V2, Table 14.1.1-2). Of the 41 patients without evidence of disease at baseline, 34 (82.9%) had no evidence of disease also as best response during the study, and 7 patients (17.1%) had progressive disease (APN311-202 V1 and V2, Tables 14.1.1-2 and 14.2.1-10).

Table 16: APN311-202 V1 and V2 - Treatment response – only patients with any measurable disease at baseline (FAS)

	Statistics	Mid evaluation	End of treatment	Best response
	N	76	73	77
No evidence of disease	n (%)	12 (15.8%)	12 (16.4%)	17 (22.1%)
Improved disease	n (%)	23 (30.3%)	23 (31.5%)	27 (35.1%)
Stable disease	n (%)	29 (38.2%)	12 (16.4%)	21 (27.3%)
Mixed response	n (%)	1 (1.3%)	0	1 (1.3%)
Progressive disease	n (%)	11 (14.5%)	26 (35.6%)	11 (14.3%)

Abbreviation: FAS = full analysis set

Percentages were based on all evaluable patients, excluding missing values.

Source data: APN311-202 V1 and V2, Table 14.2.1-4

The table below shows the overall treatment response for the different disease types (refractory/relapsed) at the end of Cycle 2 (mid evaluation) and at the end of treatment, and the best achieved overall response regardless of time point. Only results for patients with evidence of disease at baseline are presented (Table 17).

Table 17: APN311-202 V1 and V2 - Treatment response in patients with refractory and relapsed disease – only patients with any measurable disease at baseline (FAS)

	Statistics	Mid evaluation	End of treatment	Best response
Refractory Disease (Patients with Evidence of Disease at Baseline)				
	N	44	44	45
No evidence of disease	n (%)	5 (11.4%)	7 (15.9%)	9 (20.0%)
Improved disease	n (%)	15 (34.1%)	18 (40.9%)	19 (42.2%)
Stable disease	n (%)	22 (50.0%)	11 (25.0%)	15 (33.3%)
Mixed response	n (%)	0	0	0
Progressive disease	n (%)	2 (4.6%)	8 (18.2%)	2 (4.4%)

	Statistics	Mid evaluation	End of treatment	Best response
Relapsed Disease (Patients with Evidence of Disease at Baseline)				
	N	32	31	32
No evidence of disease	n (%)	7 (21.9%)	5 (17.2%)	8 (25.0%)
Improved disease	n (%)	8 (25.0%)	5 (17.2%)	8 (25.0%)
Stable disease	n (%)	7 (21.9%)	1 (3.5%)	6 (18.8%)
Mixed response	n (%)	1 (3.1%)	0	1 (3.1%)
Progressive disease	n (%)	9 (28.1%)	18 (62.1%)	9 (28.1%)

Abbreviation: FAS = full analysis set

Percentages were based on all evaluable patients, excluding missing values.

Source data: APN311-202 V1 and V2, Table 14.2.1-6, Table 14.2.1-8

Event-free survival

Event-free survival was defined as time between first day of IL-2 administration to date of relapse/progression or death. The EFS rate in the overall FAS population was estimated to be 62.7% at 1 year, 56.8% at 2 years and 51.7% at 3 years. At 5 years, the EFS rate was 48.2%. Mean time to progression was 1329 days (APN311-202 V1 and V2, Tables 14.2.2-1, 14.2.2-2, 14.2.2-3, and 14.2.2-7).

Mean time to EFS was lower in patients with relapsed neuroblastoma (852 days) than in patients with refractory neuroblastoma (1611 days). In patients with relapsed neuroblastoma, the EFS rate was 43.1% at 1 year, and 35.3% at 2 years, plateauing at 27.5% in years 4 and 5. The EFS rate in patients with refractory neuroblastoma was 77.3% at 1 year, 72.7% at 2 years, decreasing to 63.5% at 5 years (APN311-202 V1 and V2, Tables 14.2.2-5, 14.2.2-6).

Overall survival

OS was defined as time between first day of IL-2 administration to date of death. OS in the overall FAS population was 90.7% at 1 year, decreased to 73.7% at 2 years, and further decreased to 57.8% at 5 years. Mean time to death was 1682 days (APN311-202 V1 and V2, Tables 14.2.3-1, 14.2.3-2, 14.2.3-3, and 14.2.3-7).

The Kaplan-Meier estimate for OS was higher in refractory than in relapsed patients, i.e. 95.5% vs 84.3% at 1 year, 84.9% vs 58.8% at 2 years, 77.3% vs 52.9% at 3 years, 72.6% vs 44.8% at 4 years, and 67.1% vs 44.9% at 5 years. Accordingly, mean OS times were 1799 days and 1399 days for patients with refractory and relapsed disease, respectively (APN311-202 V1 and V2, Tables 14.2.3-5 and 14.2.3-6).

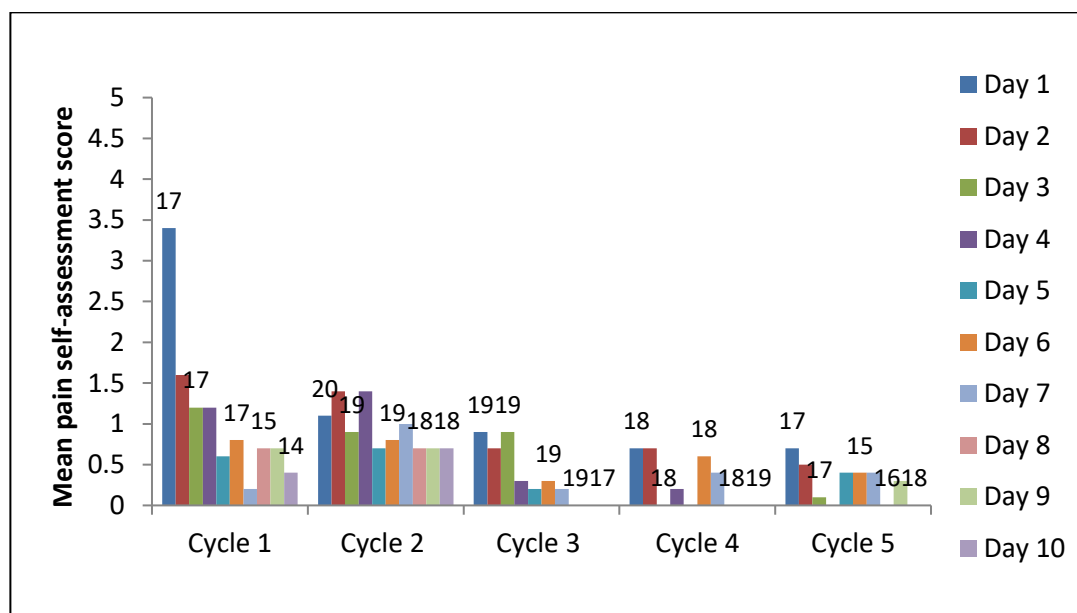
Results - safety - 202 V1 and V2 patients

Pain

For the parent and medical team pain assessments, the Modified Objective Pain Score (MOPS) scale (for children >4 years of age) was used for 67.2% of the patients and the Childhood Discomfort and Pain Scale (KUSS) (for children ≤4 years of age) was used for 32.8% of the patients. To determine patient performance, the Lansky scale (for patients ≤16 years of age) was used in 96.7% of the patients and Karnofsky performance scale (for patients >16 years of age) was used in 3.3% of the patients. For self-assessment of pain, the patients used the Faces Pain Scale - Revised (FPS-R) in 51.6% of cases and the Wong-Baker scale in 9.0% of cases. A total of 48 patients (39.3%) were not evaluable via self-assessment (APN311-202 V1 and V2, Table 14.3.5.1-1).

Figure 17 shows mean pain intensity throughout the dinutuximab beta treatment period based on all patients with pain self-assessment for each of the 10 antibody treatment days per cycle. Based on the self-assessment pain score, a trend of decreasing mean pain over time was observed. Most pain was experienced during the first day of the first cycle. From the second day and thereafter the intensity of pain was continuously decreasing. Similar results were observed for the parent assessment score for patients >4 years of age (MOPS), as shown in Figure 18 and for patients ≤4 years of age (KUSS), as shown in Figure 19.

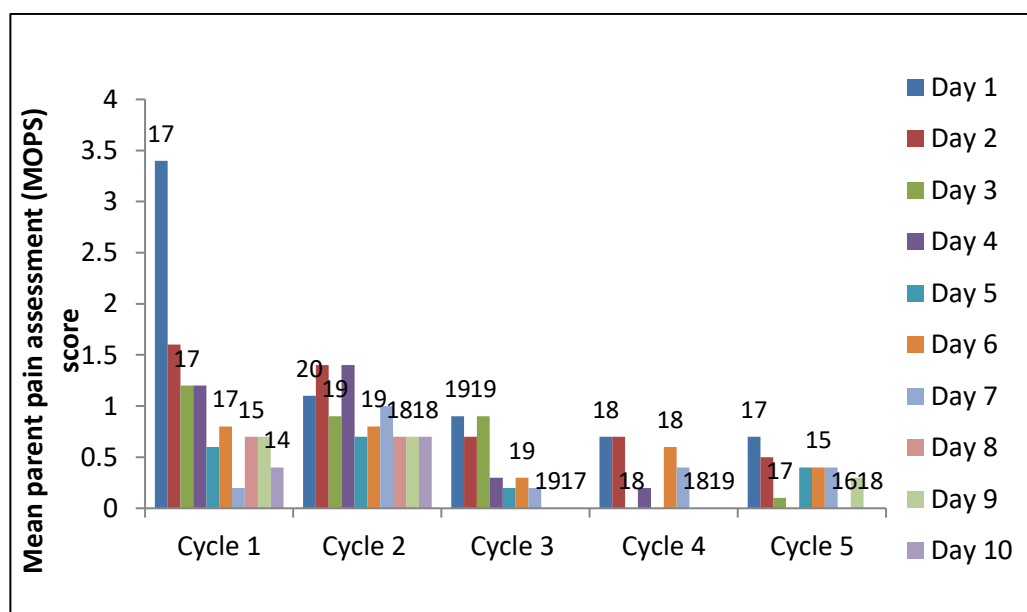
Figure 17: APN311-202 V1 and V2 - Mean pain intensity across cycles and treatment days (self-assessment)



Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8-17 represent the assessment scores after the 1st-10th day of infusion.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-3

Figure 18: APN311-202 V1 and V2 - Mean pain intensity across cycles and treatment days (self-assessment)



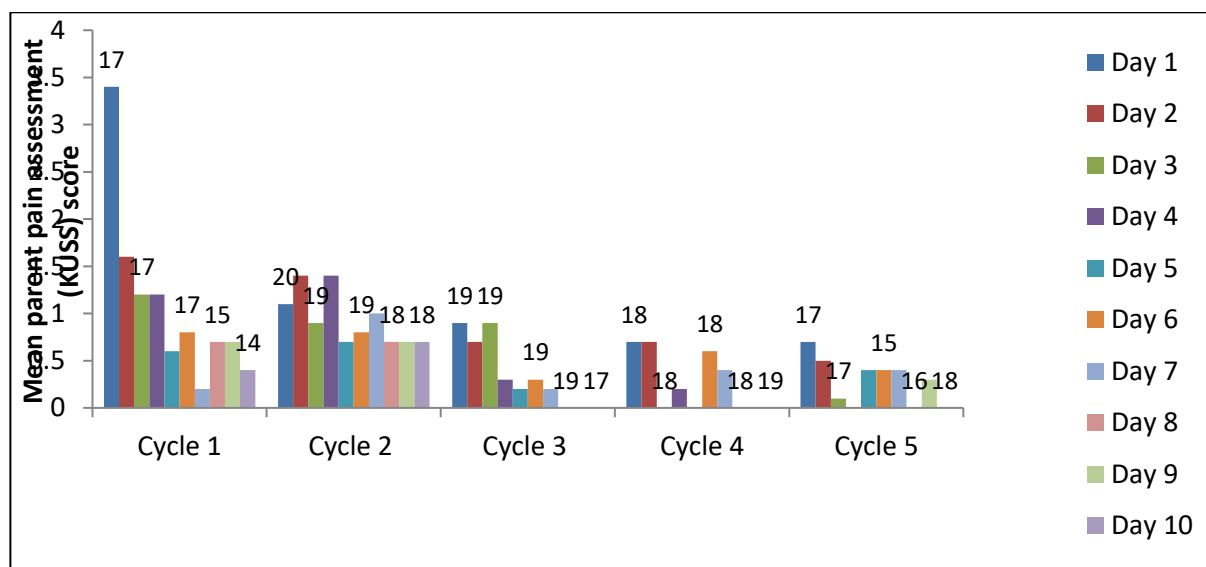
Abbreviation: MOPS = Modified Objective Pain Score.

Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8 to 17 represent the mean assessment scores after the 1st to 10th day of infusion.

Column numbers = number of patients with data available.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-5.

Figure 19: APN311-202 V1 and V2 - Mean pain intensity across cycles and treatment days (parent assessment; KUSS; ≤ 4 years)



Abbreviation: KUSS = The Childhood Discomfort and Pain Scale.

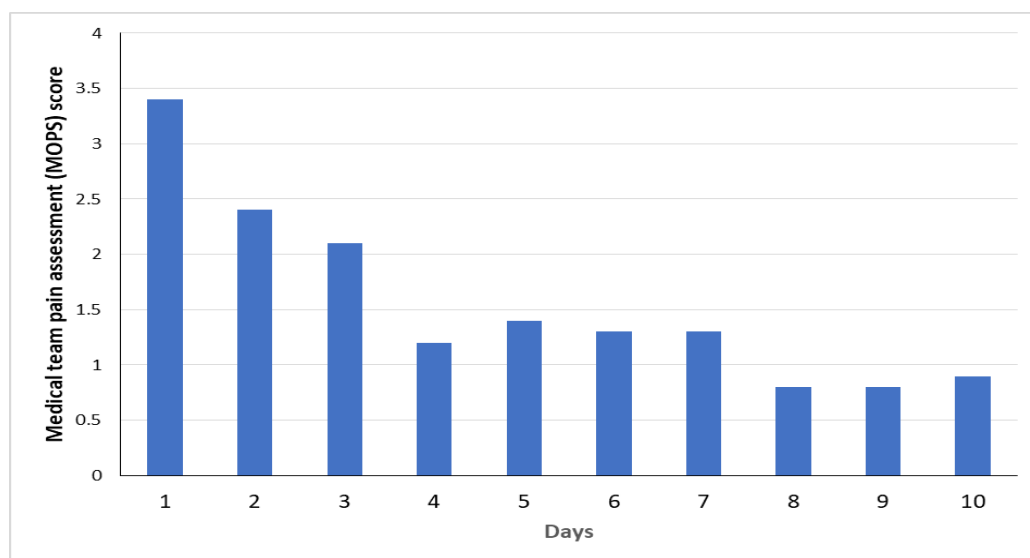
Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8 to 17 represent the mean assessment scores after the 1st to 10th day of infusion.

Column numbers = number of patients with data available.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-5.

Figure 20 and Figure 21 show the pain intensity as assessed by the medical team across the 10 treatment days with dinutuximab beta in Cycle 1 for patients >4 years of age (MOPS) and for patients ≤4 years of age (KUSS), respectively. In line with the results of the self-assessment and parent pain scores, most pain was experienced during the first day and was strongly reduced during the subsequent treatment days.

Figure 20: APN311-202 V1 and V2 - Mean pain intensity across treatment days in Cycle 1 (medical team assessment; MOPS; >4 years)

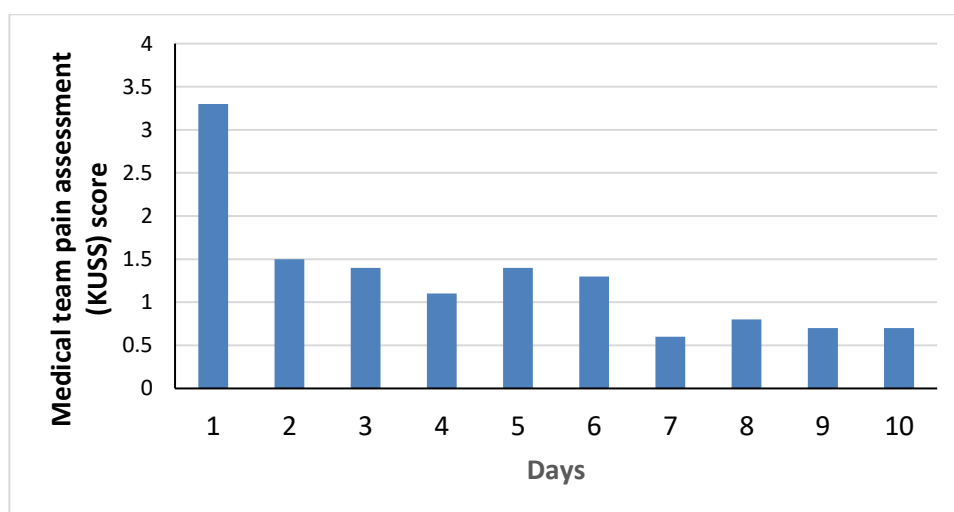


Abbreviation: MOPS = Modified Objective Pain Score

Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8-17 represent the mean assessment scores after the first to tenth day of infusion.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-8

Figure 21: APN311-202 V1 and V2 - Mean pain intensity across treatment days in Cycle 1 (medical team assessment; KUSS; >4 years)



Abbreviation: MOPS = Modified Objective Pain Score

Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8-17 represent the mean assessment scores after the first to tenth day of infusion.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-8

An overview of the frequency of patients with any pain score ≥ 7 (based on a range of 1 to 10) across all pain scales in the SAF population is provided in [Table 18](#). Overall, 51 (41.8%) of the 122 patients had a pain score of 7 or higher. The highest rate of patients with pain score ≥ 7 (33.6%) was experienced during Cycle 1. After the first cycle, at most 11% of the patients had a pain score of ≥ 7 , with only 4.4% of the patients experiencing pain score ≥ 7 in Cycle 5.

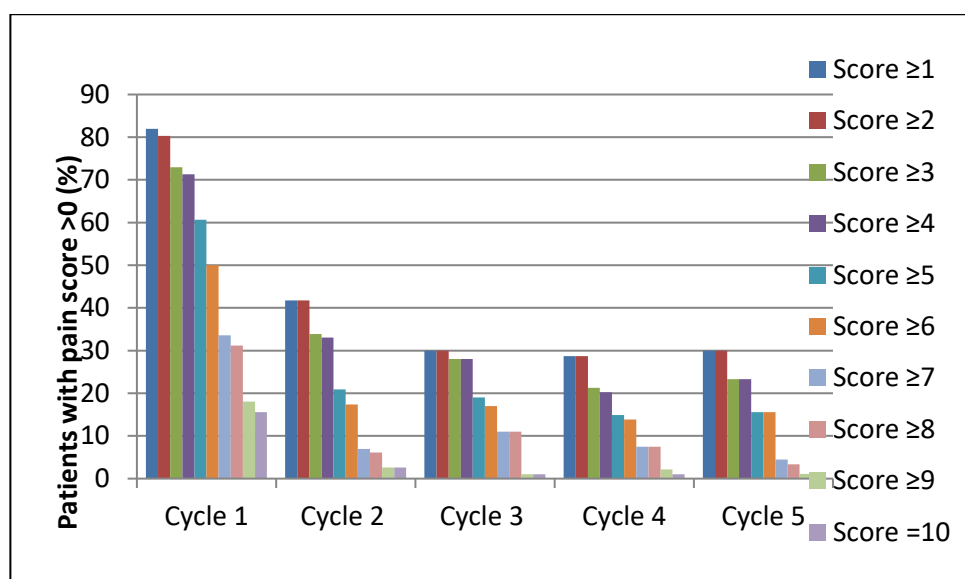
Table 18: APN311-202 V1 and V2 - Frequency of patients with any pain score ≥ 7 (SAF)

Cycle	N	Statistics	Result
1	122	n (%)	41 (33.6%)
2	115	n (%)	8 (7.0%)
3	100	n (%)	11 (11.0%)
4	94	n (%)	7 (7.45%)
5	90	n (%)	4 (4.4%)
Overall	122	n (%)	51 (41.8%)

Abbreviation: SAF = safety analysis set

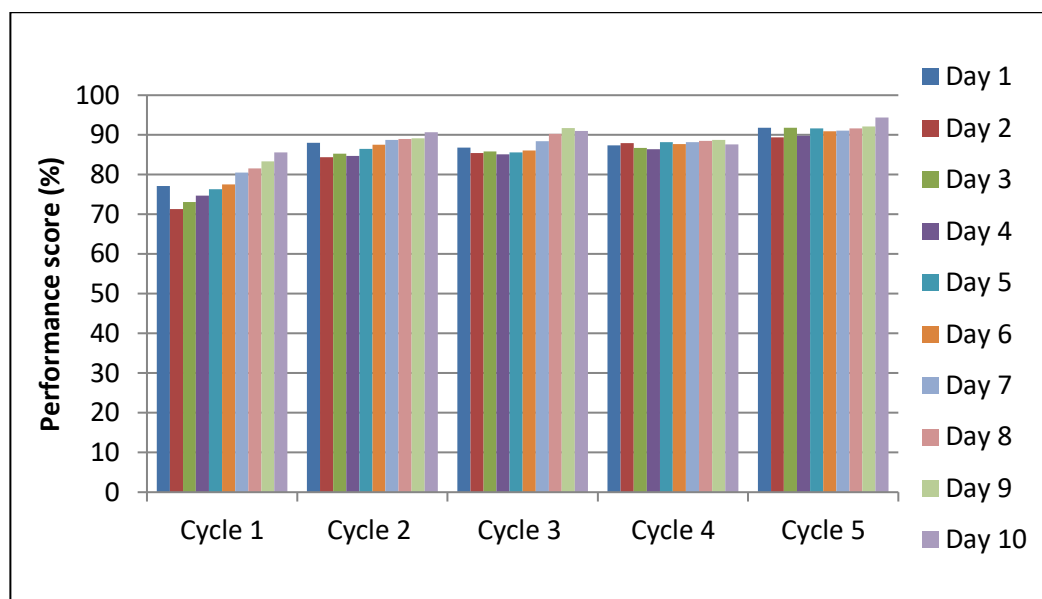
Source data: APN311-202 V1 and V2, Table 14.3.5.1-10

The percentage of patients who had any pain (scored at 1 to 10) at any day during treatment with dinutuximab beta as based on all scales (medical assessment, self-assessment, and parent assessment for both age groups) is provided in [Figure 22](#). The percentage of patients with pain in any scoring category was highest during the first treatment cycle and decreasing with further treatment cycles. The mean Lansky performance score assessed during the 10-day treatment with dinutuximab beta across the 5 treatment cycles for the 118 patients with Lansky score assessment (patients ≤ 16 years) is presented in [Figure 23](#).

Figure 22: APN311-202 V1 and V2 - Percentage of patients with pain scores of 1 to 10 per cycle (all scales)

Source data: APN311-202 V1 and V2, Table 14.3.5.1-10

Figure 23: APN311-202 V1 and V2 - Mean Performance Score Across Cycles and Treatment Days (Lansky; ≤ 16 years)



Pain was assessed on Days 8 through 18 of each cycle, 0-24 hours after dinutuximab beta infusion. Consequently, Days 8-17 represent the mean assessment scores after the 1st-10th day of infusion.

Source data: APN311-202 V1 and V2, Table 14.3.5.1-2

Adverse events

All the 122 patients (100%) included in the SAF population of this study reported at least 1 TEAE (Table 19). And, all 122 patients reported TEAEs that were considered at least possibly related to treatment with IL-2 or dinutuximab beta. For 74 patients (60.7%), SAEs were reported, of which 56 patients (45.9%) had SAEs that were considered at least possibly related to treatment with IL-2 or dinutuximab beta.

Throughout this study, 64 (52.5%) patients experienced TEAEs of severe intensity (CTCAE Grade 3) and 56 patients (45.9%) of life-threatening intensity (CTCAE Grade 4). One patient experienced pyrexia followed by sepsis which resulted in death. The sepsis was considered unrelated to dinutuximab beta and IL-2 treatment, however the pyrexia was assessed as probably related to both IL-2 and dinutuximab beta treatment (APN311-202 V1 and V2, Table 14.3.2-14).

Twenty-eight out of 122 patients (23.0%) had TEAEs which led to permanent discontinuation of study drug treatment. Nine patients (7.4%) terminated treatment early due to toxicities (APN311-202 V1 and V2, Table 14.1-3).

A total of 51 patients (41.8%) died during the analysis period. Except for 1 patient who died of sepsis during Cycle 4, all patients died during the follow-up period due to disease progression (APN311-202 V1 and V2, Table 12-6), with the exception of 1 patient, who died due to chronic GvHD after HSCT and 3 other patients for whom the cause of death was not reported. One additional patient (screening failure) died after screening but before start of Cycle 1.

Table 19: APN311-202 V1 and V2 – Overall summary of treatment-emergent adverse events (SAF)

Patients with	Number (%) of patients* (N = 122)
Any TEAE	122 (100.0%)
Any TEAE at least possibly related to IL-2 or dinutuximab beta	122 (100.0%)
Any TEAE at least possibly related to dinutuximab beta	122 (100.0%)
Any TEAE at least possibly related to IL-2	121 (99.2%)
Any serious TEAE	74 (60.7%)
Any serious TEAE at least possibly related to IL-2 or dinutuximab beta	56 (45.9%)
Any serious TEAE at least possibly related to dinutuximab beta	52 (42.6%)
Any serious TEAE at least possibly related to IL-2	44 (36.1%)
Any TEAE leading to permanent discontinuation of study drugs	28 (23.0%)
Maximum NCI CTCAE Grade	
Grade 1	-
Grade 2	2 (1.6%)
Grade 3	64 (52.5%)
Grade 4	56 (45.9%)
Any TEAE leading to death	1 (0.8%)
Deaths †	51 (41.8%)

*Percentages are based on the total of 122 patients. † All documented deaths

Abbreviations: IL-2 = interleukin-2; SAF = safety analysis set; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events; TEAE = treatment-emergent adverse event;

Possibly related AEs and SAEs: AEs and SAEs with relationship coded as 'Possibly', 'Probably', 'Definitely', and coded as 'Not assessable' or with missing relationship. 'Missing' is defined as no information added to electronic case report form (eCRF), and 'Not assessable' defined as result ticked as 'Not assessable' on eCRF.

Source data: APN311-202 V1 and V2, Tables 14.3.1-1, 14.3.1-2, 14.3.1-3, 14.3.1.4-1, 14.3.1.4-3, 14.3.1.5-1, 14.3.1.5-3, 14.3.1.6-1, and 14.3.1.6-3, and Listings 14.3.2-3, 14.3.2-4, 14.3.2-9, 14.3.2-13, and 14.3.2-14

A summary of TEAEs (regardless of relatedness) reported in $\geq 20\%$ of patients is presented in [Table 20](#) below. Most frequently reported TEAEs concerned general conditions (pyrexia, pain), investigations (ALT increased, platelet count decreased), gastrointestinal disorders (vomiting, diarrhea), respiratory system disorders (cough), vascular disorders (hypotension), and the blood and lymphatic system disorders (anemia).

Table 20: APN311-202 V1 and V2 - Treatment-emergent adverse events by MedDRA system organ class and preferred term experienced by $\geq 20\%$ of patients (SAF)

SYSTEM ORGAN CLASS	Total (N = 122)	
Preferred term	Number ¹ (%) ² of patients with any AE	Number of AEs ³
Any adverse event	122 (100)	6692
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	122 (100)	1187
Pyrexia	120 (98.4)	603
Pain	75 (61.5)	143
Fatigue	44 (36.1)	94
Chills	42 (34.4)	59
Face edema	35 (28.7)	63
Injection site reaction	34 (27.9)	94
INVESTIGATIONS	112 (91.8)	1668
ALT increased	78 (63.9)	213
Platelet count decreased	72 (59.0)	156
GGT increased	70 (57.4)	132
Weight increased	65 (53.3)	187
AST increased	60 (49.2)	182

SYSTEM ORGAN CLASS Preferred term	Total (N = 122)	
	Number¹ (%)² of patients with any AE	Number of AEs³
Neutrophil count decreased	58 (47.5)	187
White blood cell count decreased	56 (45.9)	133
Blood alkaline phosphatase increased	31 (25.4)	65
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	109 (89.3)	393
Cough	97 (79.5)	230
Hypoxia	39 (32.0)	51
GASTROINTESTINAL DISORDERS	107 (87.7)	795
Vomiting	78 (63.9)	222
Diarrhea	74 (60.7)	149
Abdominal pain	60 (49.2)	127
Nausea	55 (45.1)	106
Constipation	53 (43.4)	102
VASCULAR DISORDERS	94 (77.0)	293
Hypotension	76 (62.3)	193
Capillary leak syndrome	53 (43.4)	87
BLOOD AND LYMPHATIC SYSTEM DISORDERS	96 (78.7)	380
Anemia	90 (73.8)	201
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	97 (79.5)	389
Pruritus	58 (47.5)	109
Dry skin	51 (41.8)	107
Rash	39 (32.0)	63
INFECTIONS AND INFESTATIONS	89 (73.0)	205
Device related infection	25 (20.5)	31
METABOLISM AND NUTRITION DISORDERS	76 (62.3)	697
Hypernatremia	39 (32.0)	99
Hypertriglyceridemia	36 (29.5)	87
Hypophosphatemia	36 (29.5)	110
Hypocalcemia	36 (29.5)	116
Hypokalemia	34 (27.9)	94
CARDIAC DISORDERS	62 (50.8)	192
Sinus tachycardia	35 (28.7)	124
Tachycardia	26 (21.3)	59
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS	56 (45.9)	135
Pain in extremity	44 (36.1)	87
RENAL AND URINARY DISORDERS	56 (45.9)	80
Urinary retention	41 (33.6)	55
NERVOUS SYSTEM DISORDERS	55 (45.1)	116
Headache	25 (20.5)	46
EYE DISORDERS	43 (35.2)	63

Abbreviations: AE = adverse event; ALT = alanine aminotransferase; AST = aspartate aminotransferase; GGT = gamma-glutamyl transferase; MedDRA = Medical Dictionary for Regulatory Activities; SAF = safety analysis set.

¹: Number of patients with any adverse event (AE), each patient was counted only once within each category

²: Number of patients with any adverse event (AE) / number of all patients

³: Number of all newly occurring AEs

Source data: APN311-202 V1 and V2, Table 14.3.1-1.

A summary of treatment-related TEAEs reported in $\geq 20\%$ of patients is presented in [Table 21](#) below. Relation to study medication treatment is provided for dinutuximab beta and/or IL-2.

Confidential

Most TEAEs assessed as at least possibly related to dinutuximab beta treatment were also assessed as at least possibly related to IL-2, and with a similar frequency.

Injection site reactions, hypertriglyceridemia, and chills were considered more frequently related to IL-2 than to dinutuximab beta (injection site reaction: 27.9% vs. 4.9%; hypertriglyceridemia: 25.4% vs. 6.6%, chills: 32.0% vs. 23.8%).

Table 21: APN311-202 V1 and V2 - Treatment-emergent adverse events at least possibly related to IL-2 and/or dinutuximab beta experienced by $\geq 20\%$ patients (SAF)

	Number ¹ (%) ² of patients (N = 122) with any TEAE at least possibly related to		
SYSTEM ORGAN CLASS Preferred term	IL-2 or dinutuximab beta	IL-2	Dinutuximab beta
Overall	122 (100.0)	121 (99.2)	122 (100.0)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	121 (99.2)	119 (97.5)	120 (98.4)
Pyrexia	119 (97.5)	118 (96.7)	116 (95.1)
Pain	73 (59.8)	57 (46.7)	71 (58.2)
Fatigue	43 (35.2)	42 (34.4)	42 (34.4)
Chills	41 (33.6)	39 (32.0)	29 (23.8)
Face edema	35 (28.7)	34 (27.9)	35 (28.7)
Injection site reaction	34 (27.9)	34 (27.9)	6 (4.9)
INVESTIGATIONS	111 (91.0)	109 (89.3)	108 (88.5)
ALT increased	73 (59.8)	73 (59.8)	70 (57.4)
Platelet count decreased	67 (54.9)	66 (54.1)	65 (53.3)
GGT increased	66 (54.1)	64 (52.5)	61 (50.0)
Weight increased	65 (53.3)	64 (52.5)	63 (51.6)
AST increased	59 (48.4)	58 (47.5)	58 (47.5)
White blood cell count decreased	51 (41.8)	51 (41.8)	51 (41.8)
Neutrophil count decreased	54 (44.3)	54 (44.3)	52 (42.6)
Blood alkaline phosphatase increased	31 (25.4)	28 (23.0)	31 (25.4)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	104 (85.2)	101 (82.8)	100 (82.0)
Cough	96 (78.7)	91 (74.6)	88 (72.1)
Hypoxia	38 (31.1)	37 (30.3)	36 (29.5)
GASTROINTESTINAL DISORDERS	100 (82.0)	93 (76.2)	97 (79.5)
Vomiting	72 (59.0)	63 (51.6)	69 (56.6)
Diarrhea	68 (55.7)	56 (45.9)	59 (48.4)
Abdominal pain	57 (46.7)	47 (38.5)	55 (45.1)
Nausea	52 (42.6)	47 (38.5)	48 (39.3)
Constipation	26 (21.3)	20 (16.4)	26 (21.3)
VASCULAR DISORDERS	93 (76.2)	88 (72.1)	87 (71.3)
Hypotension	75 (61.5)	71 (58.2)	71 (58.2)
Capillary leak syndrome	53 (43.4)	52 (42.6)	50 (41.0)
BLOOD AND LYMPHATIC SYSTEM DISORDERS	86 (70.5)	85 (69.7)	85 (69.7)
Anemia	79 (64.8)	77 (63.1)	78 (63.9)
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	86 (70.5)	73 (59.8)	79 (64.8)
Pruritus	42 (34.4)	39 (32.0)	36 (29.5)
Rash	34 (27.9)	29 (23.8)	28 (23.0)
Dry skin	32 (26.2)	17 (13.9)	32 (26.2)
METABOLISM AND NUTRITION DISORDERS	72 (59.0)	70 (57.4)	72 (59.0)
Hyponatremia	38 (31.1)	34 (27.9)	34 (27.9)
Hypophosphatemia	36 (29.5)	36 (29.5)	35 (28.7)
Hypocalcemia	33 (27.0)	33 (27.0)	32 (26.2)
Hypertriglyceridemia	32 (26.2)	31 (25.4)	8 (6.6)

SYSTEM ORGAN CLASS Preferred term	Number ¹ (%) ² of patients (N = 122) with any TEAE at least possibly related to		
	IL-2 or dinutuximab beta	IL-2	Dinutuximab beta
Hypokalemia	30 (24.6)	27 (22.1)	23 (18.9)
CARDIAC DISORDERS	60 (49.2)	60 (49.2)	57 (46.7)
Sinus tachycardia	36 (29.5)	36 (29.5)	35 (28.7)
INFECTIONS AND INFESTATIONS	52 (42.6)	48 (39.3)	45 (36.9)
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS	51 (41.8)	44 (36.1)	48 (39.3)
Pain in extremity	42 (34.4)	38 (31.1)	40 (32.8)
NERVOUS SYSTEM DISORDERS	47 (38.5)	41 (33.6)	40 (32.8)
RENAL AND URINARY DISORDERS	36 (29.5)	31 (25.4)	35 (28.7)
EYE DISORDERS	36 (29.5)	22 (18.0)	35 (28.7)

Abbreviations: AE = adverse event; ALT = alanine aminotransferase; AST = aspartate aminotransferase; GGT = gamma-glutamyl transferase; IL-2 = interleukin-2; SAF = safety analysis set; TEAE = treatment-emergent adverse event.

¹: Number of subjects with any AE occurring or still present, each subject was counted only once within each category.

²: Number of subjects with any AE / number of all subjects.

AEs with relationship coded as 'Possibly', 'Probably', 'Definitely', 'Not assessable' or with missing relationship.

Source data: APN311-202 V1 and V2, Tables 14.3.1.4-1, 14.3.1.5-1, and 14.3.1.6-1.

A summary of TEAEs at least possibly related to dinutuximab beta reported in $\geq 20\%$ of patients by NCI CTCAE Grade is presented in Table 22. Most frequently reported TEAEs at least possibly related to dinutuximab beta treatment were pyrexia (95.1%), cough (72.1%), and anemia (63.9%). The majority of patients experienced pyrexia and anemia of Grade 2 or 3 severity and cough of Grade 1 or 2 severity. One patient each experienced cough and anemia of Grade 4.

The most frequently reported dinutuximab beta-related TEAEs of Grade 4 severity were platelet count decreased (18.0%) and neutrophil count decreased (11.5%). Other Grade 4 related TEAEs occurred in maximally 2 patients only. Frequently reported Grade 3 TEAEs considered related to dinutuximab beta were pyrexia (43.4%), anemia (34.4%), ALT increased, gamma-glutamyl transferase (GGT) increased, and neutrophil count increased (26.2% each).

Table 22: APN311-202 V1 and V2 - TEAEs at least possibly related to dinutuximab beta by MedDRA preferred term and NCI CTCAE grade sorted in descending order of overall frequency experienced by $\geq 20\%$ patients (SAF)

Preferred term	Overall (N = 122) N ¹ (%) ²	Number ¹ (%) ² of patients per preferred term and Maximum NCI CTCAE Grade			
		1	2	3	4
Pyrexia	116 (95.1)	14 (11.5)	49 (40.2)	53 (43.4)	-
Cough	88 (72.1)	29 (23.8)	50 (41.0)	8 (6.6)	1 (0.8)
Anemia	78 (63.9)	4 (3.3)	31 (25.4)	42 (34.4)	1 (0.8)
Hypotension	71 (58.2)	30 (24.6)	20 (16.4)	21 (17.2)	-
Pain	71 (58.2)	14 (11.5)	34 (27.9)	21 (17.2)	2 (1.6)
ALT increased	70 (57.4)	18 (14.8)	18 (14.8)	32 (26.2)	2 (1.6)
Vomiting	69 (56.6)	41 (33.6)	20 (16.4)	8 (6.6)	-
Platelet count decreased	65 (53.3)	14 (11.5)	9 (7.4)	20 (16.4)	22 (18.0)
Weight increased	63 (51.6)	31 (25.4)	29 (23.8)	3 (2.5)	-
GGT increased	61 (50.0)	8 (6.6)	20 (16.4)	32 (26.2)	1 (0.8)
Diarrhea	59 (48.4)	30 (24.6)	24 (19.7)	4 (3.3)	1 (0.8)
AST increased	58 (47.5)	32 (26.2)	13 (10.7)	13 (10.7)	-
Abdominal pain	55 (45.1)	21 (17.2)	29 (23.8)	5 (4.1)	-
Neutrophil count decreased	52 (42.6)	1 (0.8)	5 (4.1)	32 (26.2)	14 (11.5)

Preferred term	Overall (N = 122) N ¹ (%) ²	Number ¹ (%) ² of patients per preferred term and Maximum NCI CTCAE Grade			
		1	2	3	4
White blood cell count decreased	51 (41.8)	12 (9.8)	23 (18.9)	14 (11.5)	2 (1.6)
Capillary leak syndrome	50 (41.0)	6 (4.9)	30 (24.6)	12 (9.8)	2 (1.6)
Nausea	48 (39.3)	28 (23.0)	15 (12.3)	5 (4.1)	-
Fatigue	42 (34.4)	36 (29.5)	6 (4.9)	-	-
Pain in extremity	40 (32.8)	15 (12.3)	23 (18.9)	2 (1.6)	-
Hypoxia	36 (29.5)	1 (0.8)	19 (15.6)	15 (12.3)	1 (0.8)
Pruritus	36 (29.5)	12 (9.8)	23 (18.9)	1 (0.8)	-
Face edema	35 (28.7)	29 (23.8)	6 (4.9)	-	-
Hypophosphatemia	35 (28.7)	21 (17.2)	11 (9.0)	3 (2.5)	-
Sinus tachycardia	35 (28.7)	32 (26.2)	3 (2.5)	-	-
Hyponatremia	34 (27.9)	26 (21.3)	2 (1.6)	5 (4.1)	1 (0.8)
Dry skin	32 (26.2)	27 (22.1)	5 (4.1)	-	-
Hypocalcemia	32 (26.2)	11 (9.0)	20 (16.4)	-	1 (0.8)
Blood alkaline phosphatase increased	31 (25.4)	27 (22.1)	4 (3.3)	-	-
Chills	29 (23.8)	26 (21.3)	3 (2.5)	-	-
Rash	28 (23.0)	17 (13.9)	10 (8.2)	1 (0.8)	-
Constipation	26 (21.3)	16 (13.1)	8 (6.6)	2 (1.6)	-

Abbreviations: AE = adverse event; ALT = alanine aminotransferase; AST = aspartate aminotransferase; GGT = gamma-glutamyl transferase; MedDRA = Medical Dictionary for Regulatory Activities; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events; TEAE = treatment-emergent adverse events; SAF = safety analysis set.

¹: Number of patients with any AE, each patient was counted only once within each category.

²: Number of patients with any AE / number of all patients.

Source data: APN311-202 V1 and V2, Tables 14.3.1.6-1 and 14.3.1.6-2.

Pain is the main expected AE of treatment with dinutuximab beta. The most frequent pain TEAEs are presented overall and by NCI CTCAE grade in Table 23. Pain, abdominal pain, and pain in extremity were the most frequent pain events. For the majority of patients the events were of Grade 1 or 2, for about a quarter the events were of Grade 3. Three patients experienced Grade 4 pain. Those were the only pain-related events of Grade 4 and were also documented as SAEs (APN311-202 V1 and V2, Table 14.3.1.2-4).

Table 23: APN311-202 V1 and V2 - TEAEs of pain by MedDRA preferred term and NCI CTCAE grade sorted in descending order of overall frequency experienced by ≥ 3% patients (SAF)

Preferred term	Overall (N = 122) N ¹ (%) ²	Number ¹ (%) ² of patients per preferred term and Maximum NCI CTCAE Grade			
		1	2	3	4
Any pain AE	99 (81.1)	13 (10.7)	54 (44.3)	29 (23.8)	3 (2.5)
Pain	75 (61.5)	16 (13.1)	36 (29.5)	21 (17.2)	3 (2.5)
Abdominal pain	60 (49.2)	25 (20.5)	30 (24.6)	5 (4.1)	-
Pain in extremity	44 (36.1)	17 (13.9)	25 (20.5)	2 (1.6)	-
Headache	25 (20.5)	16 (13.1)	8 (6.6)	1 (0.8)	-
Oropharyngeal pain	14 (11.5)	8 (6.6)	6 (4.9)	-	-
Neuralgia	13 (10.7)	3 (2.5)	8 (6.6)	2 (1.6)	-
Back pain	11 (9.0)	2 (1.6)	5 (4.1)	4 (3.3)	-
Injection site pain	8 (6.6)	7 (5.7)	1 (0.8)	-	-
Musculoskeletal pain	6 (4.9)	2 (1.6)	3 (2.5)	1 (0.8)	-
Neck pain	6 (4.9)	4 (3.3)	1 (0.8)	1 (0.8)	-
Urinary tract pain	6 (4.9)	4 (3.3)	2 (1.6)	-	-
Bone pain	5 (4.1)	2 (1.6)	3 (2.5)	-	-

Preferred term	Overall (N = 122) N ¹ (%) ²	Number ¹ (%) ² of patients per preferred term and Maximum NCI CTCAE Grade			
		1	2	3	4
Ear pain	4 (3.3)	3 (2.5)	1 (0.8)	-	-
Arthralgia	4 (3.3)	2 (1.6)	2 (1.6)	-	-

Abbreviations: AE = adverse event; MedDRA = Medical Dictionary for Regulatory Activities; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events; TEAE = treatment-emergent adverse event, SAF = safety analysis set.

¹: Number of patients with any AE, each patient was counted only once within each category.

²: Number of patients with any AE / number of all patients.

Source data: APN311-202 V1 and V2, Tables 14.3.1.2-1 and 14.3.1.2-2.

One patient died due to sepsis which was judged to be unrelated to study treatment. An additional 50 patients died at least 30 days after the last study treatment. The vast majority of these deaths (46) were linked to progression of the pre-existing neuroblastoma (APN311-202 V1 and V2, Tables 14.3.2-13 to 14.3.2-16).

The most common severity grade across all SAEs was Grade 3 (34.4% of patients), followed by Grade 4 (18.0%). Most frequent SAEs of Grade 4 severity were platelet count decreased (4.1% of patients), sepsis (3.3%), and pain (2.5%). Pyrexia was most commonly reported as SAE (13.1%), with most events of Grade 2 (8.2%) or Grade 3 (3.3%) severity. Other frequent SAEs of Grade 3 severity were device related infection and hypotension (4.9% each) and vomiting (4.1%) (APN311-202 V1 and V2, Table 14.3.1-4).

A summary of treatment-related SAEs reported in ≥ 2 patients is presented in [Table 24](#). Relation to study medication treatment is provided for dinutuximab beta and/or IL-2. The majority of the SAEs occurring in at least 2 patients were assessed as at least possibly related to study medication treatment.

Table 24: APN311-202 V1 and V2 - Serious treatment-emergent adverse events at least possibly related to IL-2 and/or dinutuximab beta experienced by ≥ 2 patients (SAF)

SYSTEM ORGAN CLASS Preferred term	Number (%) of patients (N = 122) With any serious TEAE at least possibly related to		
	IL-2 or dinutuximab beta N ¹ (%) ²	IL-2 N ¹ (%) ²	Dinutuximab beta N ¹ (%) ²
Overall	56 (45.9)	44 (36.1)	52 (42.6)
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	15 (12.3)	10 (8.2)	13 (10.7)
Pyrexia	12 (9.8)	8 (6.6)	9 (7.4)
Pain	2 (1.6)	1 (0.8)	2 (1.6)
INVESTIGATIONS	9 (7.4)	9 (7.4)	7 (5.7)
Blood culture positive	2 (1.6)	2 (1.6)	2 (1.6)
Platelet count decreased	5 (4.1)	5 (4.1)	4 (3.3)
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	10 (8.2)	9 (7.4)	10 (8.2)
Respiratory distress	3 (2.5)	1 (0.8)	3 (2.5)
Hypoxia	4 (3.3)	4 (3.3)	4 (3.3)
GASTROINTESTINAL DISORDERS	9 (7.4)	4 (3.3)	8 (6.6)
Vomiting	3 (2.5)	1 (0.8)	3 (2.5)
Diarrhea	2 (1.6)	1 (0.8)	1 (0.8)
Nausea	3 (2.5)	2 (1.6)	3 (2.5)
Constipation	2 (1.6)	-	2 (1.6)
VASCULAR DISORDERS	11 (9.0)	10 (8.2)	11 (9.0)
Hypotension	6 (4.9)	5 (4.1)	6 (4.9)

SYSTEM ORGAN CLASS Preferred term	Number (%) of patients (N = 122) With any serious TEAE at least possibly related to		
	IL-2 or dinutuximab beta N ¹ (%) ²	IL-2 N ¹ (%) ²	Dinutuximab beta N ¹ (%) ²
Capillary leak syndrome	5 (4.1)	5 (4.1)	5 (4.1)
BLOOD AND LYMPHATIC SYSTEM DISORDERS	3 (2.5)	3 (2.5)	3 (2.5)
Anemia	2 (1.6)	2 (1.6)	2 (1.6)
METABOLISM AND NUTRITION DISORDERS	4 (3.3)	4 (3.3)	4 (3.3)
Hyponatremia	2 (1.6)	2 (1.6)	2 (1.6)
INFECTIONS AND INFESTATIONS	10 (8.2)	7 (5.7)	9 (7.4)
Device related infection	2 (1.6)	2 (1.6)	2 (1.6)
IMMUNE SYSTEM DISORDERS	4 (3.3)	1 (0.8)	4 (3.3)
Anaphylactic reaction	3 (2.5)	1 (0.8)	3 (2.5)
Hypersensitivity	2 (1.6)	1 (0.8)	2 (1.6)
NERVOUS SYSTEM DISORDERS	5 (4.1)	5 (4.1)	5 (4.1)
Encephalopathy	2 (1.6)	2 (1.6)	2 (1.6)
Somnolence	3 (2.5)	3 (2.5)	3 (2.5)
EYE DISORDERS	5 (4.1)	1 (0.8)	5 (4.1)
Mydriasis	3 (2.5)	1 (0.8)	3 (2.5)

Abbreviations: AE = adverse event; IL-2 = interleukin-2; SAF = safety analysis set; TEAE = treatment-emergent adverse events.

¹: Number of subjects with any AE occurring or still present, each subject was counted only once within each category.

²: Number of subjects with any AE/ number of all subjects.

AEs with relationship coded as 'Possibly', 'Probably', 'Definitely', 'Not assessable' or with missing relationship.

Source data: APN311-202 V1 and V2, Tables 14.3.1.4-3, 14.3.1.5-3, and 14.3.1.6-3.

A total of 28 patients experienced AEs that led to permanent discontinuation of study drug (APN311-202 V1 and V2 Tables 14.3.2-9 to 14.3.2-12.1). Almost all were assessed as at least possibly related to dinutuximab beta treatment.

Laboratory, vital signs, physical examinations, and other safety assessments

Out-of-range chemistry and hematology values occurred in the majority of patients. These abnormal values were considered not clinically significant or expected to be clinically significant by the investigator and showed no apparent treatment-related trends. There were no trends in any vital sign changes, ECG, echocardiogram, pulse oximetry or electroencephalogram. There was no clear trend with regard to physical examination findings. Physical findings affecting more than 1 patient were eye disorders, skin disorders, cough and fever.

Conclusion

A 10-day infusion schedule of 10 mg/m² dinutuximab beta (total dose 100 mg/m²) in combination with IL-2 treatment was shown to be tolerable with a reduced pain-toxicity profile of dinutuximab beta whilst maintaining immunomodulatory efficacy, as based on response rates, EFS and OS results, in patients with either primary refractory or relapsed neuroblastoma.

5.3.2.5. Study APN311-302

(Source: APN311-302 CSR with cut-off 22 Jan 2016; APN311-302 CSR addendum with cut-off 5 Sep 2016; APN311-302 Historical Control Report, Version 1.0, dated 24 Jan 2017; Ladenstein et al, 2020)

Study APN311-302 is an investigator-initiated, randomized, open-label, multicenter trial in high-risk neuroblastoma patients using standard–neuroblastoma treatment elements, i.e. intensive induction and high dose chemotherapy followed by stem cell transplantation, operation to the primary site, local site radiation followed by immunotherapy. Dinutuximab beta was given to all eligible patients, but patients were randomized for the added use of s.c. IL-2. The main objective was to test whether the addition of s.c. IL-2 to dinutuximab beta, following myeloablative therapy and autologous stem cell rescue, improves EFS in patients with high-risk neuroblastoma.

Design

Patients: Patients who had completed intensive induction chemotherapy (Rapid COJEC), with or without 2 additional cycles of topotecan, vincristine, doxorubicin (TVD), had received myeloablative therapy, followed by peripheral blood stem cell rescue (PBSCR), local radiotherapy, and presented without evidence of progression at the time of re-evaluation were eligible for study inclusion.

Treatment: The therapy cycle started with oral 13-cis-RA (160 mg/m²/day) for 14 days, every 4 weeks, after completion of local irradiation, no later than 120 days post PBSCR. Dinutuximab beta was administered as an 8-hour i.v. infusion, at a dose of 20 mg/m²/day (for patients ≤12 kg, 0.67 mg/kg/day) over 5 days, every 4 weeks over 5 cycles. The first course started in the fourth week after initiation of 13-cis-RA. s.c. IL-2 (6 x 10⁶ IU/m²/day or 0.2 x 10⁶ IU/kg/day for patients ≤12 kg) was administered over 2 x 5 days starting in Week 3.

Premedication: intravenous morphine; bolus injection followed by continuous infusion to be reduced on a daily basis as possible.

Safety and efficacy endpoints: primary efficacy endpoint: 5-year EFS. Further endpoints: cumulative incidence of relapse/progression, OS, deaths and toxicities in both studies, and EFS at 1 and 2 years, overall response, and SAEs in APN311-302.

Results - patients

A total of 844 patients were randomized and had Case Report Form data available, i.e. 466 patients in the standard maintenance treatment with isotretinoin (control population [CP]) and 378 patients in the treatment with dinutuximab beta and isotretinoin (immunotherapy population [IP]). Median follow-up was 5.8 years (inter-quartile range (IQR): 4.2 to 8.2 years). The median age of patients at diagnosis was 2.9 years (IQR: 1.8. to 3.8).

Of 844 patients, 524 patients were male and 320 patients were female. [1]

Results - efficacy

Event-free survival (EFS)

EFS was defined as the time between randomization to immunotherapy to the occurrence of disease progression, relapse, death, or occurrence of second neoplasm.

Therapy with dinutuximab beta showed a 3-year estimate for EFS of 61.2% in patients receiving IL-2 treatment and, slightly lower, 55.4% in patients without IL-2-treatment (Table 25). Mean time to progression was 985 days in patients with concomitant IL-2 and 1112 days in patients without concomitant IL-2 treatment (APN311-302 CSR Addendum Table 14.2.7.1). No difference between the dinutuximab beta and dinutuximab beta + IL-2 treatment groups were seen for 1-year and 2-year EFS (Table 25). Ladenstein et al, 2020 reported the 5-year EFS rate was 57% (95% CI: 51–62%) for patients receiving immunotherapy, compared to 42% (95% CI: 38–47%) for control population patients ($p < 0.001$).

The cumulative incidence of relapse/progression and non-relapse mortality at 5 years was 41% (95% CI: 37% to 47%) and 2% (95% CI: 1% to 4%) for the patients (Figure 24) receiving immunotherapy and 57% (95% CI: 53% to 61%) and 1% (95% CI <1% to 2%) for the control population patients ($p < 0.001$). [1]

Similar trends like in the overall population were observed in patients with and without evidence of disease at baseline. Compared with the overall population, EFS was generally lower in patients with than in patients without evidence of disease at baseline.

Overall survival

OS in patients without and with IL-2 treatment is presented in Table 25. Mean time to death was 1096 days in patients with concomitant IL-2 and 1270 days in patients without concomitant IL-2 treatment (APN311-302 CSR Addendum Section 5.6.2.3). The 5-year OS rate for the patients receiving immunotherapy was 64% (95% CI: 59–69%), compared to 50% (95% CI: 46–55%) for control population patients (Figure 25). [1]

Similar trends like in the overall population were observed in patients with and without evidence of disease at baseline. OS was generally lower in patients with than in patients without evidence of disease at baseline (Table 25).

Table 25: APN311-302 - event-free survival and overall survival

Kaplan-Meier estimate at All patients	Event-free survival		Overall survival	
	Dinutuximab beta	Dinutuximab beta + IL-2	Dinutuximab beta	Dinutuximab beta + IL-2
	(N = 180)	(N = 190)	(N = 180)	(N = 190)
1 year	72.3%	72.3%	86.3%	87.9%
2 year	63.2%	66.3%	76.0%	75.4%
3 year	55.4%	61.2%	64.1%	69.1%
p-value ^a	0.3202		0.6114	
5 year	57%	42%	64%	50%
P-value	<0.001		<0.001	
Patients with evidence of disease at baseline ^b				
1 year	66.6%	72.3%	82.9%	86.0%
2 year	58.1%	61.6%	73.1%	71.2%
3 year	45.9%	53.8%	54.2%	63.3%
p-value ^a	0.4944		0.5710	
Patients without evidence of disease at baseline ^c				
1 year	76.5%	72.6%	89.2%	88.5%
2 year	66.7%	69.5%	78.2%	77.8%
3 year	61.7%	66.2%	71.0%	72.2%
p-value ^a	0.5648		0.9571	

Data taken from D180R response to European Medicines Agency (EMA)

^a Treatment comparison at 3 years using the log-rank test adjusted for previous treatment (BuMel or CEM).

^b N = 73 (w/o IL-2), N = 76 (with IL-2). ^c N = 103 (w/o IL-2), N = 105 (with IL-2).

BuMel = busulfan and melphalan, CEM = carboplatin, etoposide, and melphalan, FAS = full analysis set, IL-2 = aldesleukin, N = number of patients in treatment group (for some analyses 1 or 2 patients were missing), w/o = without.

Source data: [1] APN311-302 CSR Addendum Tables 5-5, 5-6, and 5-9

Figure 24 APN311-302 - Analysis population comparing control population versus immunotherapy population - event-free survival

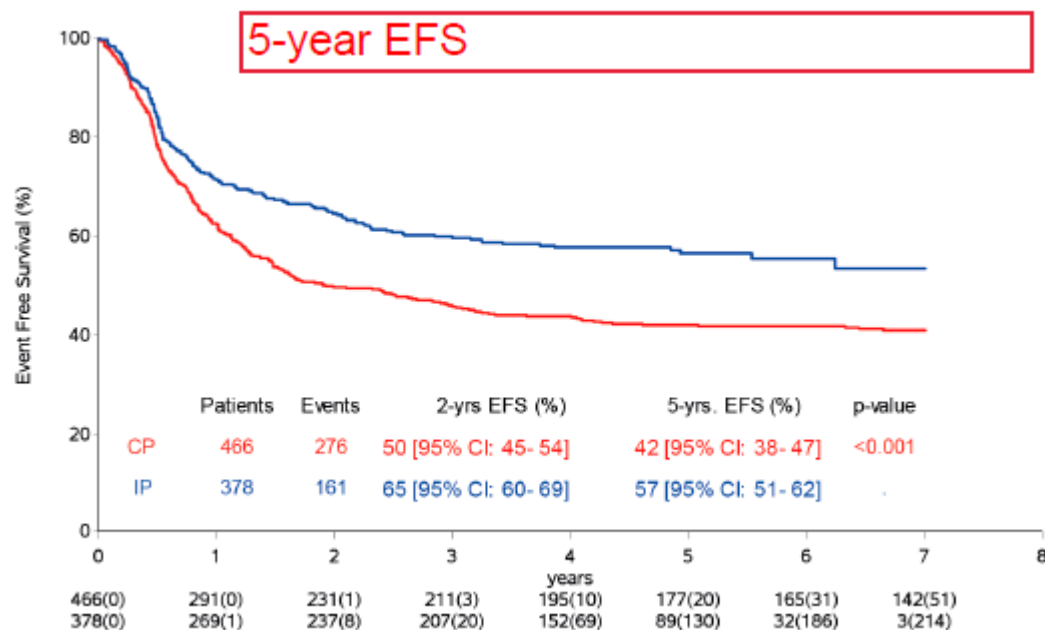
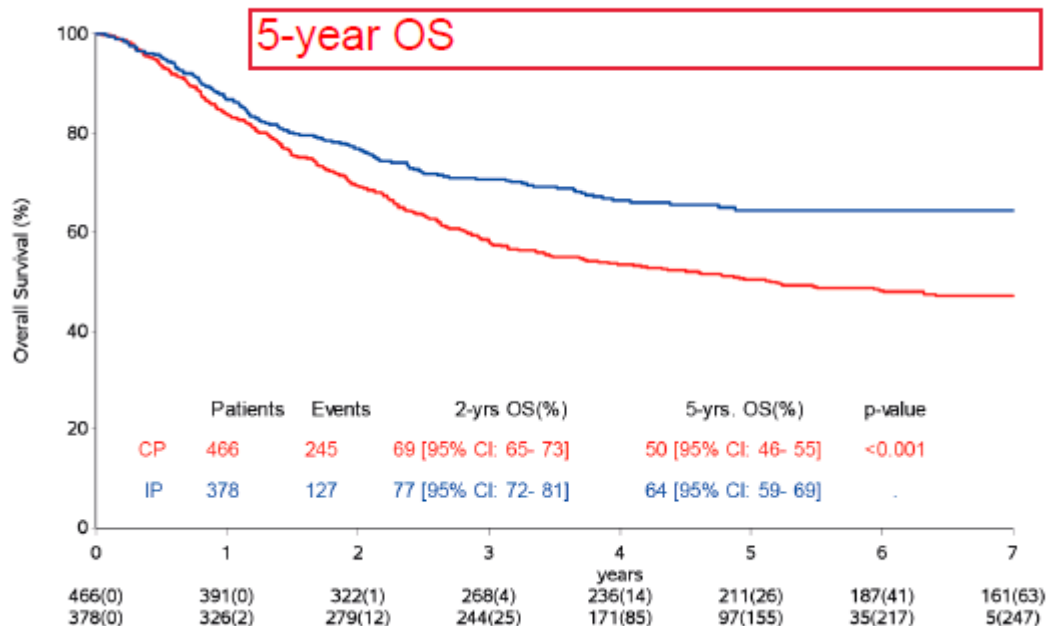


Figure 25 APN311-302 - Analysis population comparing control population versus immunotherapy population - overall survival



Overall response

In the IP, the 2-year and 5-year EFS rates for 210 patients (81 events) in CR were 68% (95% CI: 61–74%) and 61% (95% CI: 53–57%). In the CP, the 2-year and 5-year EFS rates for 258 patients in CR (144 events) were 54% (95% CI: 48–60%) and 46% (95% CI: 39–52%).

[1]

The ORR (CR+VGPR+PR) was 43.9% in patients receiving dinutuximab beta and 51.4% in patients receiving dinutuximab beta plus IL-2 (APN311-302 CSR Addendum Section 5.6.2.2).

Response to Maintenance Treatments

Thirty-nine of 108 (36%) CP patients with evaluable diseases prior to maintenance (67 very good partial response [VGPR] and 41 PR) responded; of whom, 35/108 (32%) achieved CR after isotretinoin. In contrast, 64/130 (49%) of IP patients with evaluable diseases (85 VGPR and 45 PR) prior to immunotherapy responded; of whom, 52/130 (40%) achieved CR after immunotherapy ($p = 0.226$). [1]

Historical control analysis

Study APN311-302 reflects one period of the ongoing HR-NBL-1/SIOPEN study. The HR-NBL-1/SIOPEN study is an open, multicenter randomized Phase III therapy optimization study. The study includes 3 main study phases: an induction phase, a consolidation (myeloablative therapy [MAT]) phase, and a maintenance phase. During the latter, patients receive immunotherapy. The study included several randomization steps. The first, the R1 randomization phase, started in 2002 and compared the MAT regimens busulfan and melphalan (BuMel) to carboplatin, etoposide, and melphalan (CEM). The R2 randomization phase (study APN311-302), only started in 2009, i.e. several years after the initiation of the R1 phase. The R2 phase compared ch14.18/CHO immunotherapy following MAT with or without aldesleukin (IL-2). Thus, patients included in the R1 randomization phase who received standard of care neuroblastoma treatment, including MAT but no immunotherapy, form a valid historical control group to patients receiving immunotherapy in addition to standard of care neuroblastoma treatment.

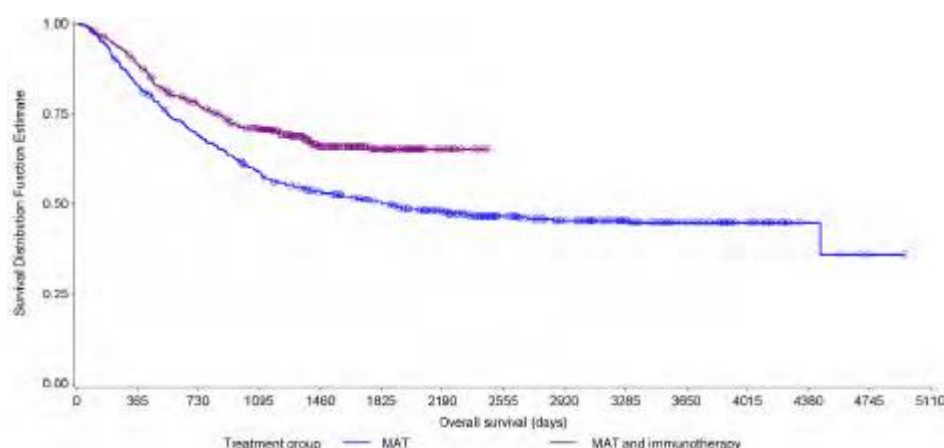
The historical control group for comparison to APN311-302 patients thus consisted of patients who were solely part of the R1 randomization of study HR-NBL-1/SIOPEN. Patients were eligible for the historical comparison, if they had received first-line treatment, and received treatment with BuMel or CEM MAT chemotherapy and had no disease progression after MAT nor after radiotherapy.

The historical control analysis evaluated OS. The analysis set included 450 patients in the historical (MAT) group and 370 in the MAT + immunotherapy group.

In the MAT group, OS was 83% at 1 year, 69% at 2 years, 59% at 3 years, and 50% at 5 years. In the MAT + immunotherapy group, OS was 89% at 1 year, 78% at 2 years, 71% at 3 years, and 65% at 5 years.

The difference in OS between the MAT group and the MAT and immunotherapy group was statistically significant ($p < 0.0001$), in favor of MAT and immunotherapy (Figure 26).

The mean OS time was substantially longer in MAT patients (2447.1 days) than in the MAT+immunotherapy patients (1359.4 days). This is likely due to a longer follow-up time for the MAT group.

Figure 26: APN311-302 versus historical control - overall survival

Overall survival was defined as time from the auxiliary starting point to death from any cause. Patients without an event were censored at the date of their last follow-up evaluation.
APN311-302 Historical Control Report Figure 5-1

Results - safety

Overall, 86.3% of patients received the planned number of 6 treatment cycles (dinutuximab beta: 89.1%; dinutuximab beta plus IL-2: 83.6%) (APN311-302 CSR Table 14.1.5.1).

Treatment-emergent adverse events

Study APN311-302 documented AEs mainly as National Cancer Institute Common Terminology Criteria (NCI CTC) toxicities. Only toxicities and SAEs were reported.

Most frequent AEs reported in more than 80% of patients included hematological AEs, infections, and changes in the patient's general condition. (Table 26).

Adverse events were generally more frequent with IL-2 than without IL-2, in particular non-hematological AEs (86% vs 66%), platelets (61% vs 34%), granulocytes (42.6% vs 26.2%), infections (34% vs 26%), fever (40% vs 14%), and pain related to dinutuximab beta (26% vs 16%). [1]

Table 26: APN311-302 -Adverse events (reported in different groups)

Adverse events	Control Group		dinutuximab beta without IL-2		dinutuximab beta + IL-2	
Non-hematological toxicity	317	(15)	186	(66)	192	(86)
General condition	314	(2)	185	(16)	192	(41)
hemoglobin	313	(4)	186	(42)	191	(66)
WBC	313	(6)	186	(26)	191	(36)
Granulocytes	313	(7)	186	(33)	191	(58)
Platelets	313	(9)	186	(34)	191	(61)
Infection	315	(6)	185	(26)	191	(34)
Fever	314	(2)	185	(14)	190	(40)
Pain related to dinutuximab beta	-	-	122	(16)	122	(26)

IL-2 = aldesleukin, N = number of patients, WBC = white blood cells.

Note: Columns in bold show the values for the combined Grade 3 and 4 toxicities

Source: [1]

Apart from hematological toxicities the most frequent AEs of Grade 3/4 in severity included toxicities regarding the patient's general condition, pain related to the ch14.18/CHO, infections, WBC, granulocytes, platelets, and elevated SGOT and SGPT levels. [1]

Serious adverse events

Most common SAEs reported in the SOC were 'infections and infestations', 'gastrointestinal disorders', 'general disorders and administration site conditions', in particular pyrexia, 'immune disorders' such as hypersensitivity, and 'respiratory, thoracic and mediastinal disorders' (Table 27).

Table 27: APN311-302 - frequent serious adverse events (reported in more than 2% of patients in at least 1 group)

SYSTEM ORGAN CLASS Preferred term	Number (%) of Patients			
	dinutuximab beta (N = 183)		dinutuximab beta+IL-2 (N = 183)	
Overall	50	(27.3)	84	(45.9)
Infections and infestations	16	(8.7)	22	(12.0)
Sepsis	3	(1.6)	4	(2.2)
Bacteremia	1	(0.5)	4	(2.2)
Infection	1	(0.5)	4	(2.2)
Gastrointestinal disorders	10	(5.5)	21	(11.5)
Diarrhea	5	(2.7)	7	(3.8)
Vomiting	4	(2.2)	5	(2.7)
Abdominal pain	2	(1.1)	5	(2.7)
General dis. and administration site cond.	8	(4.4)	19	(10.4)
Pyrexia	2	(1.1)	15	(8.2)
Pain	4	(2.2)	2	(1.1)
Immune system disorders	5	(2.7)	15	(8.2)
Hypersensitivity	4	(2.2)	13	(7.1)
Respiratory, thoracic and mediastinal dis.	9	(4.9)	11	(6.0)
Cough	1	(0.5)	4	(2.2)
Vascular disorders	3	(1.6)	12	(6.6)
Capillary leak syndrome	1	(0.5)	8	(4.4)
Nervous system disorders	3	(1.6)	11	(6.0)
Seizure	-		4	(2.2)

13-cond. = condition, dis. = disorder, IL-2 = aldesleukin, N = number of patients.

One patient experienced an anaphylactic shock 30-40 min after start of the first dinutuximab beta infusion. The patient is recorded as having not received any dinutuximab beta and was therefore excluded from the safety analysis.

Source data: APN311-302 CSR Table 14.3.2.1

Death

Overall, 100 patients died during the study, 51 patients in the dinutuximab beta arm and 49 in the dinutuximab beta + IL-2 arm. Among these, 98 deaths were due to disease progression, while 2 deaths were due to other reasons: 1 patient without IL-2 treatment developed acute respiratory distress causing a multiorgan failure leading to death; 1 patient with IL-2 treatment

experienced pulmonary hypertension with congestive heart failure in Cycle 1 as a consequence of CLS (APN311-302 Tables 14.3.8.1 and 14.3.8.2).

Conclusion

The 3-year EFS with a short-term dinutuximab beta infusion (8-hour infusion for 5 consecutive days, +13-cis-RA) of 61% and 55% with and without IL-2, respectively was higher than EFS rates published for 13-cis-RA only for maintenance treatment for first-line patients (46% [28]). In addition, immunotherapy with dinutuximab beta led to comparable 3-year EFS than treatment with standard triple therapy i.e. ch14.18/SP2/0, IL-2 and GM-CSF (63% [29]). In patients without evidence of disease at baseline, the 3-year EFS with dinutuximab beta was even higher i.e. 69% with and 65% without IL-2. Addition of IL-2 to dinutuximab beta did not show further substantial clinical improvements. On the other hand, IL-2 treatment increased the occurrence of toxicities and reported SAEs, which negatively affected treatment completion.

SIOPEN recommend the use of dinutuximab beta without cytokine (s.c.IL-2) according to the LTI schedule. [1]

5.3.2.6. Study APN311-304

(Source: APN311-304 CSR, draft interim clinical study report version 0.1 dated 18 Dec 2020)

APN311-304 was a Phase II, single arm, open-label, multicenter study designed to evaluate the anti-tumor activity in patients with primary refractory or relapsed neuroblastoma. The primary objective of the study was to evaluate the anti-tumor activity of dinutuximab beta continuous infusion in patients with primary refractory or relapsed neuroblastoma.

Design

Patients: patients with primary relapsed or recurrent neuroblastoma.

Treatment: dinutuximab beta at dose of 10 mg/m²/day was administered as an i.v. infusion, for a period of 10 days/cycle for 5 cycles.

Premedication: i.v. morphine; bolus injection (as required) followed by continuous infusion. Gabapentin starting 3 days before the start of the dinutuximab beta infusion

Safety and efficacy endpoints: primary endpoint: response rate; secondary endpoints: progression-free survival (PFS), OS, PK and pharmacodynamics parameters, immunogenicity, genotyping, safety, and tolerability.

Results - patients

40 patients received treatment: 28 male (70.0%) and 12 female (30.0%) patients (CSR APN311-304 Tables 14.1.2-1, 14.1.2-2, 14.1.2-3). Mean age at baseline was 6.4 years with a minimum of 1 year and a maximum of 19 years (CSR APN311-304, Section 11.2.1).

INSS stage was 4 in 36 patients (90.0%), 3 in 2 patients (5.0%), 1 in 2 patients (5.0%) (CSR APN311-304, Tables 14.1.2-1, 14.1.2-2, 14.1.2-3). At study entry, 20 patients (50.0%) presented with relapsed disease and 18 patients (45.0%) with refractory disease. Twenty-eight of the 40 patients (70.0%) completed all 5 treatment cycles. The mean (standard deviation) number of treatment cycles completed was 4.2 (1.4).

Results - efficacy

Primary efficacy endpoint

For the 38 patients in the FAS, 10 patients (26.3%) showed a CR and PR at End of Study (Table 28). The response rate was statistically significantly higher than 10% ($p = 0.0034$, exact binomial test [one-sided] for response rate $\geq 10\%$). For the 35 patients in the per-protocol set, the response rate at End of Study was comparable (28.6%, $p = 0.0017$).

Table 28: APN311-304 – Treatment response at end of study (FAS)

	Statistics	N=38	p-value
Response (CR + PR)	N (%)	10 (26.3%)	0.0034
No response	N (%)	28 (73.7%)	

Abbreviation: CR = complete response, FAS = full analysis set, PD = progressive disease, PR = partial response, SD = stable disease.

A patient is considered as responder, if the patient's overall response assessment demonstrates CR or PR. Patients with SD, PD, not evaluable or missing overall response assessment are considered as non-responder.

An exact binomial test (one-sided) for response rate $\geq 10\%$

Source data: APN311-304, Table 14.2-4.1.1

The best response rate (best overall tumor response observed at any time point for each individual patient) was 36.8%. For the 38 patients in the FAS, 4 patients (10.5%) had a CR as best response and 10 patients (26.3%) had a PR as best response (Table 29).

Table 29: APN311-304 – Overall treatment response by time point (FAS)

	Statistics	Best response
	N	38
Complete response	N (%)	4 (10.5%)
Partial response	N (%)	10 (26.3%)
Stable disease	N (%)	17 (44.7%)
Progressive disease	N (%)	7 (18.4%)

Abbreviation: FAS = full analysis set

The best overall tumor response observed at any time point for each individual patient is considered the patient's best overall response.

Source data: APN311-304, Table 14.2-4.2.2.

Secondary efficacy endpoint

Progression-Free Survival was defined as the number of days from start of treatment (i.e. first day of the first cycle) until disease progression evaluated by any of the 3 methods (CT/MRI, mIBG or BM examination), or until death. The number of censored observations and the mean time to progression is presented in Table 30. Of the 38 patients, 15 patients experienced disease progression and for 23 patients, the observation was censored.

Table 30: APN311-304 - Progression-free survival (FAS)

Progression-free Survival	Statistics	N=38	95% CI
Disease progression	N (%)	15 (39.5%)	
Censored	N (%)	23 (60.5%)	
Mean time to progression ¹	Days (SE)	288.5 (23.9)	
Quartile estimates			
75%	KM estimate (days)	-	394.0 -
50%	KM estimate (days)	394.0	246.0 -
25%	KM estimate (days)	168.0	64.0 - 267.0

Abbreviations: CI = confidence interval; FAS = full analysis set; KM = Kaplan-Meier; SE = standard error

¹ The mean survival time and its standard error were underestimated because the largest observation was censored and the estimation was restricted to the largest event time.

Source data: APN311-304, Table 14.2-4.2.5

Overall Survival was defined as the number of days from start of treatment (i.e. first day of the first cycle) until death. Overall OS results in the FAS are summarized in [Table 31](#). Of the 38 patients, 1 patient died 36 days after the start of treatment. For the remaining 37 patients the observations were censored. Because of the large number of censored observations, no reliable mean OS could be determined.

Table 31: APN311-304 - Overall survival (FAS)

Overall Survival	Statistics	N=38
Death	N (%)	1 (2.6%)
Censored	N (%)	37 (97.4)

Abbreviations: FAS = full analysis set

Source data: AON311-304, Table 14.2-4.2.8.

Urinary catecholamine metabolites

Evaluation of disease was performed by measuring urinary catecholamine metabolites at Screening, at Mid-evaluation (2 weeks prior to the end of Cycle 2), at end of treatment, at the 12-week FU visit, and at End of Study (date of the last data collection).

Results of the analysis of urinary catecholamine metabolites homovanillic acid and vanillylmandelic acid are presented in APN311-304, Table 11-3.

Immunophenotyping

The analysis of immunophenotyping included the measurement of cytotoxic NK cells (CD16⁺/CD56⁺ dim NK cells), regulatory T-cells (CD4⁺/CD25⁺/CD127⁻) and activated granulocytes and monocytes (CD64). Measurements were performed on Day 1 (prior to the start of the dinutuximab beta infusion) and Day 8 of each treatment cycle.

Results of the immunophenotyping analysis are presented in APN311-304, Table 14.3-4.3.2 and Table 14.2-4.3.2.

Results - safety

Adverse events

All patients reported at least one AE and 39 patients (97.5%) had TEAEs that were considered at least possibly related to treatment with dinutuximab beta (Table 32). For 24 patients (60.0%), serious TEAEs were reported, of which 8 patients (20.0%) had serious TEAEs that were considered at least possibly related to treatment with dinutuximab beta. One patient (2.5%) had a TEAE which led to permanent withdrawal from study drug treatment. Throughout this study, 26 (65.0%) patients experienced TEAEs of severe intensity (CTCAE Grade 3) and 1 patient (2.5%) of fatal intensity (CTCAE Grade 5). No patient experienced a Grade 4 event.

Table 32: APN311-304 - Overall summary of treatment-emergent adverse events (SAF)

Patients with	Number of (%) patients with TEAEs	
	(N=40)	Number of TEAEs
Any TEAE	40 (100.0%)	1433
Any related TEAE	39 (97.5%)	1019
Any serious TEAE	24 (60.0%)	42
Any related serious TEAE	8 (20.0%)	10
Any TEAE leading to permanent withdrawal from study drug	1 (2.5%)	1
Maximum NCI CTCAE Grade		
Mild (Grade 1)	40 (100.0%)	1153
Moderate (Grade 2)	39 (97.5%)	217
Severe (Grade 3)	26 (65.0%)	62
Fatal (Grade 5)	1 (2.5%)	1
Any TEAE leading to death	1 (2.5%)	1

Abbreviations: NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events; SAF = safety analysis set; TEAE = treatment-emergent adverse event.

A related TEAE was defined as a TEAE assessed as definitely, probably, possibly related or with relationship assessment missing.

Source data: APN311-304 Table 14.3-5.1.1 and Table 14.3-5.1.5

The most frequently reported AEs were weight increased, sinus tachycardia, pyrexia, and abdominal pain (Table 33).

Table 33: APN311-304 - Frequent adverse events (reported in ≥20% of patients)

SYSTEM ORGAN CLASS	Total (N = 40)	
Preferred term	Number ¹ (%) ² of patients with TEAEs	Number of TEAEs
ANY ADVERSE EVENT	40 (100.0)	1433
GASTROINTESTINAL DISORDERS	40 (100.0)	224
Abdominal pain	32 (80.0)	70
Constipation	20 (50.0)	35
Vomiting	19 (47.5)	37
Diarrhea	18 (45.0)	31
Nausea	12 (30.0)	21
INVESTIGATIONS	39 (97.5)	144
Weight increased	38 (95.0)	112

SYSTEM ORGAN CLASS	Total (N = 40)	
Preferred term	Number¹ (%)² of patients with TEAEs	Number of TEAEs
C-reactive protein increased	8 (20.0)	8
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	38 (95.0)	211
Pyrexia	34 (85.0)	85
Face edema	23 (57.5)	44
Pain	16 (40.0)	21
Fatigue	13 (32.5)	17
Edema peripheral	9 (22.5)	11
Edema	8 (20.0)	8
CARDIAC DISORDERS	37 (92.5)	147
Sinus tachycardia	35 (87.5)	139
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	34 (85.0)	133
Pruritus	21 (52.5)	32
Dry skin	18 (45.0)	44
Rash	13 (32.5)	18
Petechiae	8 (20.0)	11
INFECTIONS AND INFESTATIONS	31 (77.5)	108
Rhinitis	23 (57.5)	53
Infection	8 (20.0)	15
VASCULAR DISORDERS	31 (77.5)	100
Hypotension	25 (62.5)	60
Capillary leak syndrome	12 (30.0)	14
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS	30 (75.0)	80
Pain in extremity	25 (62.5)	46
Back pain	8 (20.0)	14
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	29 (72.5)	111
Cough	26 (65.0)	63
Oropharyngeal pain	8 (20.0)	16
EYE DISORDERS	27 (67.5)	62
Periorbital edema	12 (30.0)	23
NERVOUS SYSTEM DISORDERS	21 (52.5)	31
RENAL AND URINARY DISORDERS	10 (25.0)	16
BLOOD AND LYMPHATIC SYSTEM DISORDERS	10 (25.0)	14
METABOLISM AND NUTRITION DISORDERS	10 (25.0)	14
PSYCHIATRIC DISORDERS	8 (20.0)	10

Abbreviations: TEAE = treatment-emergent adverse event.

¹: Number of patients with any TEAE, each patient was counted only once within each category

²: Number of patients with any TEAE / number of all patients

Source data: APN311-304, Table 14.3-5.1.2

Grade 3 TEAEs were reported for 65.0% of patients. Most Grade 3 events occurred in 1 patient each. Frequent Grade 3 events reported in $\geq 10\%$ of patients were infection (7 [17.5%]), device related infection (4 [10.0%]), and pain in extremity (4 [10.0%]).

Most frequently reported related TEAEs were weight increased (92.5% of patients), sinus tachycardia (85.0%), pyrexia (82.5%), and abdominal pain (80.0%). All other related TEAEs were reported in less than 80% of the patients on a preferred term level (APN311-304, Section 12.2.3). One TEAE leading to death (reported term: neoplastic meningitis; cause of

death: massive CNS infestation triggered by neuroblastoma) was reported for 1 patient (APN311-304, Section 12.3.1.1). One patient experienced a TEAE that resulted in permanent discontinuation of study drug. The patient experienced a severe (Grade 3) fracture (reported as a serious TEAE), which was considered unrelated to DB by the investigator.

The most frequent SAEs were infection in 8 patients (20.0%), pyrexia in 4 patients (10.0%), and device related infection in 3 patients (7.5%). The SAE of pyrexia was judged to be dinutuximab beta-related (APN311-304, Table 14.3.5.1-3).

Laboratory, vital signs, and other safety assessments

There were no trends regarding a dose-relationship. Three patients experienced clinically significant out-of-range hematology values during a first or second treatment cycle, concerning changes in hemoglobin, platelets, and white blood cells. Twelve patients presented with clinically significant chemistry values, mainly during the first treatment cycle, concerning mainly changes in C-reactive protein and sodium. (APN311-304, Section 12.4).

No clinically significant changes were reported in vital signs, chest X-ray, ECG measurements, electroencephalogram recordings, ECHO, and pulmonary function tests. With the exception of 1 abnormal result at end of treatment, all abnormal results were not clinically significant. Clinically significant physical examination findings mainly concerned general appearance and abdomen, and in addition skin, eyes, cardiac, throat, lymph nodes, and extremity (APN311-304, Section 12.6).

Conclusion

A 10-day infusion schedule of 10 mg/m² dinutuximab beta (total dose 100 mg/m²) was shown to be tolerable with a reduced pain-toxicity profile of dinutuximab beta while maintaining clinical activity in patients with primary refractory or relapsed neuroblastoma.

5.3.2.7. Adverse events of special interest

Typical side effects of dinutuximab beta or other anti-GD2 antibodies include pain, hypersensitivity and infusion reactions, and CLS. To further characterize these adverse reactions, the safety database of dinutuximab beta was searched for specific terms associated with these reactions, using modified Standardized MedDRA Queries and/or pre-specified algorithms.

5.3.2.7.1. Undesirable Effects

The safety of dinutuximab beta has been evaluated in 514 patients with high-risk and relapsed/refractory neuroblastoma, who received it as a continuous infusion (98) or as repeated daily infusions (416). It was combined with 13-cis retinoic in most patients and with IL-2 in 307 patients.

The most common adverse reactions were pyrexia (88%) and pain (77%) that occurred despite analgesic treatment. Other frequent adverse reactions were hypersensitivity (63%), vomiting (57%), diarrhea (51%), CLS (40%) and hypotension (39%).

Description of selected adverse reactions

Hypersensitivity

The most frequent hypersensitivity reactions included hypotension (39%), urticaria (18%) and bronchospasm (4%). Cytokine release syndrome was also reported in 32% of the patients. Serious anaphylactic reactions occurred in 3.5% of the patients.

Pain

Pain typically occurs during the first infusion of dinutuximab beta and decreases over the treatment courses. Most commonly, patients reported abdominal pain, pain in the extremities, back pain, chest pain, or arthralgia.

Capillary leak syndrome

Overall, 10% of CLS were severe (Grade 3 to 4) and their frequency decreased over the treatment courses.

Eye problems

These included impaired visual accommodation that is correctable with eyeglasses, as well as mydriasis (13%), blurred vision (3%), or photophobia (3%), which were usually reversible after treatment discontinuation. Severe eye disorders were also reported including ophthalmoplegia (2%) and optic atrophy.

Peripheral neuropathy

Both motor and sensory peripheral neuropathies have been reported, overall in 9% of the patients. Most events were of Grade 1 to 2 and resolved.

Safety profile with and without IL-2

The combination of dinutuximab beta with IL-2 increases the risk of adverse drug reactions compared to dinutuximab beta without IL-2, especially for pyrexia (92% vs. 79%), CLS (50% vs. 25%), pain related to dinutuximab beta (75% vs. 63%), hypotension (43% vs. 26%), and peripheral neuropathy (14% vs. 7%), respectively (Summary of Product Characteristics [SmPC] dated 19 Dec 2019) [31].

5.4 Marketing experience

The European Commission granted a marketing authorization valid throughout the European Union for dinutuximab beta Apeiron on 8 May 2017 and the name of the product was changed to dinutuximab beta EUSA on 4 Aug 2017 and then from dinutuximab beta EUSA to the tradename QARZIBA[®] on 27 Nov 2017 with EUSA Pharma as Marketing Authorization Holder.

The Periodic Safety Update Report 6 (Periodic Safety Update Report [PSUR] 6) covers all relevant safety data received by the MAH for the period from 09 Nov 2019 to 08 Nov 2020 and cumulatively since the approval on 08 May 2017.

The number of patients treated since the product was approved in the European Economic Area (EEA) is estimated based on a standard calculated individual patient dose. Since product launch, a total of 1,719,630 mg of QARBIZA[®] have been distributed in the EEA, corresponding to an estimation of 4,647.6 individual treatments supplied. Additionally, in the

rest of the world, a QARZIBA[®] is available on a named-patient basis including managed access programs and global access program. Since the EU MA approval, 1,429.05 individual treatments have been supplied through the named-patient basis.

Based on the review of data received during this PSUR period, the known safety profile of dinutuximab beta remains unchanged.

5.5 Risk-benefit assessment

Neuroblastoma, the most common extracranial solid tumor in children, remains one of the major challenges in pediatric oncology. Despite the introduction of novel treatment strategies, the outcome of high-risk neuroblastoma patients remains poor [4,5] and new effective adjuvant therapeutics are needed to further improve outcome in this disease.

Dinutuximab beta is a mouse-human chimeric monoclonal IgG1 antibody targeting disialoganglioside GD2 which is expressed on virtually all neuroblastoma cells as well as on other high-risk tumors.

Efficacy

Dinutuximab beta either given as 5 daily short-term infusions (8-hours) or continuous infusion over 24 hours for 10 days demonstrated significant clinical benefit in terms of ORR, EFS, and OS both in patients with high-risk neuroblastoma during first-line treatment and in the relapsed/refractory setting.

With the continuous infusion regimen, ORR at end of treatment in relapsed/refractory patients was 29% (relapsed disease) and 41% (refractory disease), with EFS rates at 2 years between 29% and 56% and OS rates at 2 years between 42% and 78%. In historical controls, the OS rate at 3 years was only between 21% and 24%.

In first-line patients, the 8-hour infusion resulted in an ORR of 51% with and 45% without IL-2. EFS at 3 years with and without IL-2 were 62% and 66%, and OS at 3 years 71% and 72%, respectively.

In the relapsed/refractory patient population after haploidentical SCT, the 8-hour infusion resulted in an ORR of 65% with EFS and OS rates of 40% and 55% at 3 years.

The efficacy was independent of disease status at baseline, although patients who had received immunotherapy as adjuvant treatment following first-line therapy had greater benefits than those with relapsed or refractory neuroblastoma. However, compared to multimodality treatment for relapsed/refractory neuroblastoma without immunotherapy, treatment including dinutuximab beta showed significant advantages. Efficacy was independent of IL-2 co-administration. Monotherapy with dinutuximab beta showed similar survival outcomes as combination treatment with IL-2 and was comparable to therapy with a similar anti-GD2 antibody, ch14.18/SP2/0 in combination with i.v. IL-2 and GM-CSF in the population of complete responders after multimodality treatment. [2]

Safety

Dinutuximab beta is associated with considerable but manageable toxicities. The most prominent adverse reaction with dinutuximab beta is neuropathic pain, and skilled pain management including the use of morphine is a prerequisite when dinutuximab beta is

administered. Further dinutuximab beta-related frequent toxicities include pyrexia, hypersensitivity reactions, pruritus, CLS, cytokine release syndrome, hypotension, and tachycardia. A continuous infusion regimen of dinutuximab beta over 10 days, i.e. a lower dinutuximab beta concentration per hour of infusion, leads to an improved pain-toxicity profile and omission of IL-2 as co-treatment reduces side effects and supports compliance. Administration of the compound must be performed by well-trained physicians with experience in treating oncologic pediatric patients and severe allergic reactions. Patient safety during and after administration must be closely monitored.

Conclusion

In conclusion, dinutuximab beta has a favorable benefit-risk profile for the treatment of high-risk neuroblastoma in patients aged 12 months and above, who have previously received induction chemotherapy and achieved at least a PR, followed by myeloablative therapy and stem cell transplantation, as well as patients with history of relapsed or refractory neuroblastoma, with or without residual disease (PSUR 6 [09 Nov 2019 to 08 Nov 2020]).

6. Reference safety information

The European Commission granted a marketing authorization valid throughout the European Union for dinutuximab beta on 8 May 2017. Section 4.8 of the SmPC forms the Reference Safety Information for all pharmacovigilance activities for the product in the post-marketing environment. Use of this data for assessment of adverse event reports from clinical studies will also ensure consistent assessment of expectedness for cases, and also help to ensure that all healthcare professionals using dinutuximab beta have the same information on the safety of the product. This investigator's brochure has been updated to include the tabulated information on expected adverse reactions contained in Section 4.8 of the current SmPC (dated 19 Dec 2019).

Life-threatening events the nature, severity or outcome of which are consistent with the information given below will not be considered unexpected. Events with a fatal outcome will be considered unexpected.

Summary of the safety profile

The safety of dinutuximab beta has been evaluated in 514 patients with high-risk and relapsed/refractory neuroblastoma, who received it as a continuous infusion (98) or as repeated daily infusions (416). It was combined with 13-cis retinoic in most patients and with IL-2 in 307 patients.

The most common adverse reactions were pyrexia (88%) and pain (77%) that occurred despite analgesic treatment. Other frequent adverse reactions were hypersensitivity (63%), vomiting (57%), diarrhea (51%), CLS (40%) and hypotension (39%).

Adverse reactions are summarized in the table below. These adverse reactions are presented by MedDRA system organ class and frequency. Frequency categories are defined as very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), and uncommon ($\geq 1/1,000$ to $< 1/100$). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System organ class	Very common	Common	Uncommon
Infections and infestations	infection (including pneumonia, skin infection, herpes virus infection, myelitis, encephalomyelitis), device related infection	sepsis	
Blood and lymphatic system disorders	anemia, leukopenia, neutropenia, thrombocytopenia	lymphopenia	disseminated intravascular coagulation, eosinophilia
Immune system disorders	hypersensitivity, cytokine release syndrome	anaphylactic reaction	serum sickness
Metabolism and nutrition disorders	fluid retention	decreased appetite, hypoalbuminemia, hyponatremia, hypokalemia, hypophosphatemia, hypomagnesemia, hypocalcemia, dehydration	
Psychiatric disorders		agitation, anxiety	
Nervous system disorders	headache	peripheral neuropathy, seizure, paraesthesia, dizziness, tremor	intracranial pressure increased, posterior reversible encephalopathy syndrome
Eye disorders	mydriasis, pupillotonia, eye edema (eyelid, periorbital)	ophthalmoplegia, papilledema, accommodation disorder, blurred vision, photophobia	
Cardiac disorders	tachycardia	cardiac failure, left ventricular dysfunction, pericardial effusion	
Vascular disorders	hypotension, capillary leak syndrome	hypertension	hypovolemic shock, veno-occlusive disease

System organ class	Very common	Common	Uncommon
Respiratory, thoracic and mediastinal disorders	hypoxia, cough	bronchospasm, dyspnea, respiratory failure, lung infiltration, pulmonary edema, pleural effusion, tachypnoea, laryngospasm	
Gastrointestinal disorders	vomiting, diarrhea, constipation, stomatitis	nausea, lip edema, ascites, abdominal distension, ileus, dry lips	enterocolitis
Hepatobiliary disorders			hepatocellular injury
Skin and subcutaneous tissue disorders	pruritus, rash, urticaria	dermatitis (including exfoliative), erythema, dry skin, hyperhidrosis, petechiae, photosensitivity reaction	
Musculoskeletal and connective tissue disorders		muscle spasms	
Renal and urinary disorders		oliguria, urinary retention, hyperphosphaturia, hematuria, proteinuria	renal failure
General disorders and administration site conditions	pyrexia, chills, pain*, peripheral edema, face edema	injection site reaction	
Investigations	increased weight, increased transaminases, increased gamma glutamyl transferase, increased blood bilirubin increased blood creatinine	decreased weight, decreased glomerular filtration rate, hypertriglyceridemia, prolonged activated partial thromboplastin time, prolonged prothrombin time, prolonged thrombin time	

*includes abdominal pain, pain in extremity, musculoskeletal pain, chest pain, arthralgia

7. Summary of data and guidance for the investigator

7.1 The investigational product

Mode of action

Dinutuximab beta is a chimeric monoclonal antibody produced in CHO cells targeting the disialoganglioside GD2 antigen highly expressed by neuroectodermal tumors such as neuroblastoma, melanoma cells, and several other tumors.

“First generation” antibodies targeting the disialoganglioside GD2 antigen were entirely mouse products causing a quick immunological response. Dinutuximab beta belongs to the “second generation” anti-GD2 antibodies, chimeric products, and consists of approximately 30% mouse protein and 70% human protein.

Dinutuximab beta mediates CDC against GD2-bearing cells. Furthermore, dinutuximab beta targets cytotoxic immune effector cells towards GD2-expressing tumor cells (ADCC).

Dosage and administration

Dinutuximab beta must be administered by a healthcare professional prepared to manage severe allergic reactions including anaphylaxis. Infusion of dinutuximab beta must be initiated in an environment where full resuscitation services are immediately available.

Dinutuximab beta is prepared and administered as a short-term or continuous infusion as described by the respective study protocol. In case of STI, dinutuximab beta is administered as 8-hour i.v. infusion over 5 consecutive days, in case of i.v. continuous infusions, dinutuximab beta is administered over 10 consecutive days.

The daily dose of dinutuximab beta is calculated individually, depending on the body surface area or body weight of patients:

- according to a recent determination of their body surface area, at a rate of 10 mg/m²/day. The total dose of dinutuximab beta administered per course is 100 mg/m² body surface.
In some protocols dosing may be different for patients ≤12 kg e.g.:
- Patients ≤12 kg and >5 kg: according to their body weight, at a rate of 0.33 mg/kg/day. The total dose of dinutuximab beta administered per cycle is 3.3 mg/kg body weight.
- Patients ≤5 kg are dosed according to their body weight, at a rate of 0.22 mg/kg/day. The total dose of dinutuximab beta administered per cycle is 2.2 mg/kg body weight.

Please refer to the protocol for exact dosing.

Dinutuximab beta with a specified strength of 4.5 mg/mL is pre-diluted aseptically to the patient-specific concentration/dose with 0.9% sodium chloride solution, containing 1% human albumin. After dilution, the dinutuximab beta-containing solution is administered via i.v., using a commercially available pump.

The solution for infusion can be prepared freshly on a daily basis, or once for the total dinutuximab beta amount needed for up to 5 days of continuous infusion. The amount of solution to be infused for one of the consecutive 10 days within a treatment course should be 48 mL, and 240 mL for a 5-day dose, i.e. it is recommended to prepare 50 mL solution in a

50 mL syringe or, respectively, 250 mL in an infusion bag suitable for the employed infusion pump i.e. an overfill of 2 mL (syringe) or 10 mL (infusion bag) is included in consideration of respective dead volumes of the infusion systems.

The solution is prepared by first adding the required amount of human albumin (e.g. human albumin 20%) and swirling to coat the internal plastic surfaces. This is to prevent non-specific binding of the antibody to plastic surfaces. Then add about one third of the required amount of the sodium chloride solution. Mix the solution gently by inverting the infusion container 10 times. Using a large bore needle (e.g. 18-gauge), the calculated amount of dinutuximab beta is drawn from the vials and added to the infusion container. The solution is mixed by gently inverting the infusion container 10 times. The infusion container must not be shaken, to avoid frothing or excessive shearing of the protein solution. Then physiological sodium chloride solution is added up to the final volume of 50 mL or 250 mL and mixed by gently inverting the container 10 times.

The solution for infusion must be prepared under sterile conditions. The solution must not be exposed to direct sunlight or heat. The diluted dinutuximab beta solution should be maintained at $5^{\circ}\text{C} \pm 3^{\circ}\text{C}$ and administration should be started as soon as possible, but at least within 24 hours after dilution (storage time). The diluted solution can be administered via a peripheral or central i.v. line. Other i.v. co-administered agents should be delivered via a separate infusion line and should not be mixed with dinutuximab beta. As for any other parenteral drug product the infusion container should be inspected visually for particulates prior to administration. It is recommended that a $0.22\text{ }\mu\text{m}$ in-line filter is used during infusion. Refer to the respective study protocol for study specific filtration instructions.

Any medical device suitable for continuous infusion of the prepared antibody solution at the required infusion rate can be applied, e.g. syringe infusion pumps, infusors, electronic ambulatory infusion pumps (e.g. pump for pain management, pump for total parenteral nutrition) etc. Elastomeric pumps are not considered suitable in combination with in-line filters.

Dinutuximab beta may be given in combination with s.c. IL-2 as dictated in the respective study protocol. IL-2 is prepared and administered as stipulated by the study protocol.

If possible, the patient should receive the planned total dose of dinutuximab beta continuously within each course. However, based on the physician's evaluation of the severity of dinutuximab beta-related adverse reactions, patients may undergo a dose reduction of 50% or a temporary interruption of the infusion. For specific dose modification schedules, please refer to the study protocol.

Storage

Store in a refrigerator (2°C to 8°C).

Chemical and physical in-use stability of dinutuximab beta has been demonstrated for up to 48 hours at 25°C (50 mL syringe) and for up to 7 days at 37°C (250 mL infusion bag), after cumulative storage in a refrigerator (2°C to 8°C) for 72 hours.

7.2 Non-clinical safety data

General toxicology

Dinutuximab beta has been administered to male and female juvenile guinea pigs as well as male and female young cynomolgus monkeys as repeat-dose regimens that exceeded the recommended clinical dose. Findings of note included decrease in thymus weight as well as bone marrow changes (atrophy affecting myeloid and erythroid precursor cell lines). The bone marrow changes were slight to severe and recovered after cessation of dosing. No effects on cardiovascular functions (ECG, blood pressure) were observed in monkeys.

Other

No non-clinical studies to evaluate the potential of dinutuximab beta to cause carcinogenicity, genotoxicity or developmental and reproductive toxicity have been conducted. In the repeat-dose toxicity studies in guinea pigs and cynomolgus monkeys, no adverse effects of dinutuximab beta were observed on reproductive organs at exposure levels above clinical levels.

7.3 Pharmacokinetics in humans

Distribution

After 10 days continuous i.v. of 10 mg/m²/day dinutuximab beta (equal to a total dose of 100 mg/m²/cycle), mean plasma C_{max} levels (around 12 µg/mL) were reached on the last day of infusion.

Mean plasma C_{max} levels, observed during an 8-hours short-term infusion (20 mg/m²/day on 5 consecutive days) were slightly higher (16.5 µg/mL) and were reached on the 5th day of infusion.

Biotransformation

Dinutuximab beta is a protein for which the expected metabolic pathway is degradation to small peptides and individual amino acids by ubiquitous proteolytic enzymes. Classical biotransformation studies have not been performed.

Elimination

The half-life after LTI was about 190 hours, i.e. 8 days.

7.4 Clinical particulars

Clinical efficacy

Clinical trials provide evidence that all variants of the anti-GD2 monoclonal antibody are useful in passive immunotherapy of cancer. Therapeutic responses in patients with GD2-positive tumors have been obtained in clinical studies done with the murine variant of 14.18 (14.G2a), and the human-murine chimeric antibody produced in SP2/0 cells (ch14.18/SP2/0). Main results for dinutuximab beta are summarized below.

Continuous dinutuximab beta infusion for 10 days in combination with s.c. IL-2

Fifty-four (54) patients with relapsed or refractory neuroblastoma received 10 mg/m²/day dinutuximab beta given by continuous 10-day i.v. infusion in 5-week treatment courses,

concurrently with s.c. IL-2 treatment (study APN311-303). The primary efficacy outcome measure was EFS, defined as time from the onset of the immunotherapy to the occurrence of relapse, PD, secondary malignancy, or death. EFS and OS rates were 52% and 92% at 1 year, 35% and 75% at 2 years, and 31% and 66% at 3 years, respectively. Median time to progression was 20 months. ORR in patients with evidence of disease at baseline was 41% (with 14% CR and 27% PR).

In a second study using the same treatment schedule in 122 patients with relapsed or refractory neuroblastoma, ORR in patients with no evidence of disease at baseline was 22.1% (27.3% SD, 14.3% PD). EFS and OS were 62.7% and 90.7% at 1 year, 56.8% and 73.7% at 2 years, and 51.7% and 67.0% at 3 years, 49.1% and 60.9% at 4 years, and 48.2% and 57.8% at 5 years, respectively (study APN311-202 V1 and V2). Of 77 patients with detectable disease at baseline, 17 (22.1%) had no evidence of disease as best treatment response and 27 (35.1%) had improved disease. Therefore, the best response rate was 57.1%. The mean duration of response was 122.5 days (median: 118.0 days). Mean time to progression was 1329 days. The best response was higher in patients with refractory baseline disease status (62.2%) than in patients with relapsed baseline disease status (50.0%) (study APN311-202 V1 and V2).

In a third study using the same treatment schedule, 40 patients treated with primary refractory or relapsed neuroblastoma. For the 38 patients in the FAS, 10 patients (26.3%) showed a CR and PR at End of Study. The best response rate (best overall tumor response observed at any time point for each individual patient) was 36.8% (APN311-304).

Short-term (8 hours) dinutuximab beta infusion for 5 days in combination with IL-2

In another study, in which dinutuximab beta was administered in combination with subcutaneous IL-2 after haplo-SCT in patients with relapsed neuroblastoma, 50% of patients with evidence of disease after haplo-SCT had CR as best response (study APN311-201). The median time (from initial diagnosis to haploidentical stem cell transplantation) to progression was 33 months. EFS and OS were 72.2% and 86.4% at 1 year, 51.2% and 68.6% at 2 years, and 47% and 57.9% at 3 years, respectively.

Short-term (8 hours) dinutuximab beta infusion for 5 days with and without IL-2 as part of first-line therapy

In a Phase III study, dinutuximab beta (20 mg/m²/day for 5 days), with or without additional subcutaneous IL-2 treatment was investigated as part of the first-line therapy of 385 high-risk neuroblastoma patients (study APN311-302). Addition of IL-2 did not improve EFS or OS. EFS and OS were 72% and 86% at 1 year, 55% and 64% at 3 years, and 42% and 50% at 5 years, respectively, without IL-2 and 72% and 88% at 1 year and 61% and 69% at 3 years, and 57% and 64% at 5 years, respectively, with IL-2.

Adverse reactions

Serious adverse reactions were derived from 6 clinical studies of dinutuximab beta, i.e. 5 studies in patients with relapsed/refractory neuroblastoma and 1 study in patients with first-line high-risk neuroblastoma.

Special warnings and precautions

Pain

Neuropathic pain/pain usually occurs at the beginning of antibody treatment. Premedication with analgesics, including i.v. opioids, prior to each dose of dinutuximab beta is required. A triple therapy, including non-opioid analgesics (according to WHO guidelines), gabapentin and opioids, is recommended for pain treatment. The individual dose may vary widely.

Non-opioid analgesics should be used permanently during dinutuximab beta treatment e.g. paracetamol or ibuprofen according to institutional standard.

The patient should be primed with gabapentin 10 mg/kg/day orally, starting 3 days prior to the start of continuous dinutuximab beta infusion. The daily dose of gabapentin is increased to 10 mg/kg orally, twice daily, the next day, and to 10 mg/kg orally, 3 times daily, the day before the first day of dinutuximab beta infusion and thereafter. The maximum single dose of gabapentin is 300 mg. This dosing schedule should be maintained for as long as required by the patient. Oral gabapentin shall be tapered off after weaning off i.v. morphine infusion, at the latest after dinutuximab beta infusion therapy has stopped.

Treatment with opioids is administered as standard alongside dinutuximab beta. The first infusion day and cycle usually requires a higher dose than subsequent days and cycles. Nevertheless, before initiation of a continuous i.v. infusion, a bolus infusion of 0.02 to 0.05 mg/kg/hour morphine should be started 2 hours before dinutuximab beta infusion. Subsequently, an i.v. morphine dosing schedule of 0.03 mg/kg/hour is recommended concomitantly with dinutuximab beta infusion. With daily infusions of dinutuximab beta, morphine infusions should be continued at a decreased rate (e.g. 0.01 mg/kg/h) for 4 hours after the end of dinutuximab beta infusion. With continuous infusion, in response to the patient's pain perception, it may be possible to wean off morphine over 5 days by progressively decreasing its dosing rate (e.g. to 0.02 mg/kg/hour, 0.01 mg/kg/hour, 0.005 mg/kg/hour). If continuous morphine infusion is required for more than 5 days, treatment should be gradually reduced by 20% per day after the last day of dinutuximab beta infusion.

After weaning off i.v., in case of severe neuropathic pain, oral morphine sulphate (0.2 to 0.4 mg/kg every 4 to 6 hours) can be administered on demand. For moderate neuropathic pain, oral tramadol may be administered.

Refer to the study protocol for detailed instructions concerning pain management.

Hypersensitivity reactions

Severe infusion related reactions, including cytokine release syndrome and anaphylactic and hypersensitivity reactions, may occur despite the use of premedication. Occurrence of a severe infusion related reaction (including cytokine release syndrome) requires immediate discontinuation of dinutuximab beta therapy and may necessitate emergency treatment.

Cytokine release syndrome frequently manifests itself within minutes to hours of initiating the first infusion and is characterized by systemic symptoms such as fever, hypotension and urticaria.

Anaphylactic reactions may occur as early as within a few minutes of the first infusion with dinutuximab beta and are commonly associated with bronchospasm and urticaria.

Antihistamine premedication (e.g. diphenhydramine) should be administered by i.v. injection approximately 20 minutes before starting each dinutuximab beta infusion. It is recommended that antihistamine administration is repeated every 4 to 6 hours, upon careful assessment of the benefit/risk ratio by the attending oncologist.

Patients should be closely monitored for anaphylaxis and allergic reactions, particularly during the first and second treatment courses.

Intravenous antihistamine, epinephrine (adrenaline) and prednisolone for i.v. administration should be immediately available at the bedside during administration of dinutuximab beta to manage life-threatening allergic reactions. It is recommended that treatment for such reactions include prednisolone administered by i.v. bolus, and epinephrine administered by i.v. bolus every 3 to 5 minutes as necessary, according to clinical response. In case of bronchial and/or pulmonary hypersensitivity reaction, inhalation with epinephrine (adrenaline) is recommended and should be repeated every 2 hours, according to clinical response.

Neurological disorders of the eye

Eye disorders may occur as dinutuximab beta binds to optic nerve cells. No dose modification is necessary in the case of an impaired visual accommodation that is correctable with eyeglasses, as long as this is judged to be tolerable. Treatment with dinutuximab beta must be interrupted in patients who experience Grade 3 vision toxicity (i.e. subtotal vision loss per toxicity scale). In case of any eye problems, patients should be referred promptly to an ophthalmology specialist.

Peripheral neuropathy

Occasional occurrences of peripheral neuropathy have been reported with dinutuximab beta. Cases of motor or sensory neuropathy lasting more than 4 days must be evaluated and non-inflammatory causes, such as disease progression, infections, metabolic syndromes and concomitant medication, should be excluded. Treatment should be permanently discontinued in patients experiencing any objective prolonged weakness attributable to dinutuximab beta administration. For patients with moderate (Grade 2) neuropathy (motor with or without sensory), treatment should be interrupted and may be resumed after neurologic symptoms resolve.

Capillary leak syndrome

Capillary leak syndrome is characterized by a loss of vascular tone and extravasation of plasma proteins and fluid into the extravascular space. Capillary leak syndrome may occur in combination with dinutuximab beta. It usually develops within hours after initiation of treatment, while clinical symptoms (i.e. hypotension, tachycardia) are reported to occur after 2 to 12 hours. Careful monitoring of circulatory and respiratory function is required.

Systemic infections

Patients are likely to be immunocompromised as a result of prior therapies. As they typically have a central venous catheter in situ, they are at risk of developing systemic infection. Patients should have no evidence of systemic infection and any identified infection should be under control before starting therapy.

Hematological toxicities

Occurrence of hematological toxicities has been reported, such as erythropenia, thrombocytopenia or neutropenia. Grade 4 hematological toxicities, improving to at least Grade 2 or baseline values by start of next treatment course, do not require dose modification.

Laboratory abnormalities

Regulatory monitoring of liver function and electrolytes is recommended.

Immunogenicity

The development of ADAs is a class effect of monoclonal chimeric antibodies. Overall, measurable ADA titers were noted in 65 of the patients (62%) examined (N = 105). Given the limitation of the bioanalytical methods, data are currently insufficient to properly evaluate the impact of the formation of ADAs on PK and pharmacodynamic parameters, as well as on the efficacy and safety of dinutuximab beta.

Fertility, pregnancy, and lactation

There are no data on pregnant women. Due to the expression of the dinutuximab beta target (GD2) on neuronal tissues especially during embryofetal development, the cytotoxic potential of dinutuximab beta (CDC/ADCC) and the potential of placental transfer of antibodies, dinutuximab beta may cause fetal harm when administered to pregnant women.

To date, no animal data have been generated concerning teratogenicity or embryotoxicity. Therefore, dinutuximab beta should not be used during pregnancy.

There are no data on lactating women. It is unknown whether dinutuximab beta is excreted in human milk. Breastfeeding should be discontinued during treatment with dinutuximab beta and for 6 months after the last dose.

The effects of dinutuximab beta on fertility in humans are unknown. In animals, dedicated fertility studies have not been conducted, but adverse effects on reproductive organs were not observed in toxicity studies, performed in guinea pig and cynomolgus monkey.

Dinutuximab beta should not be used in women of childbearing potential not using contraception. It is recommended that women of childbearing potential use contraception for 6 months after discontinuation of treatment with dinutuximab beta.

Drug and other interactions

No interaction studies have been performed. A risk for indirect reduction of CYP activity due to higher tumor necrosis factor α and IL-6 levels and, therefore, interactions with concomitantly used medicinal products, cannot be excluded.

Corticosteroids

Due to immunosuppressive activity, the concomitant treatment with corticosteroids is not recommended within 2 weeks prior to the first treatment course and until 1 week after the last treatment course with dinutuximab beta, except for life-threatening conditions.

Vaccinations

Avoid vaccinations during administration of dinutuximab beta until 10 weeks after last treatment course, due to immune stimulation through dinutuximab beta and possible risk for rare neurological toxicities.

Intravenous immunoglobulin

Concomitant use of i.v. immunoglobulins is not recommended as they may interfere with dinutuximab beta-dependent cellular cytotoxicity.

Special populations

Only patients with adequate renal and hepatic function have been investigated to date.

The safety and efficacy of dinutuximab beta in children aged <12 months have not yet been established. No data are available.

Contraindications

Hypersensitivity to the active substance or to any of the excipients of dinutuximab beta.

Acute Grade 3 or 4 GvHD, or extensive chronic GvHD.

Overdose

No cases of dinutuximab beta overdose have been reported. In the case of overdose, patients should be carefully observed for signs or symptoms of adverse reactions and supportive care administered, as appropriate.

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9. Changes from the previous version

The present investigator's brochure replaces the previous Version 3.0. Compared with Version 3.0, this version (4.0) contains the following major changes:

1.2.2 Efficacy	Updated based on the text in Section 5.0
5.1 Overview on clinical studies with dinutuximab beta	Updated based on the status (completed and ongoing) of all studies.
5.3.2.3 Study APN311-102	This section was deleted as study was removed and there is no access to the data
5.3.2.4 Study APN311-201	Updated with CSR dated 21 Dec 2020
5.3.2.5 Study APN311-202	Updated with CSR dated 28 Jul 2020
5.3.2.6 Study APN311-302	Updated with Publication Ladenstein et al, 2020
5.3.2.7 Study APN311-304	Added this section based on the CSR dated 18 Dec 2020
5.3.2.8 Adverse events of special interest	Updated with SmPC dated 19 Dec 2019
5.4 Marketing Experience, 5.5 Risk-benefit assessment	Updated with PSUR dated 08 Nov 2020
7.4 Clinical Particulars	Updated the text based on the CSRs - APN311-202 V1 and V2, APN311-201, APN311-302, and APN311-304
8 References	Updated based on the references cited in the new text

